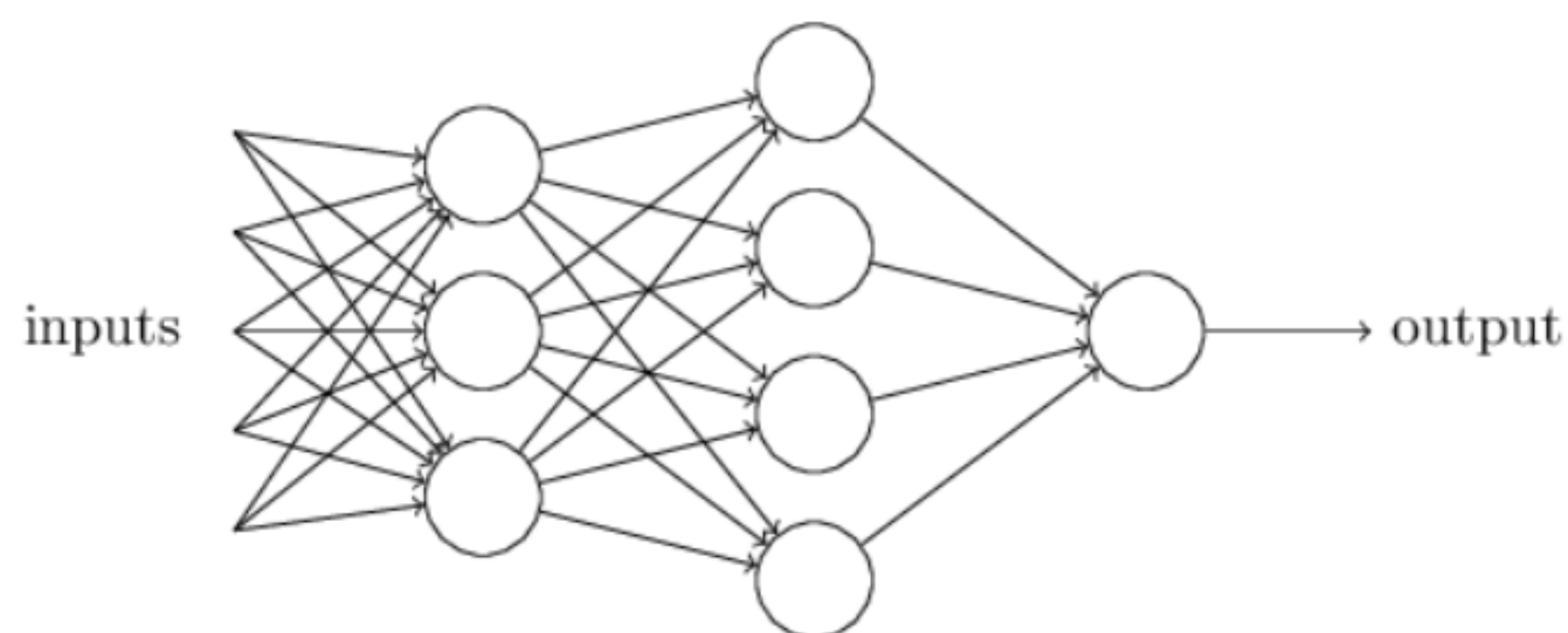
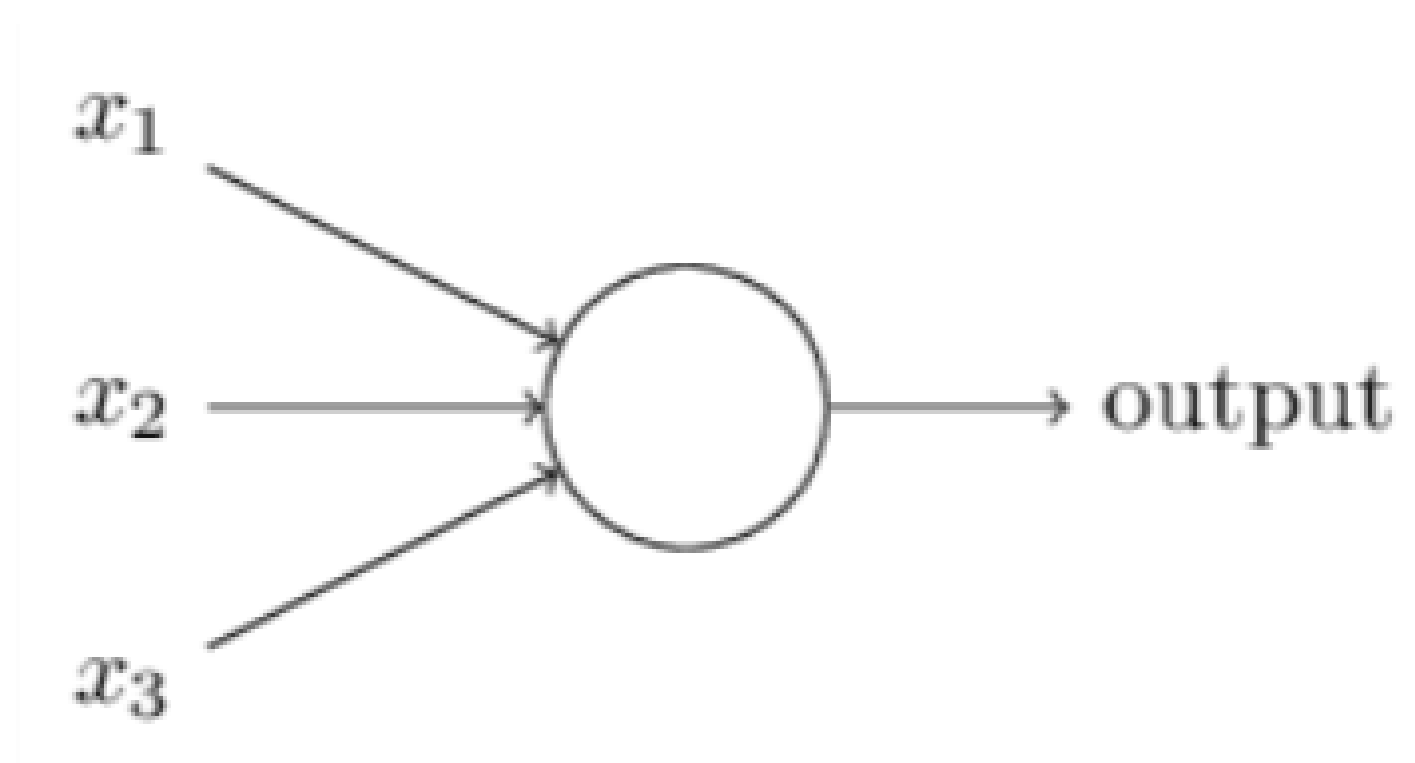


Introduction of Neural Network & Deep Learning

馬誠佑

2024/10/17

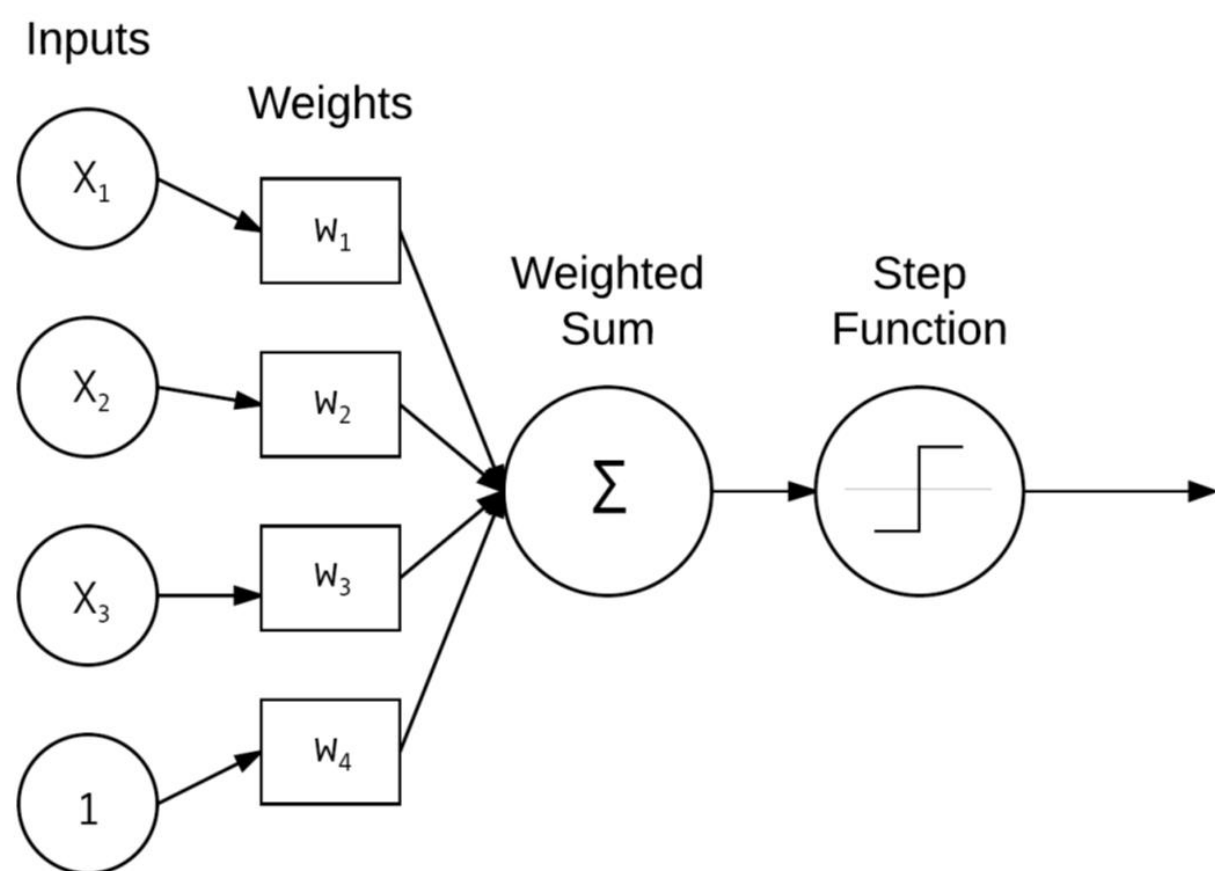
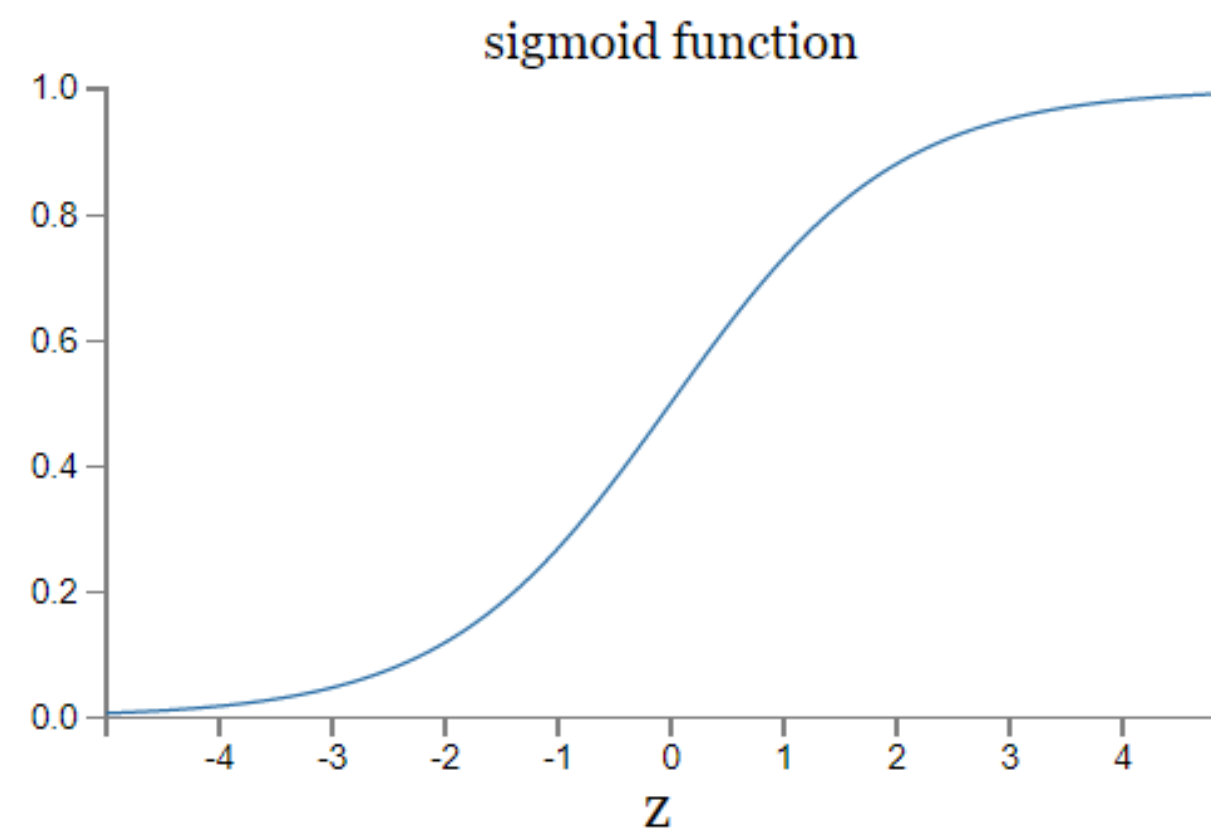
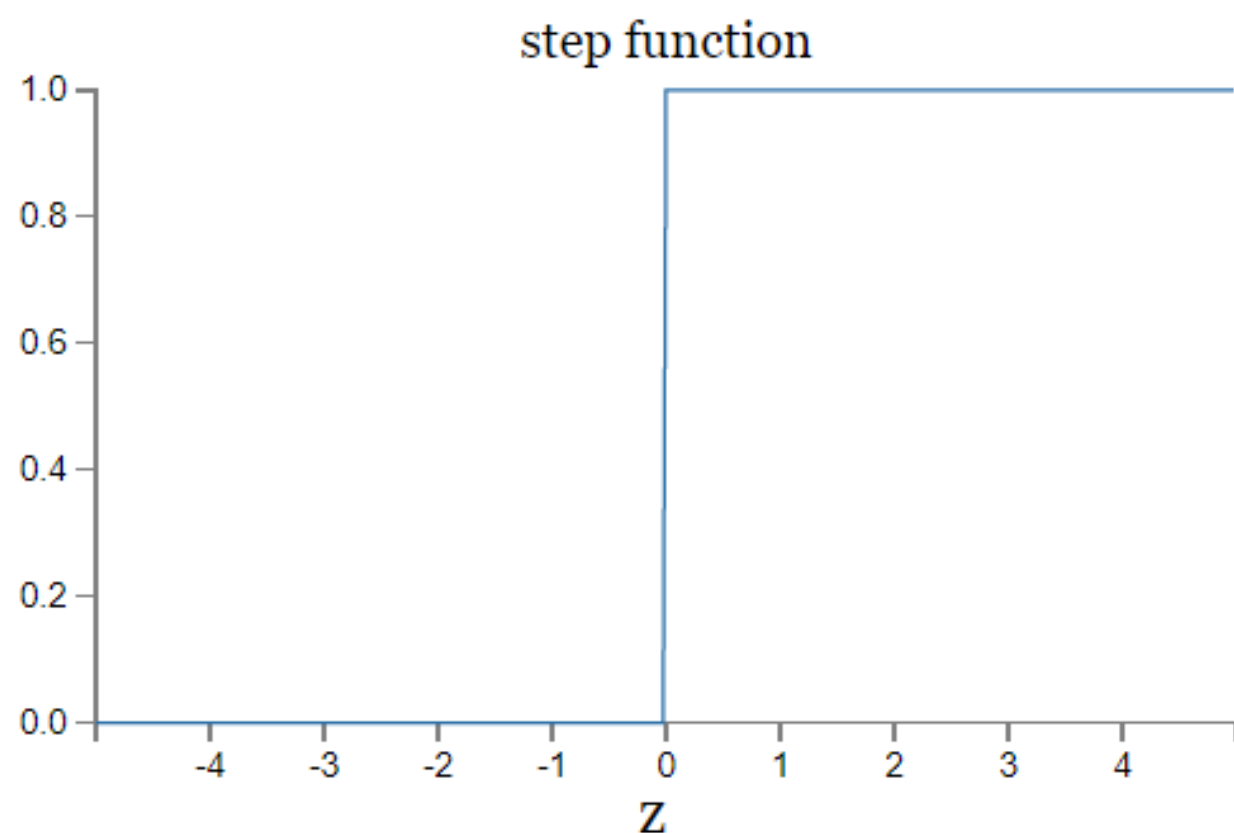
Neural Network 深度學習模型是一連串幾何轉換總和



$$\text{output} = \begin{cases} 0 & \text{if } \sum_j w_j x_j \leq \text{threshold} \\ 1 & \text{if } \sum_j w_j x_j > \text{threshold} \end{cases}$$

$$\text{output} = \begin{cases} 0 & \text{if } w \cdot x + b \leq 0 \\ 1 & \text{if } w \cdot x + b > 0 \end{cases}$$

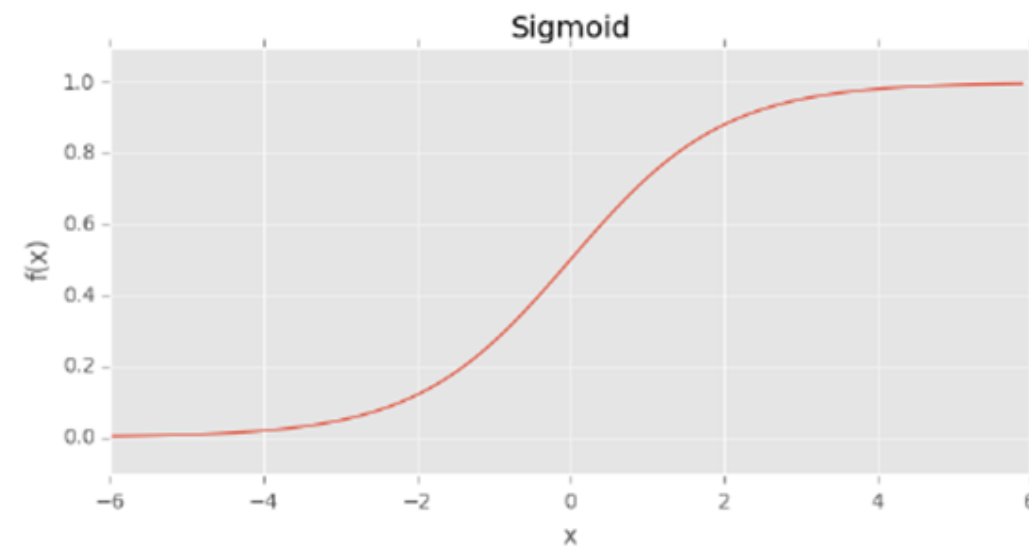
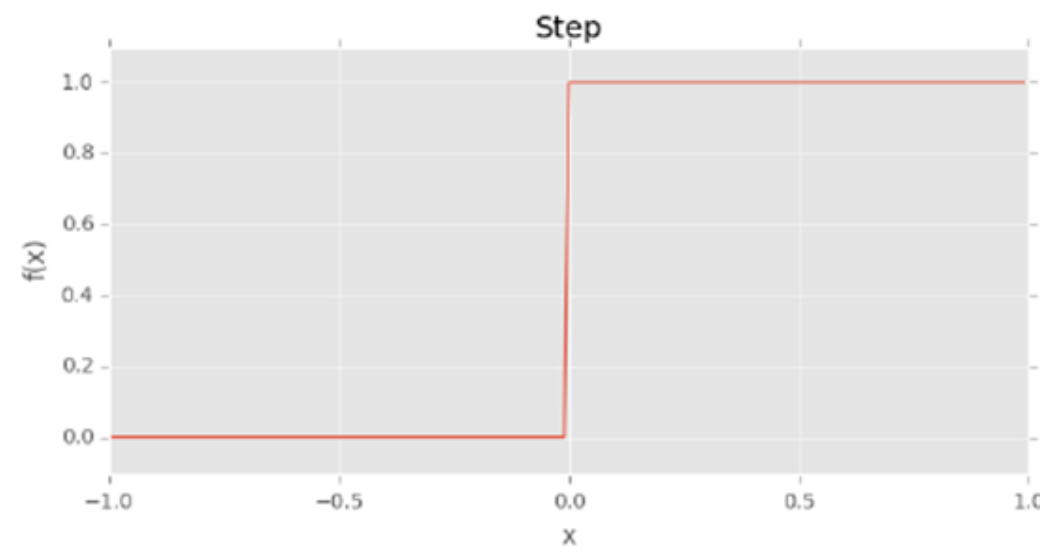
Activation Function



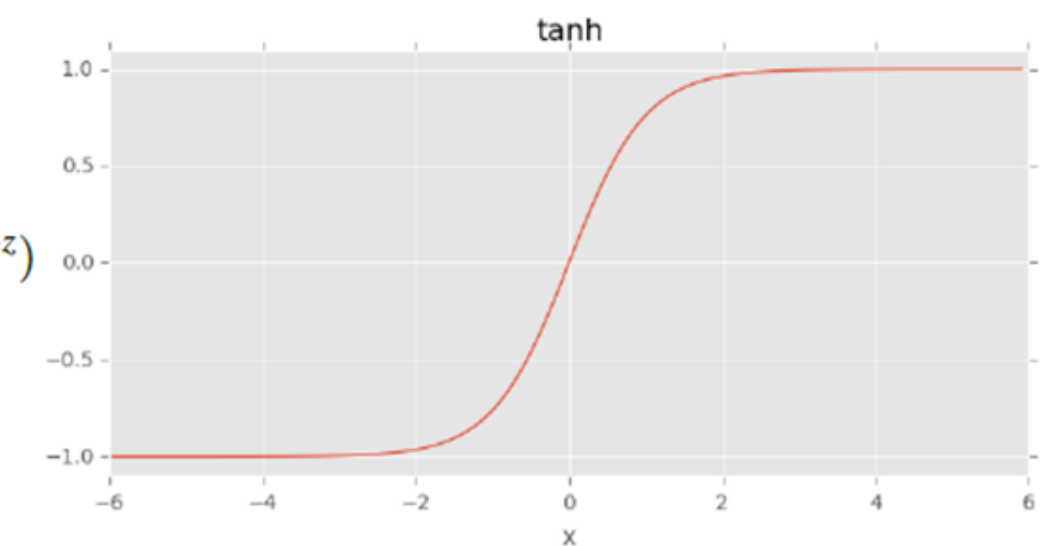
$$\sigma(z) \equiv \frac{1}{1 + e^{-z}}.$$

$$\frac{1}{1 + \exp(-\sum_j w_j x_j - b)}.$$

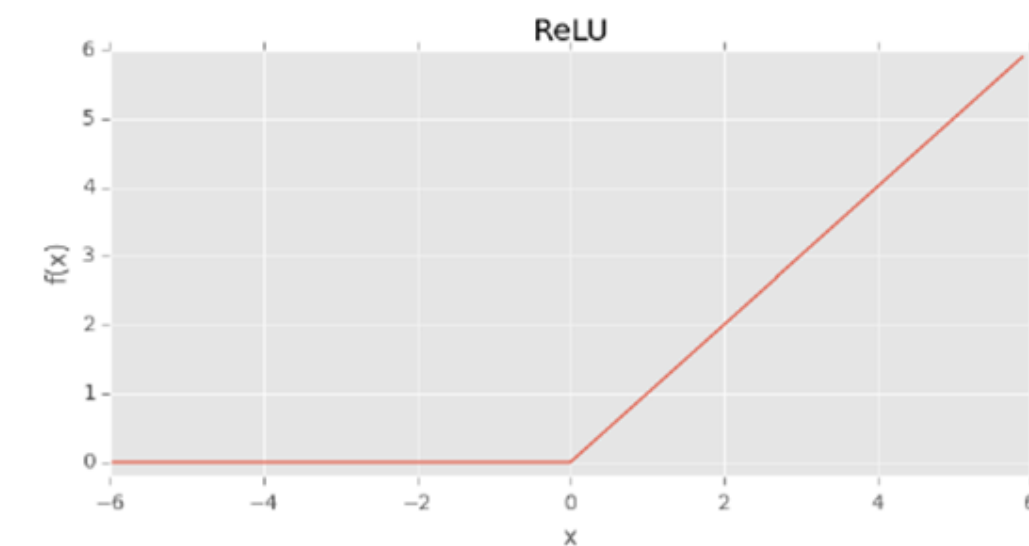
Activation Functions



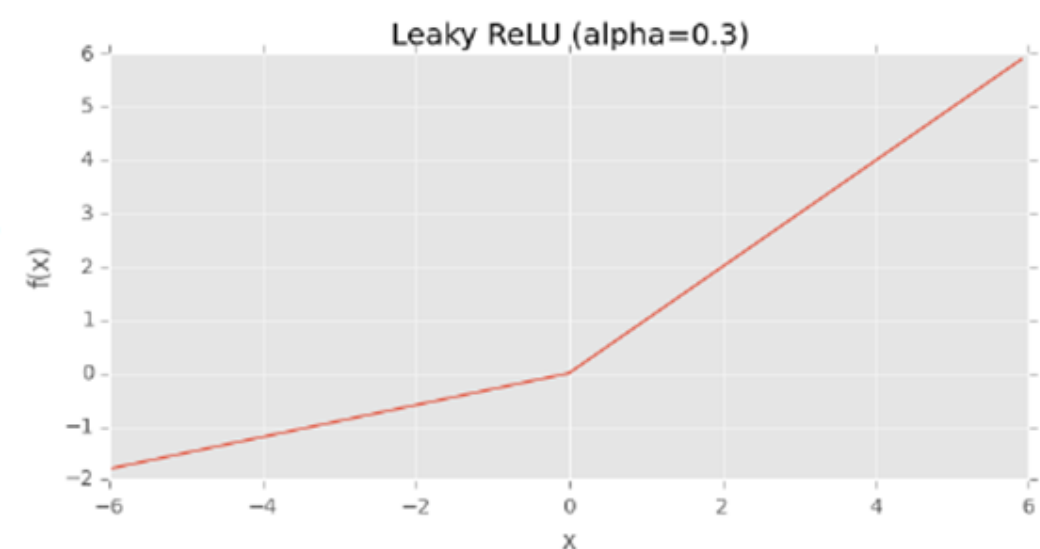
$$s(t) = 1/(1 + e^{-t})$$



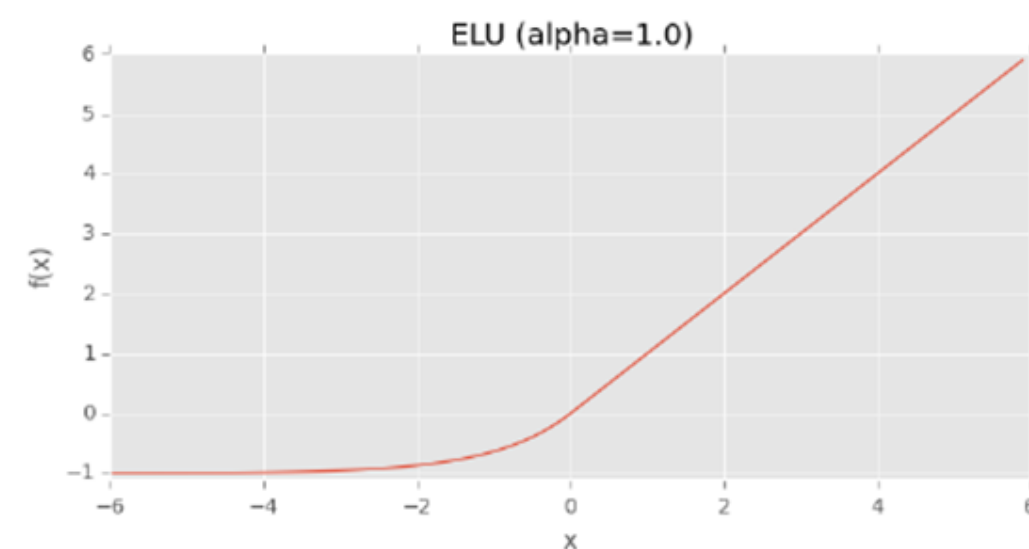
$$f(z) = \tanh(z) = (e^z - e^{-z}) / (e^z + e^{-z})$$



$$f(x) = \max(0, x)$$

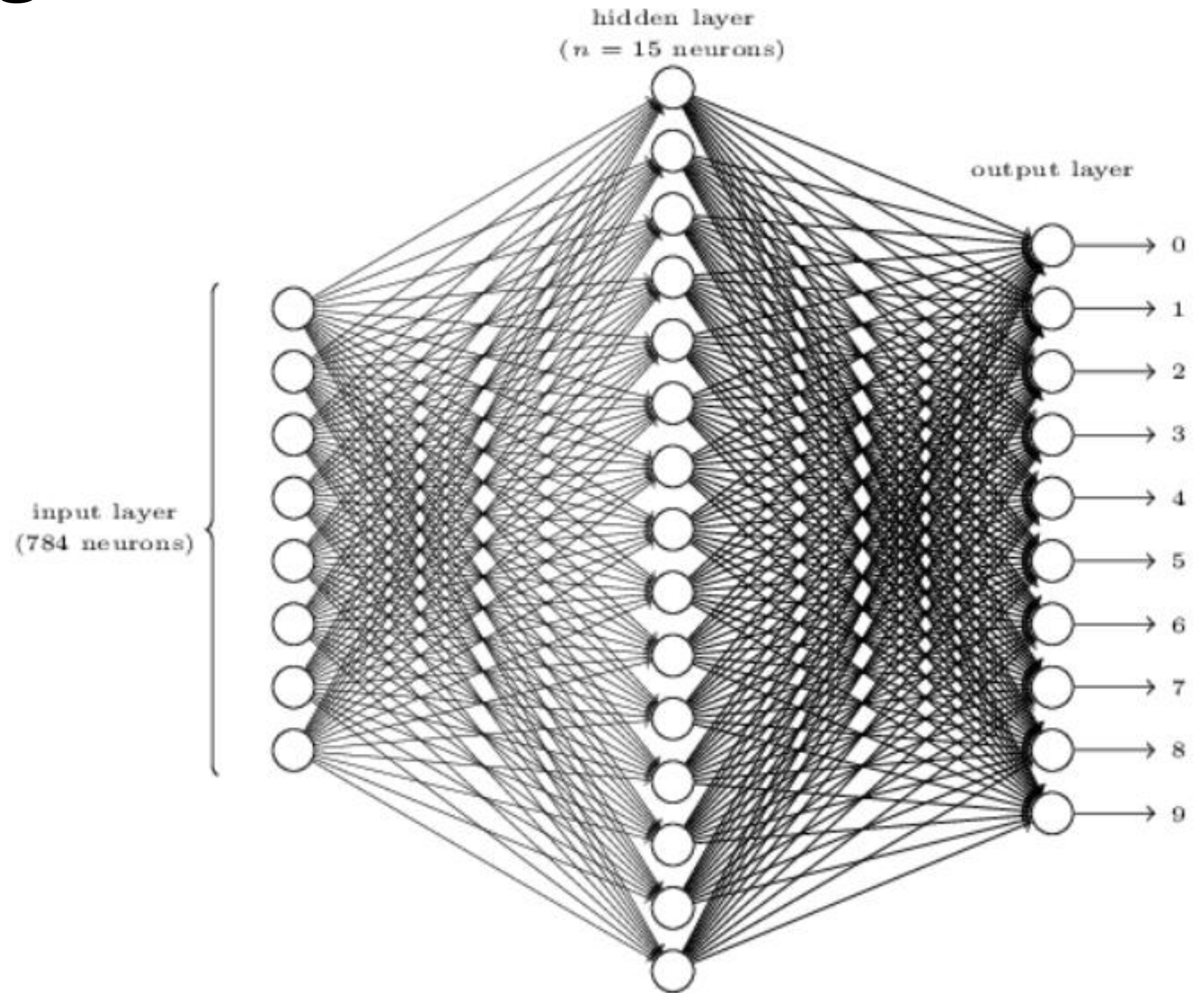
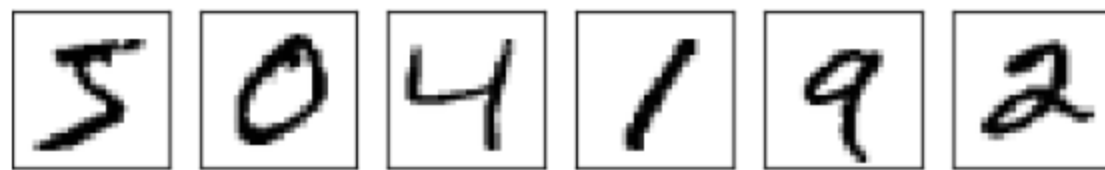


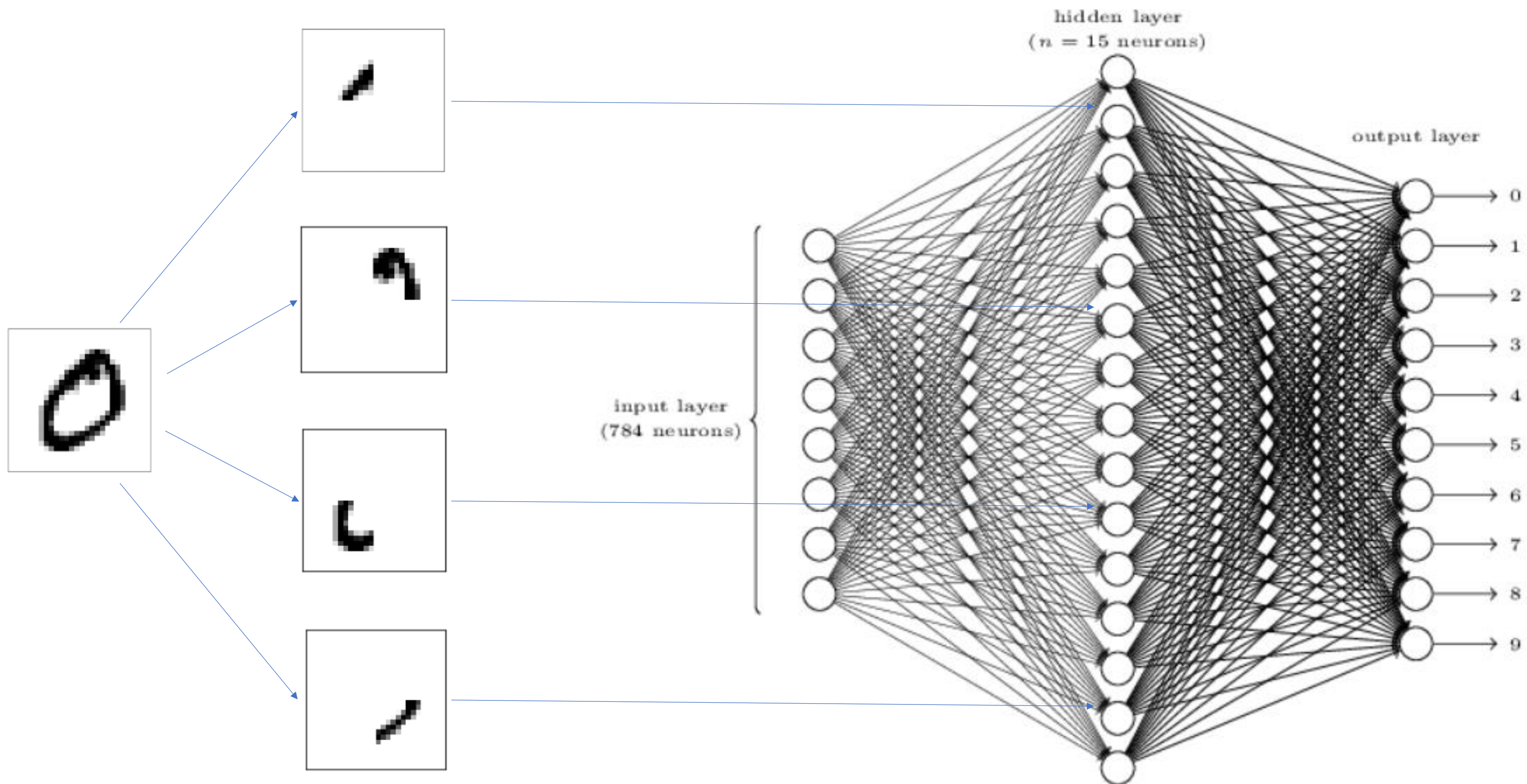
$$f(\text{net}) = \begin{cases} \text{net} & \text{if } \text{net} \geq 0 \\ \alpha \times \text{net} & \text{otherwise} \end{cases}$$



$$f(\text{net}) = \begin{cases} \text{net} & \text{if } \text{net} \geq 0 \\ \alpha \times (\exp(\text{net}) - 1) & \text{otherwise} \end{cases}$$

Hand Written Digits Classification





Loss Function (Cost Function)

- *Quadratic Cost Function (Mean Squared Error, MSE)*

$$C(w, b) \equiv \frac{1}{2n} \sum_x \|y(x) - a\|^2$$

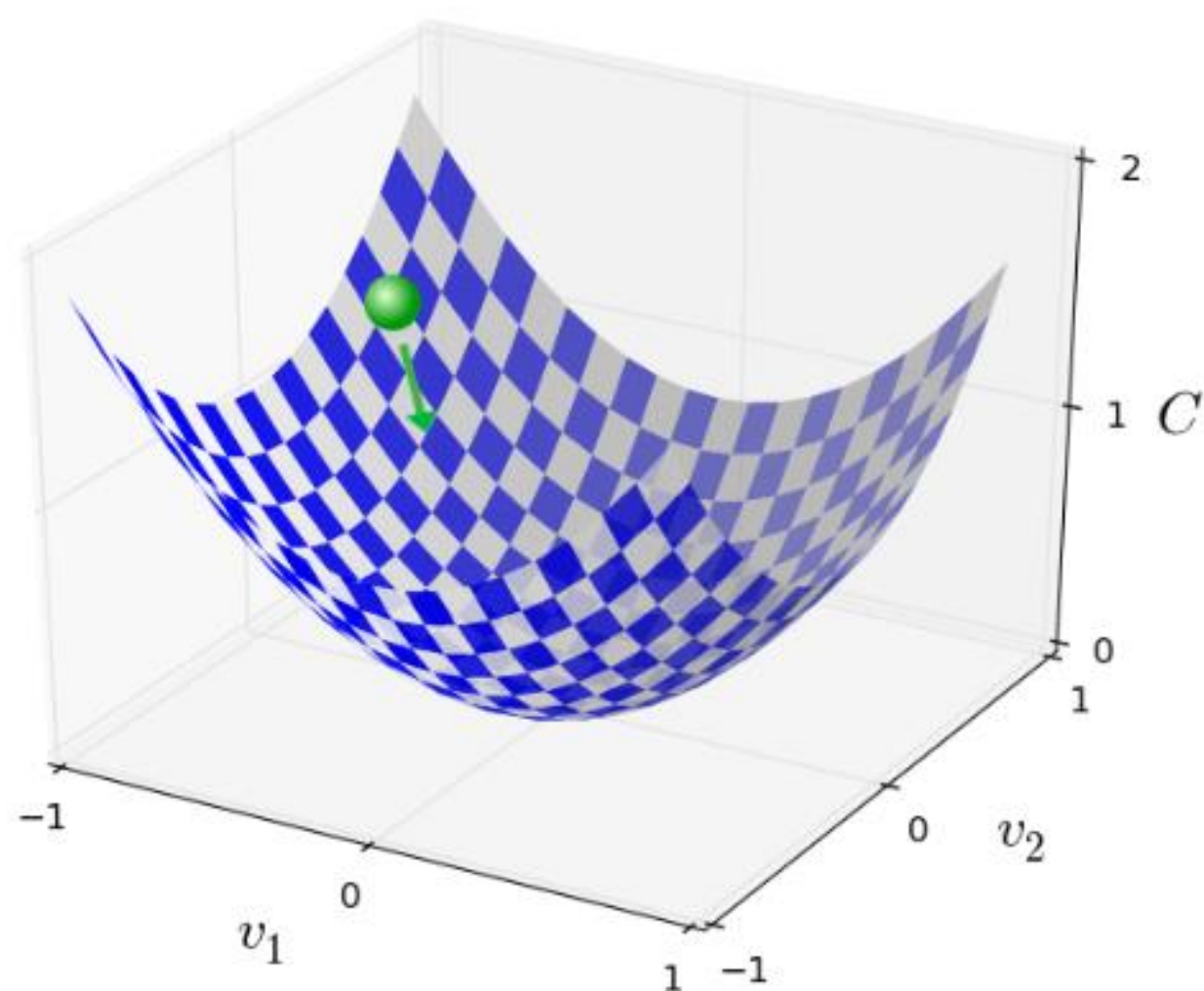
- *Cross Entropy*

$$H(p, q) = - \sum_x p(x) \log q(x)$$

- *Focal Loss*

$$Loss(x, class) = -\alpha_{class} \left(1 - \frac{e^{x[class]}}{\sum_j e^{x[j]}}\right)^\gamma \log \left(\frac{e^{x[class]}}{\sum_j e^{x[j]}}\right)$$

Optimizer - Gradient Descent



$$\Delta C \approx \frac{\partial C}{\partial v_1} \Delta v_1 + \frac{\partial C}{\partial v_2} \Delta v_2$$

$$\nabla C \equiv \left(\frac{\partial C}{\partial v_1}, \frac{\partial C}{\partial v_2} \right)^T.$$

$$\Delta C \approx \nabla C \cdot \Delta v. \quad \Delta v = -\eta \nabla C,$$

$$\Delta C \approx -\eta \nabla C \cdot \nabla C = -\eta \|\nabla C\|^2$$

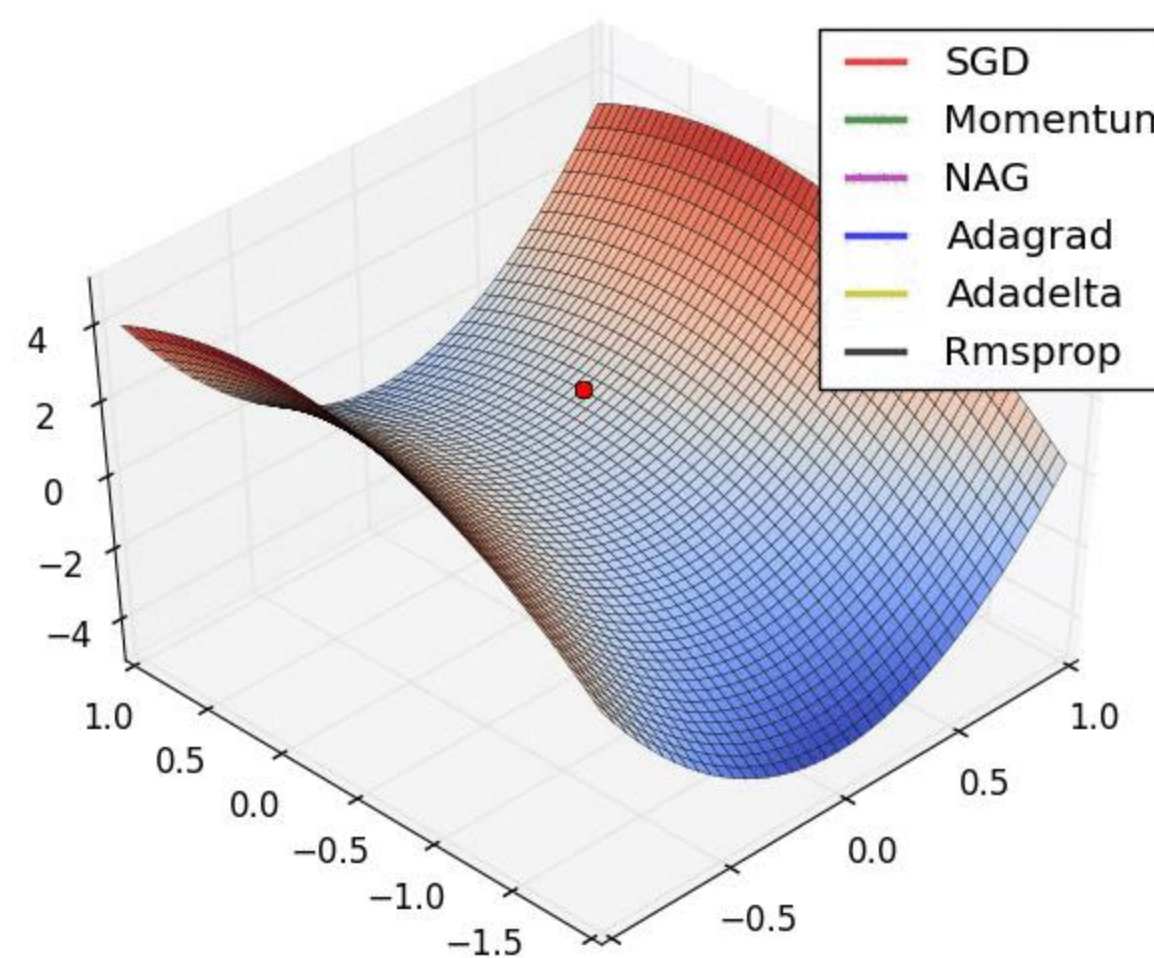
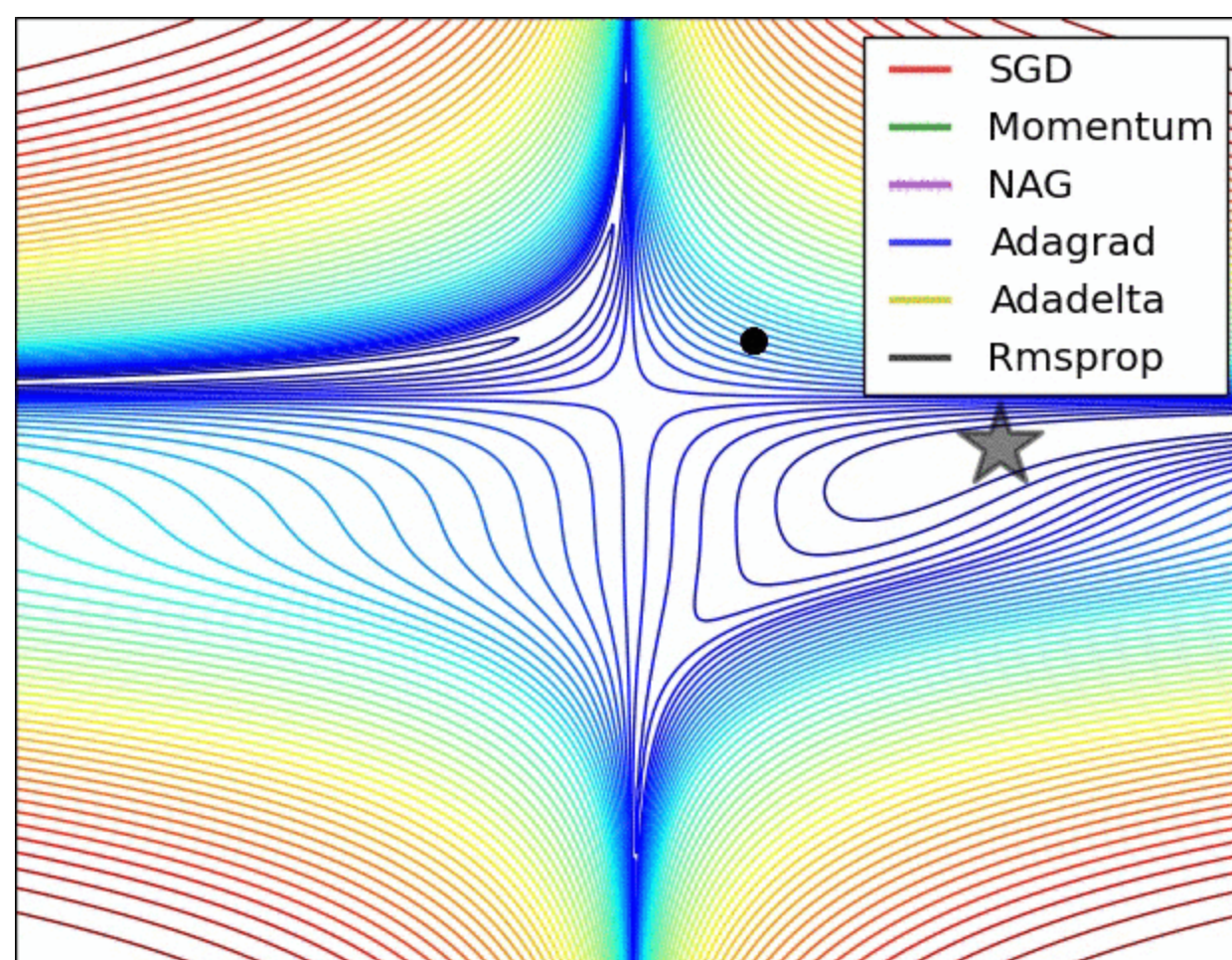
$$v \rightarrow v' = v - \eta \nabla C.$$

$$w_k \rightarrow w'_k = w_k - \eta \frac{\partial C}{\partial w_k}$$

$$b_l \rightarrow b'_l = b_l - \eta \frac{\partial C}{\partial b_l}.$$

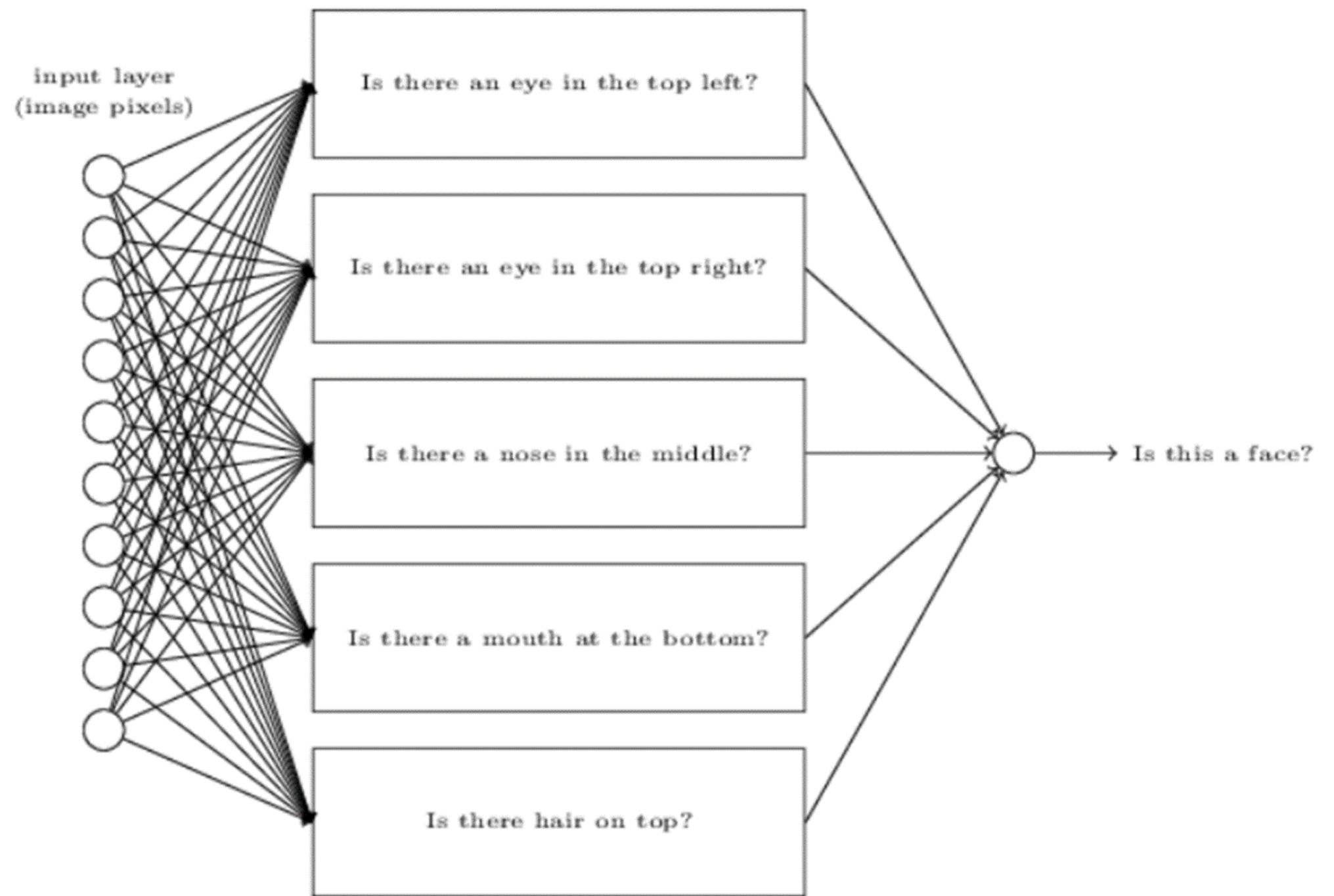
Optimizer (backpropagation)

Optimizer	特點
SGD	- 有機會跳出目前局部收斂進而進到另一個局部收斂而得到最小值，而得到全局最小值 - 需自行設定learning rate，較難選擇到合適的learning rate - 會造成loss function有嚴重的震蕩 - 需要較長時間收斂至最小值
Momentum	- 能夠在相關方向加速SGD，抑制SGD的嚴重震蕩，進而加快收斂 - 需自行設定learning rate與β，有可能會使參數的移動方向偏移梯度下分的方向，進而導至沒有那麼快速的收斂
AdaGrad	- 能夠自動調整learning rate，進而調整收斂 - 適合處理稀疏梯度 - 依然需要人工設置一個全局的learning rate - 後期，分母梯度平方的累加會越來越大，會使梯度趨近於0，使得訓練結束
Adam	- 結合了AdaGrad與Momentum的優點 - 適用於大數據集和高維空間的資料 - 目前最常使用的一個Optimizer



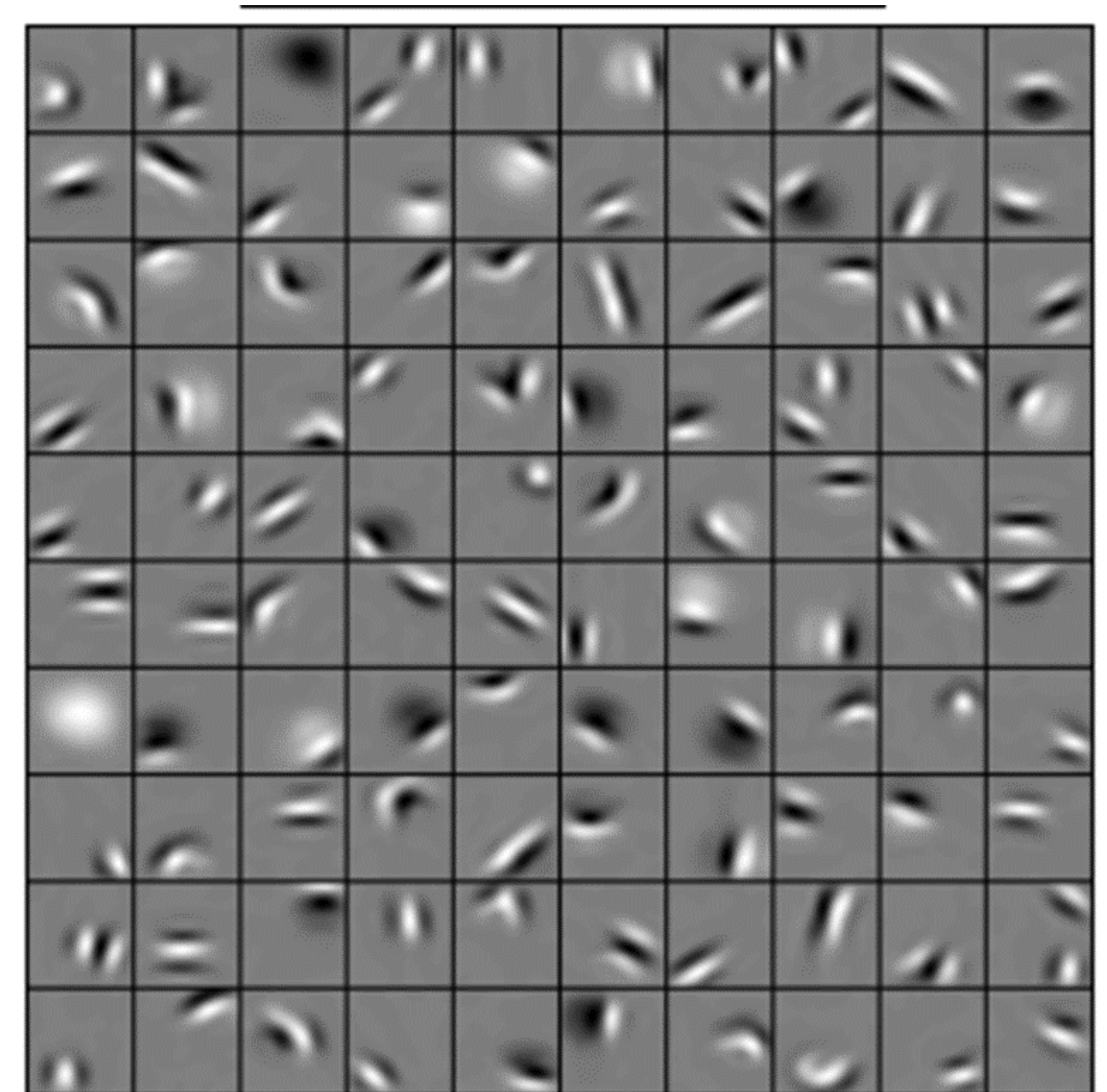
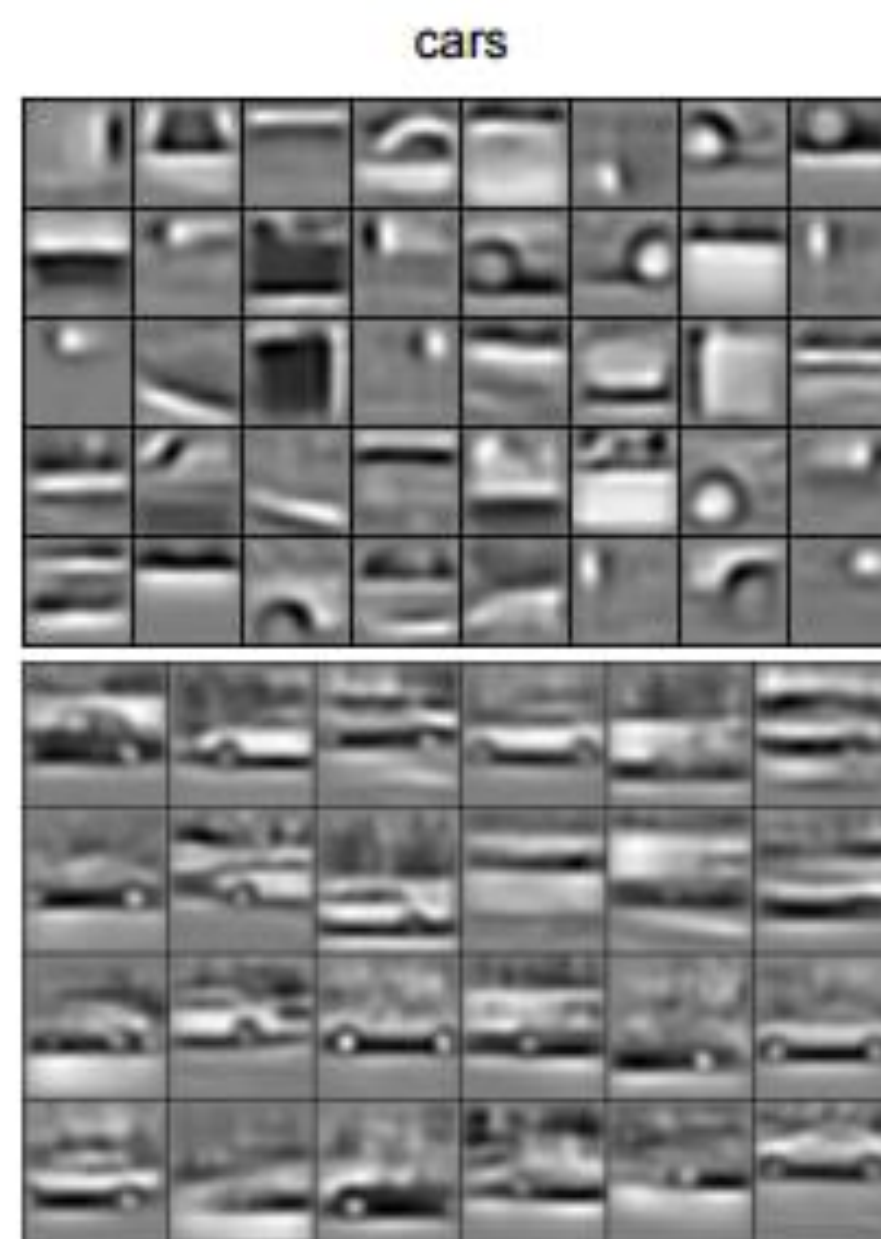
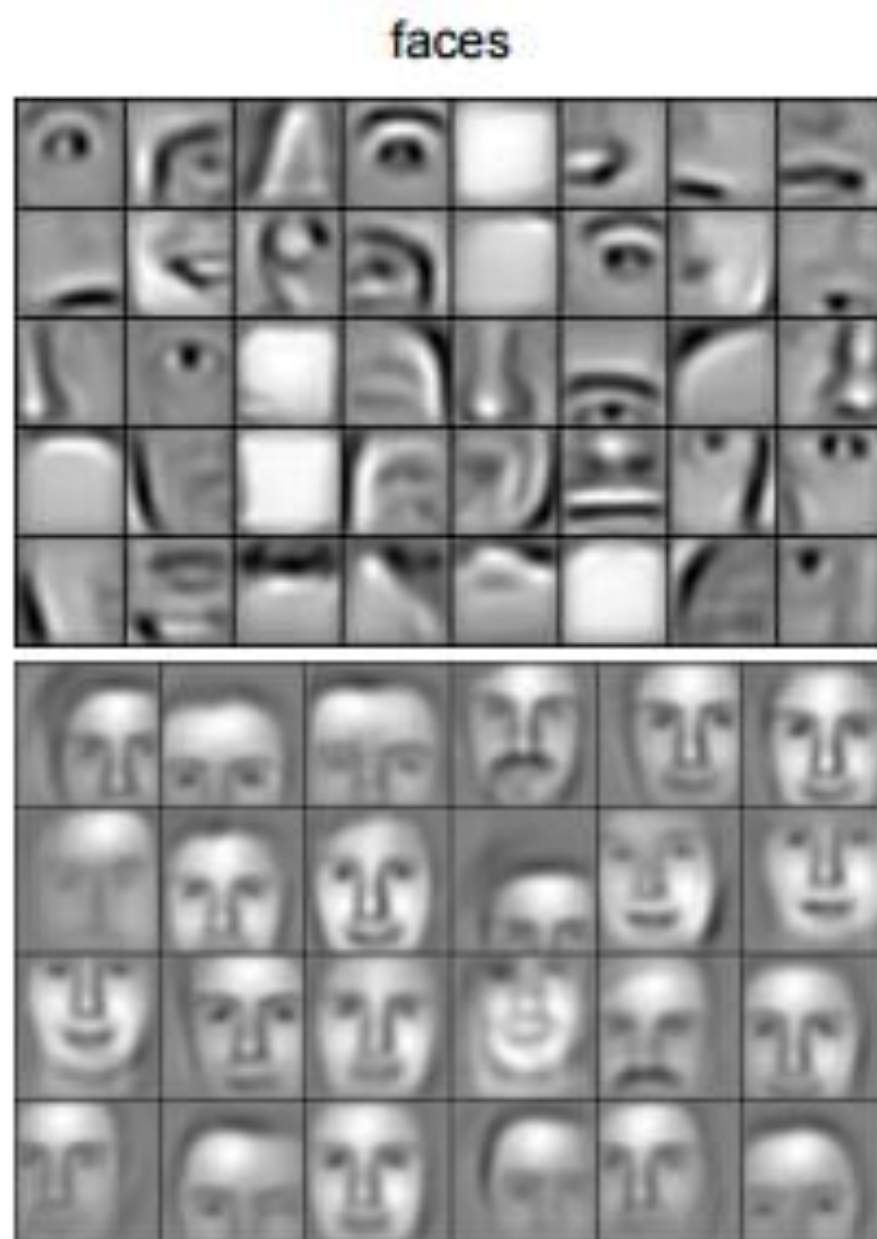
Summary

- Architecture
- Activation Function
- Loss Function
- Optimizer
- Mini Batch



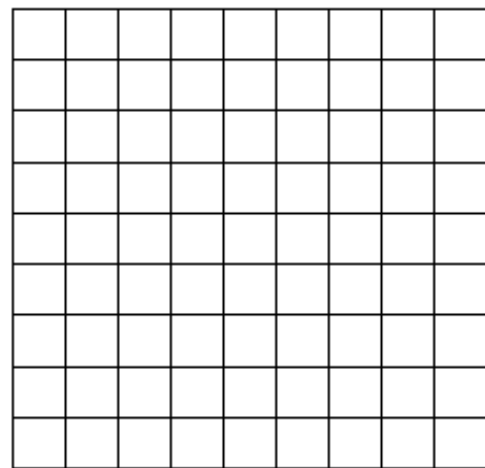
[NN Play Ground](#)

Convolutional Neural Network

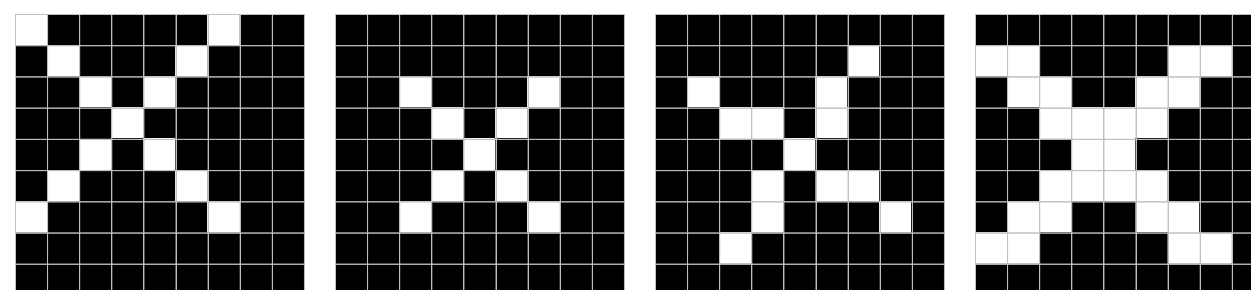
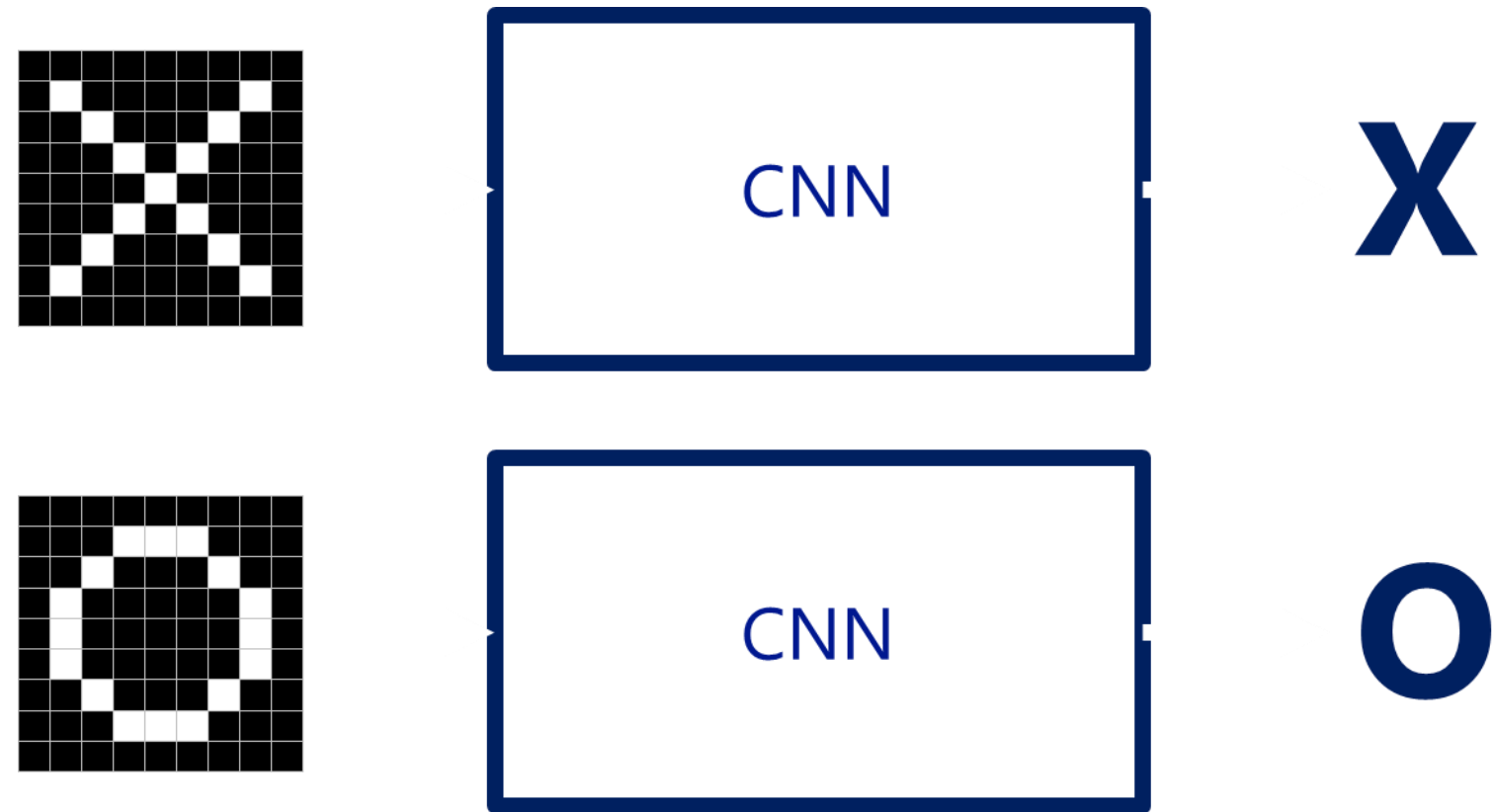


A Toy ConvNet: X's and O's

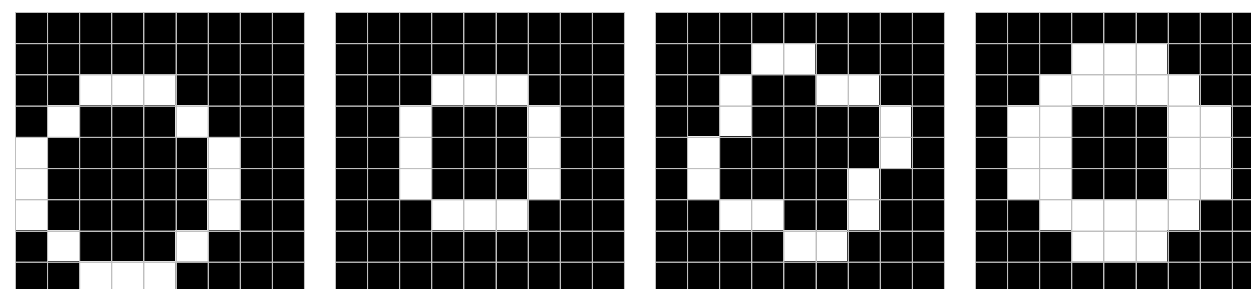
A two-dimensional
array of pixels



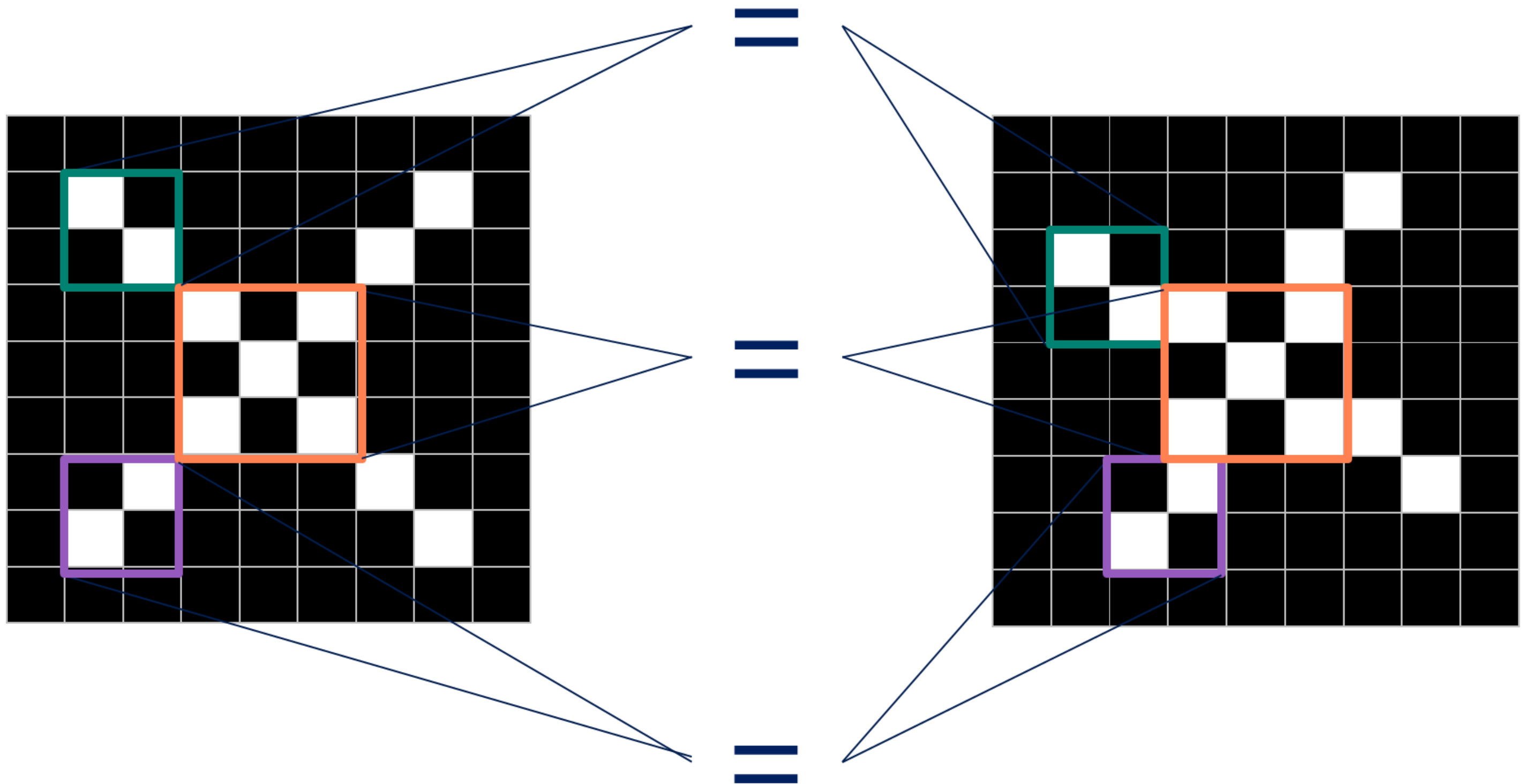
X or **O**



translation scaling rotation weight



ConvNets Match Pieces of The Image



Features Match Pieces of The Image

1	-1	-1
-1	1	-1
-1	-1	1

1	-1	1
-1	1	-1
1	-1	1

-1	-1	1
-1	1	-1
1	-1	-1

1	-1	-1
-1	1	-1
-1	-1	1

1	-1	1
-1	1	-1
1	-1	1

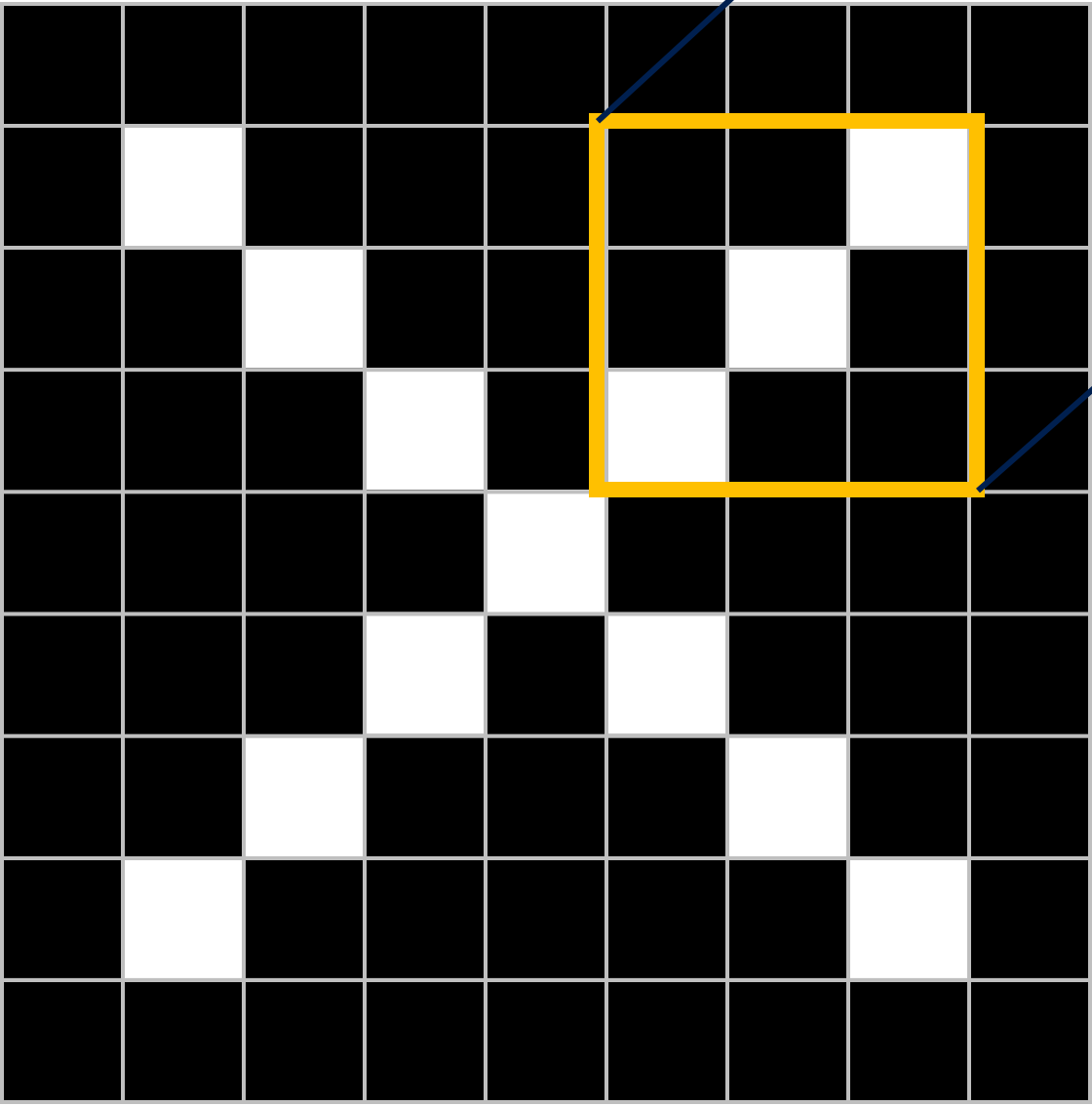
-1	-1	1
-1	1	-1
1	-1	-1

A 10x10 grid with a blue square highlighting a 3x3 subgrid. The subgrid contains white squares at (1,1), (1,3), (2,2), (3,1), and (3,3), with black squares elsewhere. Blue lines extend from the corners of the subgrid to the top-left and bottom-right corners of the grid.

1	-1	-1
-1	1	-1
-1	-1	1

1	-1	1
-1	1	-1
1	-1	1

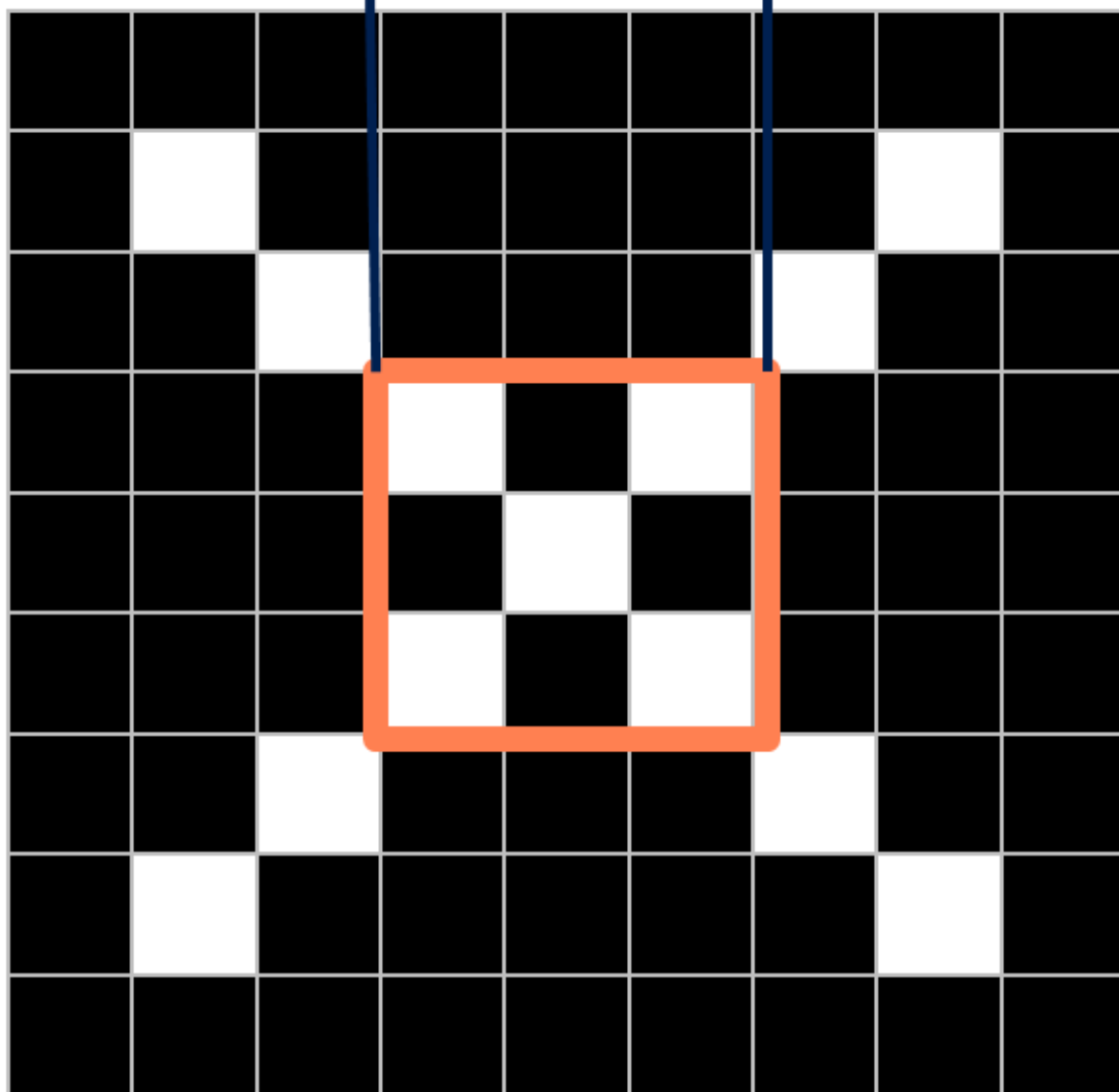
-1	-1	1
-1	1	-1
1	-1	-1



1	-1	-1
-1	1	-1
-1	-1	1

1	-1	1
-1	1	-1
1	-1	1

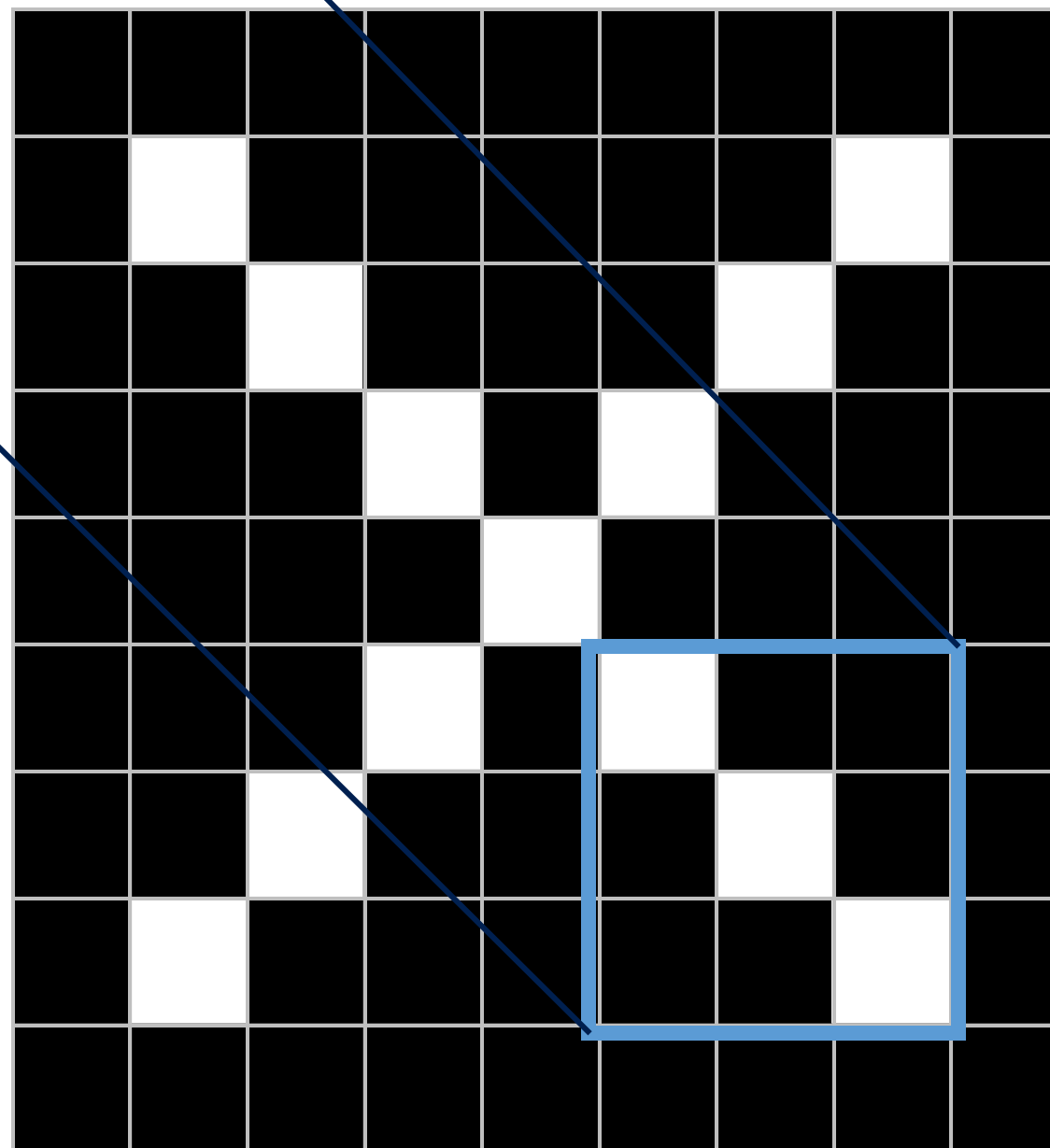
-1	-1	1
-1	1	-1
1	-1	-1



1	-1	-1
-1	1	-1
-1	-1	1

1	-1	1
-1	1	-1
1	-1	1

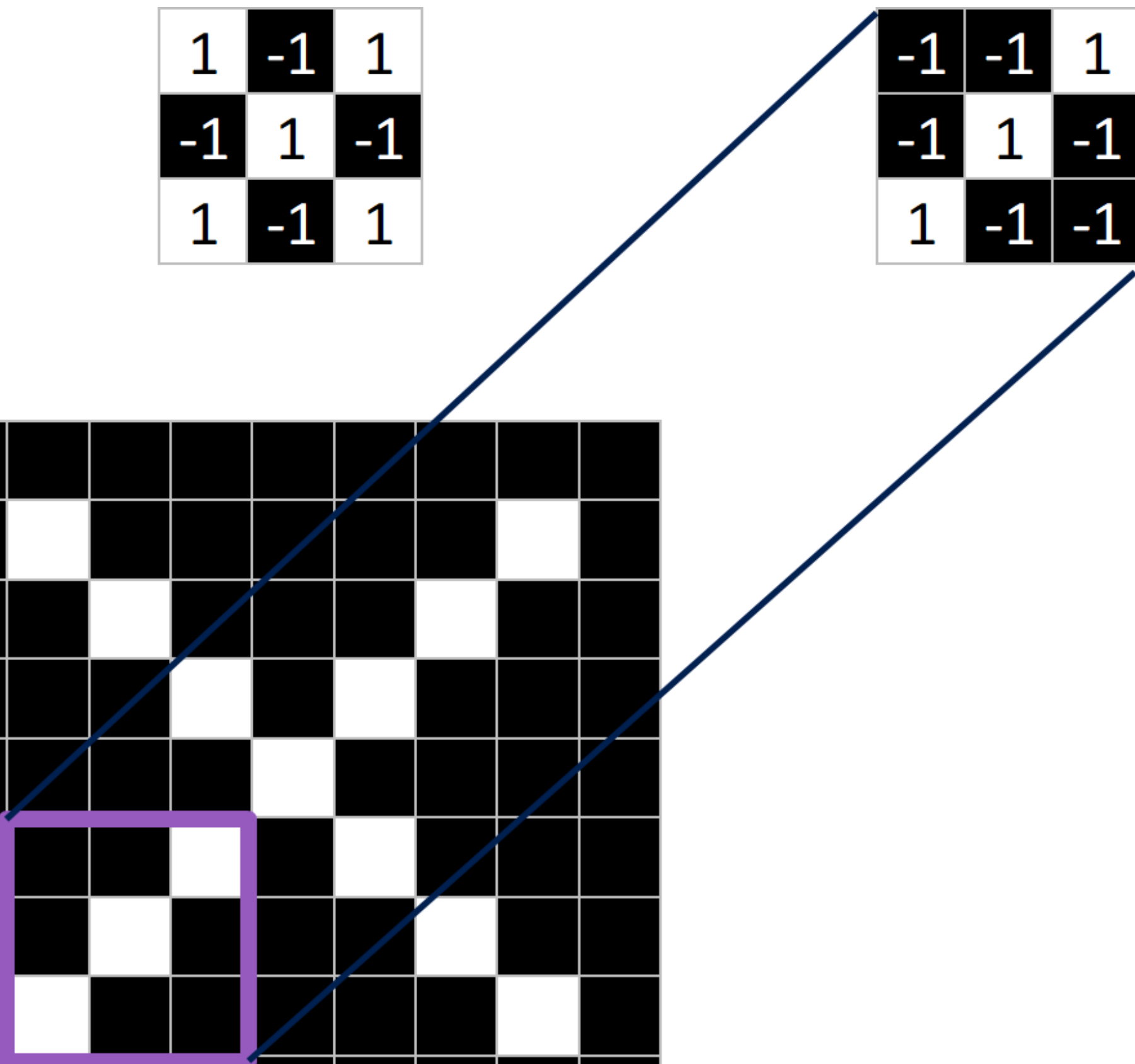
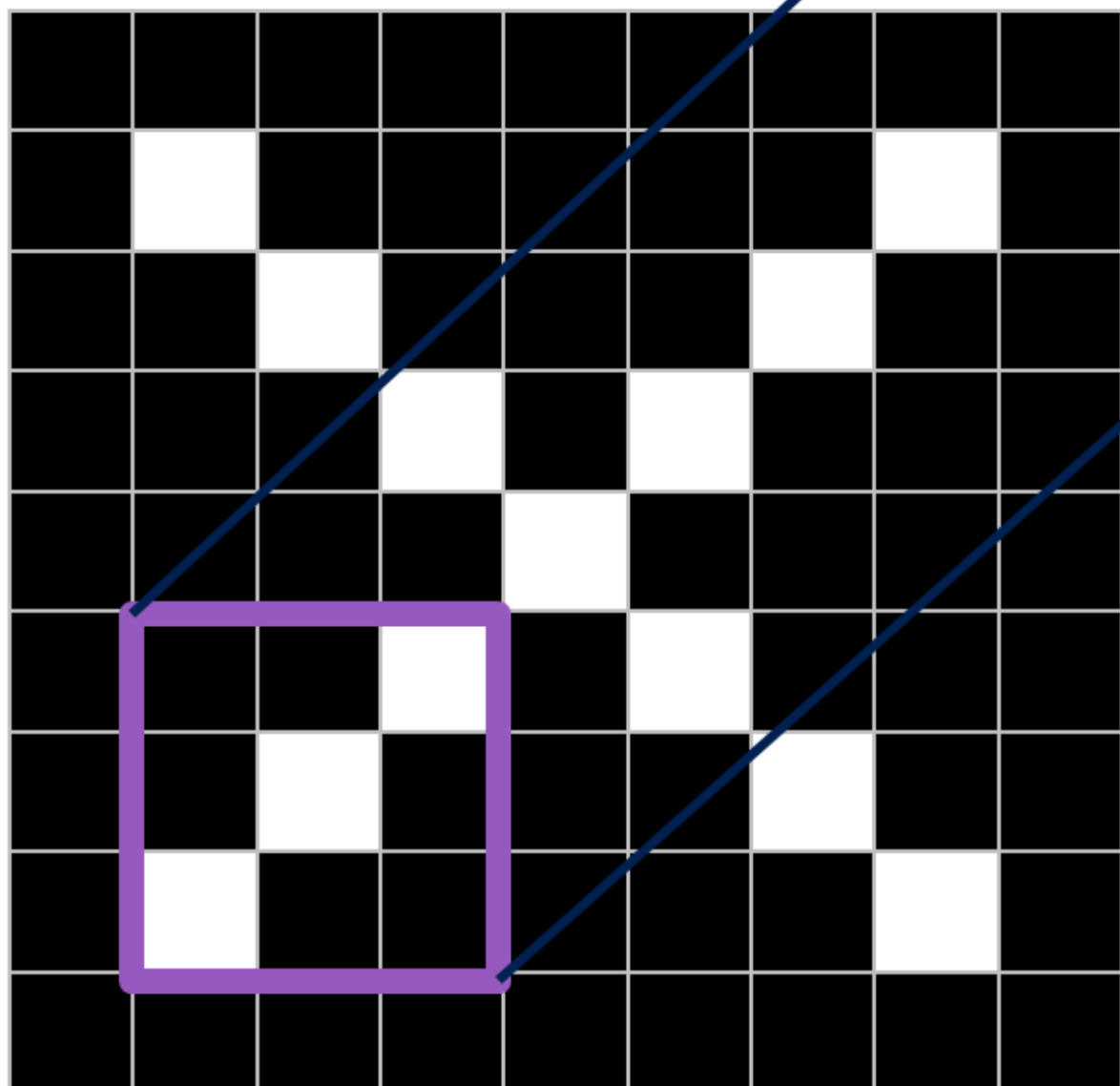
-1	-1	1
-1	1	-1
1	-1	-1



1	-1	-1
-1	1	-1
-1	-1	1

1	-1	1
-1	1	-1
1	-1	1

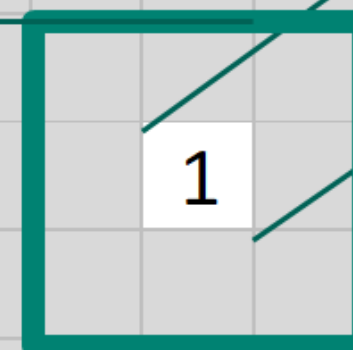
-1	-1	1
-1	1	-1
1	-1	-1



1	-1	-1
-1	1	-1
-1	-1	1

1	1	1
1	1	1
1	1	1

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1



1	-1	-1
-1	1	-1
-1	-1	1

1	1	-1
1	1	1
-1	1	1

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1

		1						



-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1



1	-1	-1
-1	1	-1
-1	-1	1

=

0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1



1	-1	1
-1	1	-1
1	-1	1

=

0.33	-0.55	0.11	-0.11	0.11	-0.55	0.33
-0.55	0.55	-0.55	0.33	-0.55	0.55	-0.55
0.11	-0.55	0.55	-0.77	0.55	-0.55	0.11
-0.11	0.33	-0.77	1.00	-0.77	0.33	-0.11
0.11	-0.55	0.55	-0.77	0.55	-0.55	0.11
-0.55	0.55	-0.55	0.33	-0.55	0.55	-0.55
0.33	-0.55	0.11	-0.11	0.11	-0.55	0.33

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1

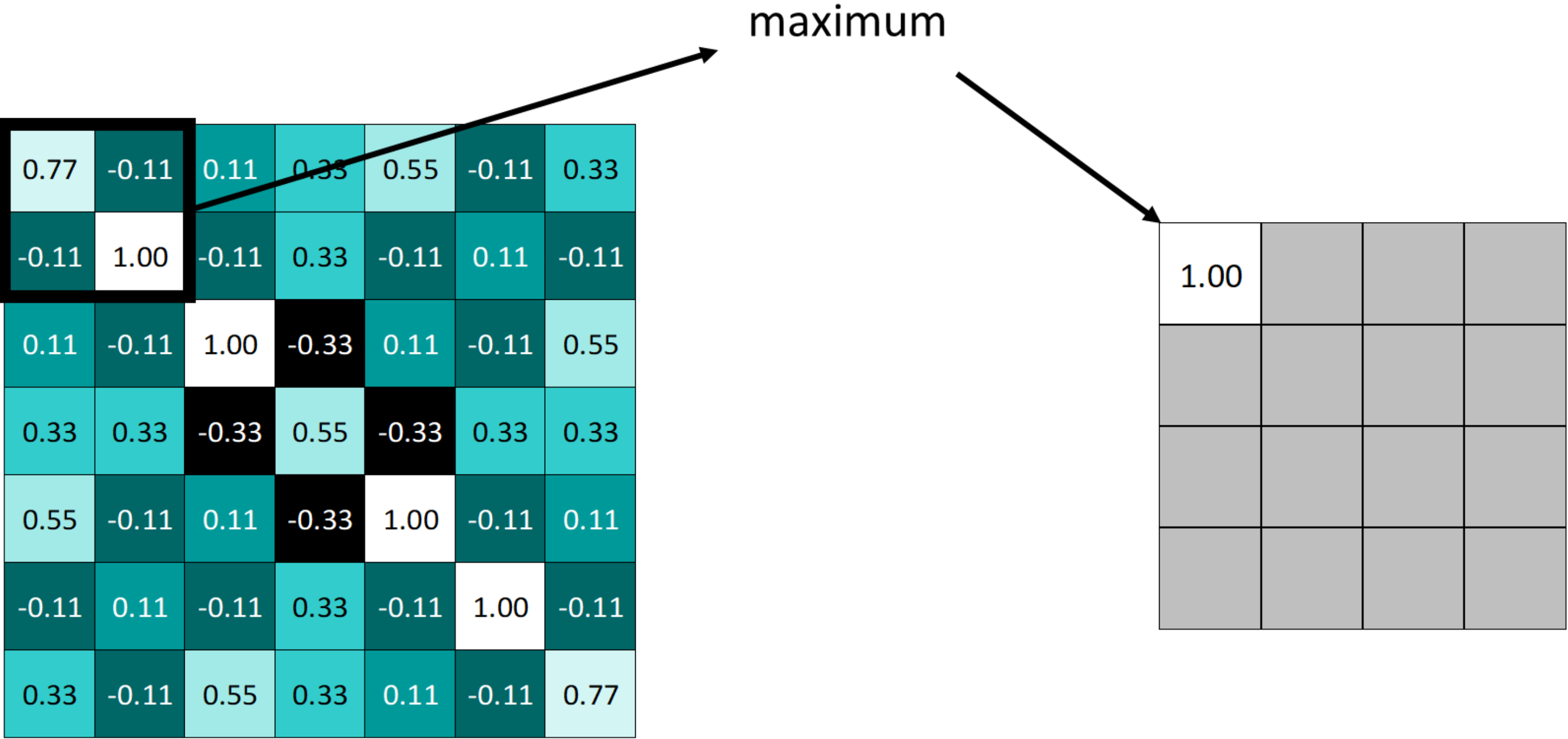


-1	-1	1
-1	1	-1
1	-1	-1

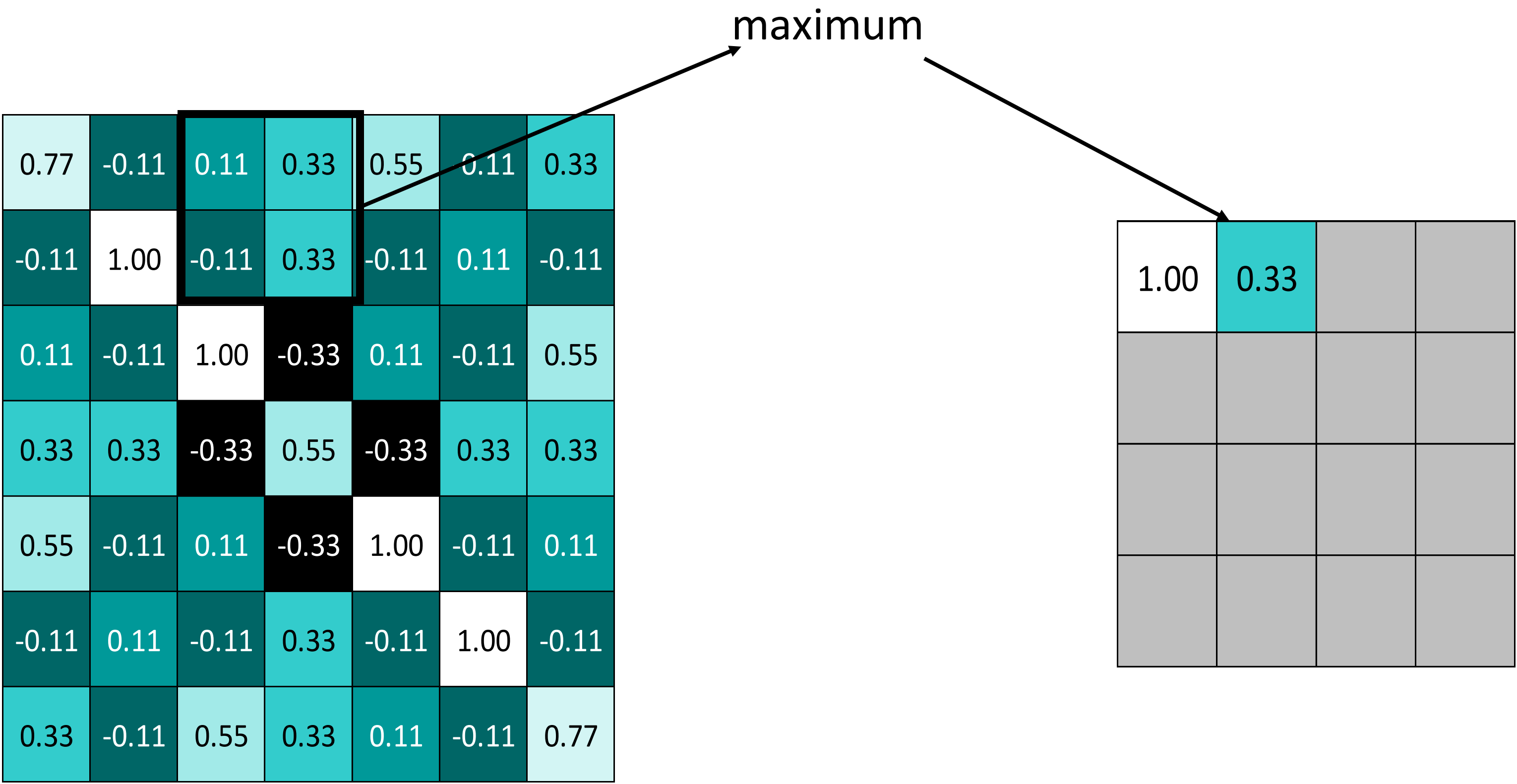
=

0.33	-0.11	0.55	0.33	0.11	-0.11	0.77
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.77	-0.11	0.11	0.33	0.55	-0.11	0.33

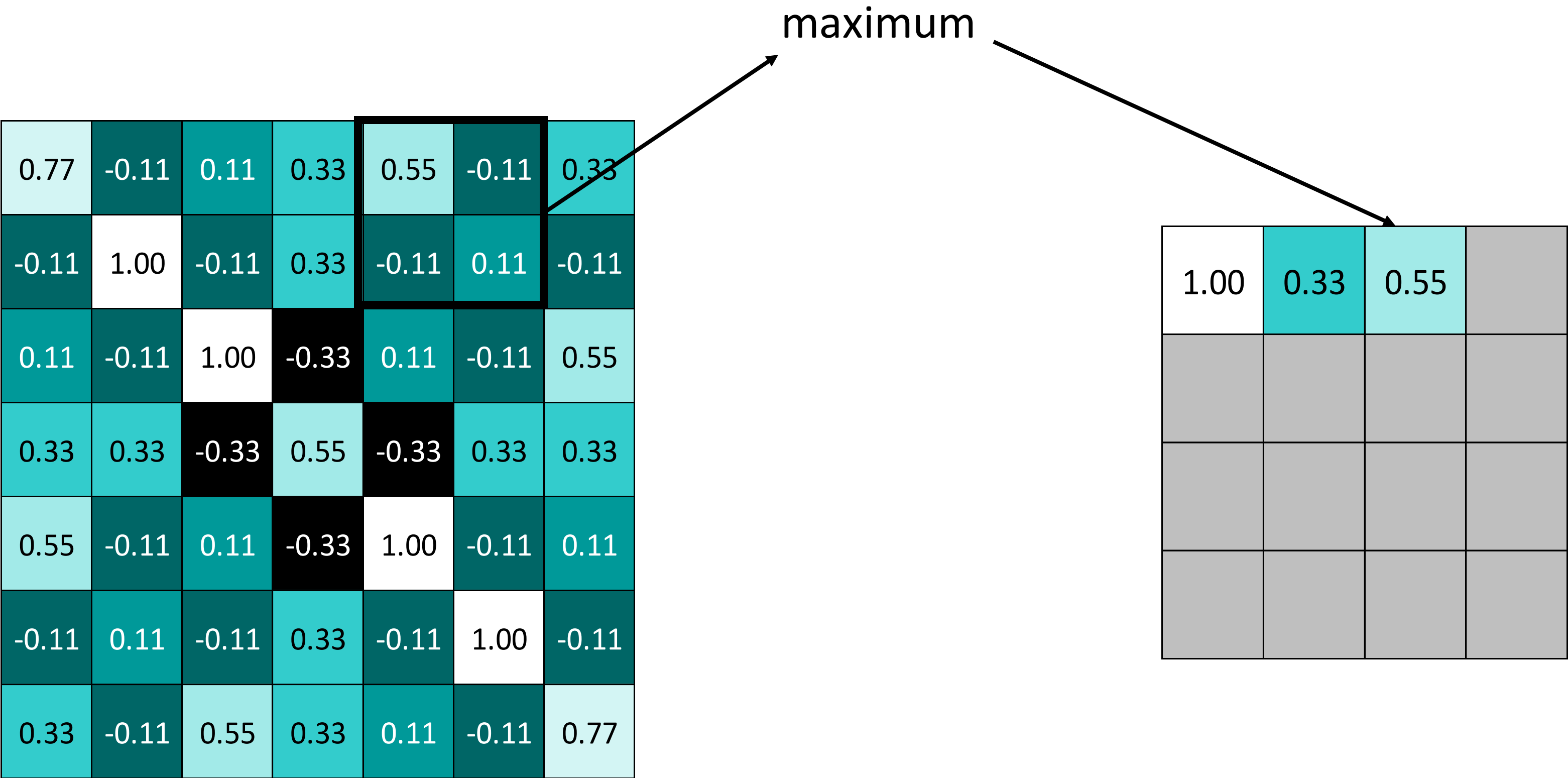
Max Pooling



Max Pooling



Max Pooling



Max Pooling

0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77

max pooling

1.00	0.33	0.55	0.33
0.33	1.00	0.33	0.55
0.55	0.33	1.00	0.11
0.33	0.55	0.11	0.77

Max Pooling

0.77	-0.11	0.11	0.33	0.55	-0.11	0.33
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.33	-0.11	0.55	0.33	0.11	-0.11	0.77

0.33	-0.55	0.11	-0.11	0.11	-0.55	0.33
-0.55	0.55	-0.55	0.33	-0.55	0.55	-0.55
0.11	-0.55	0.55	-0.77	0.55	-0.55	0.11
-0.11	0.33	-0.77	1.00	-0.77	0.33	-0.11
0.11	-0.55	0.55	-0.77	0.55	-0.55	0.11
-0.55	0.55	-0.55	0.33	-0.55	0.55	-0.55
0.33	-0.55	0.11	-0.11	0.11	-0.55	0.33

0.33	-0.11	0.55	0.33	0.11	-0.11	0.77
-0.11	0.11	-0.11	0.33	-0.11	1.00	-0.11
0.55	-0.11	0.11	-0.33	1.00	-0.11	0.11
0.33	0.33	-0.33	0.55	-0.33	0.33	0.33
0.11	-0.11	1.00	-0.33	0.11	-0.11	0.55
-0.11	1.00	-0.11	0.33	-0.11	0.11	-0.11
0.77	-0.11	0.11	0.33	0.55	-0.11	0.33



1.00	0.33	0.55	0.33
0.33	1.00	0.33	0.55
0.55	0.33	1.00	0.11
0.33	0.55	0.11	0.77

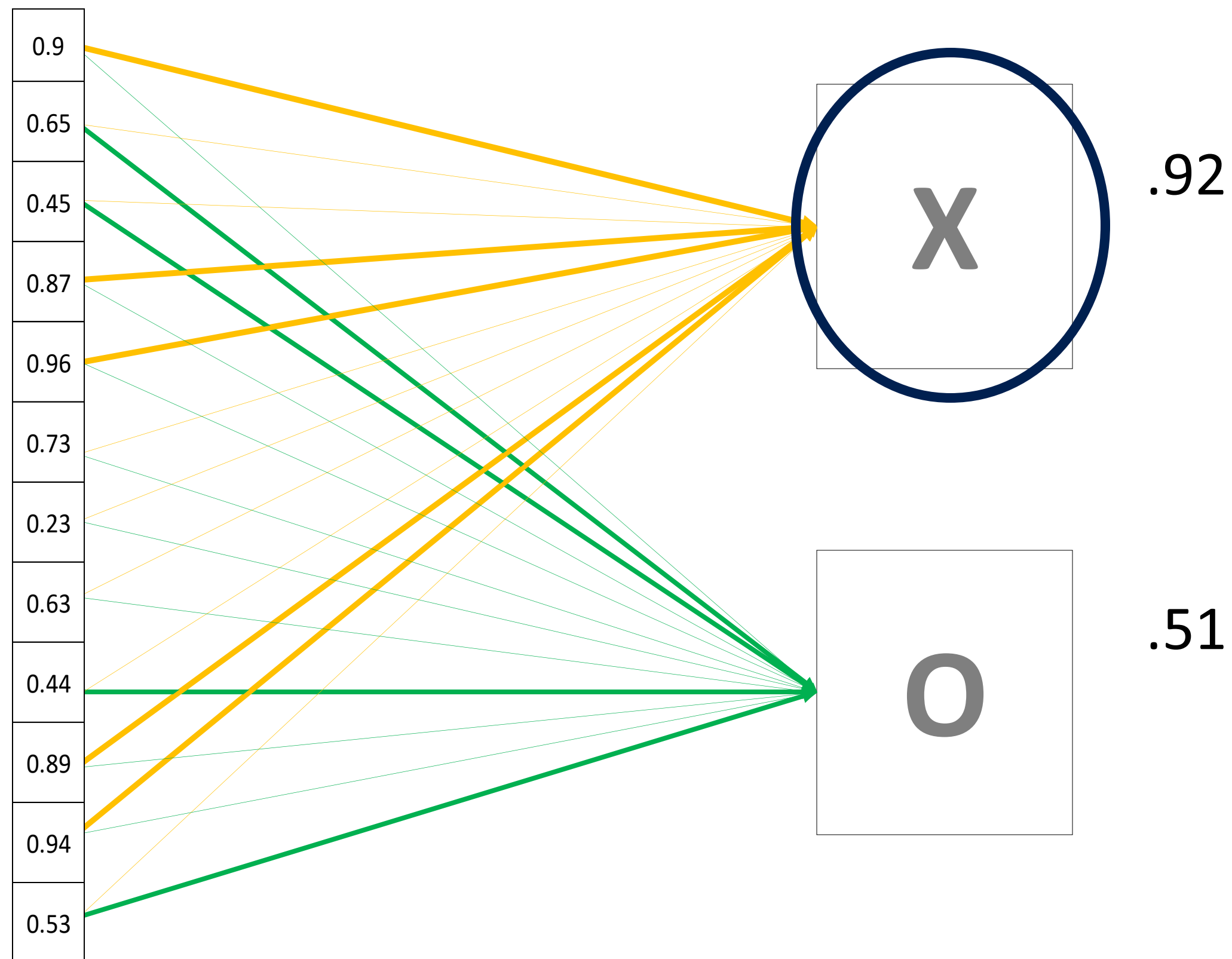
0.55	0.33	0.55	0.33
0.33	1.00	0.55	0.11
0.55	0.55	0.55	0.11
0.33	0.11	0.11	0.33

0.33	0.55	1.00	0.77
0.55	0.55	1.00	0.33
1.00	1.00	0.11	0.55
0.77	0.33	0.55	0.33

Pooling: Shrinking The Image Stack

- Pick a window size (usually 2 or 3).
- Pick a stride (usually 2).
- Walk your window across your filtered images.
- From each window, take the maximum value.

Fully Connected Layer



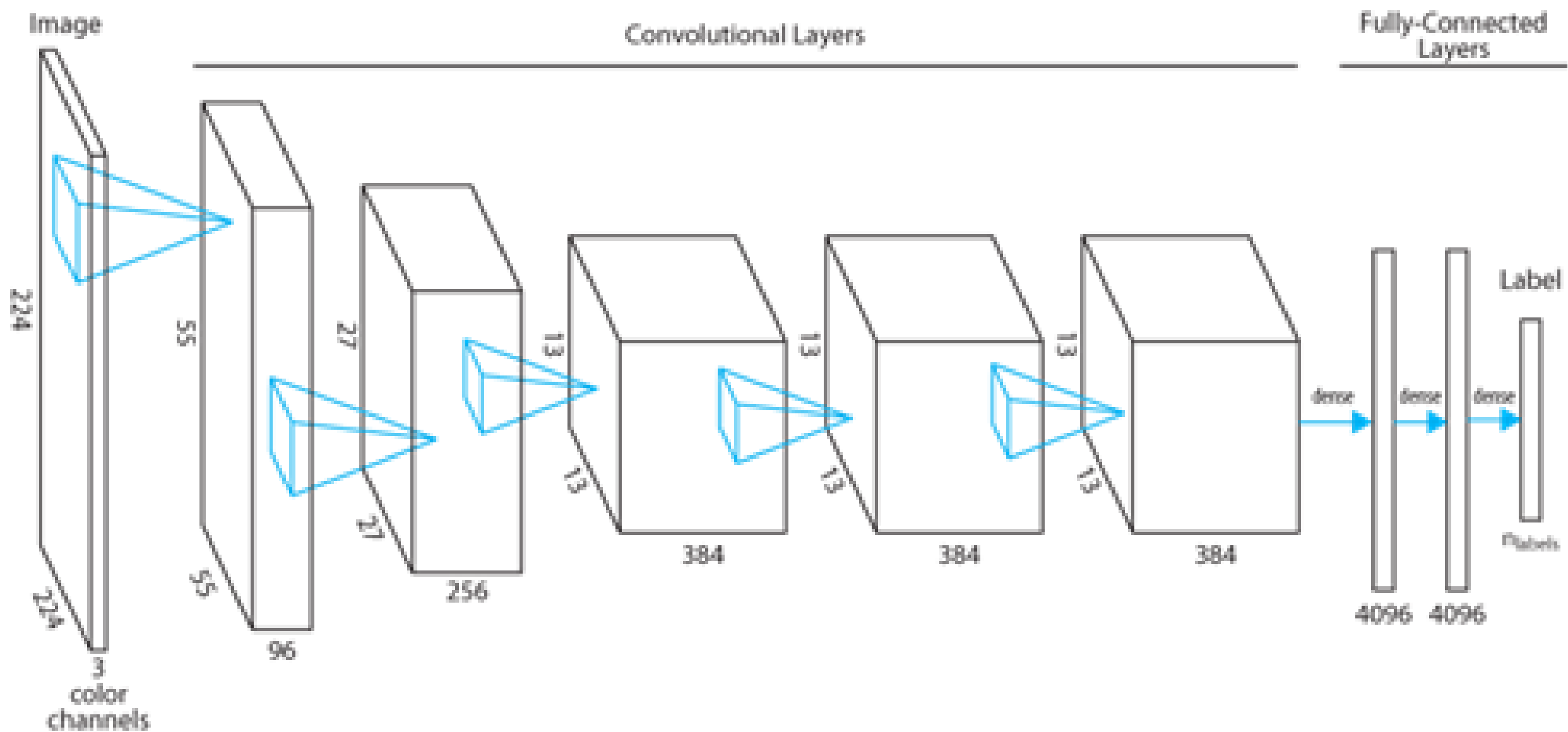
Hyperparameters

- Convolution
 - Number of features (Kernel/Filter)
 - Size of features (Kernel/Filter)
- Pooling
 - Window size
 - Window stride
- Fully Connected
 - Number of neurons (Nodes)

Architecture

- How many layers of each type ?
- In what order?
- How do all layers connect to each other?

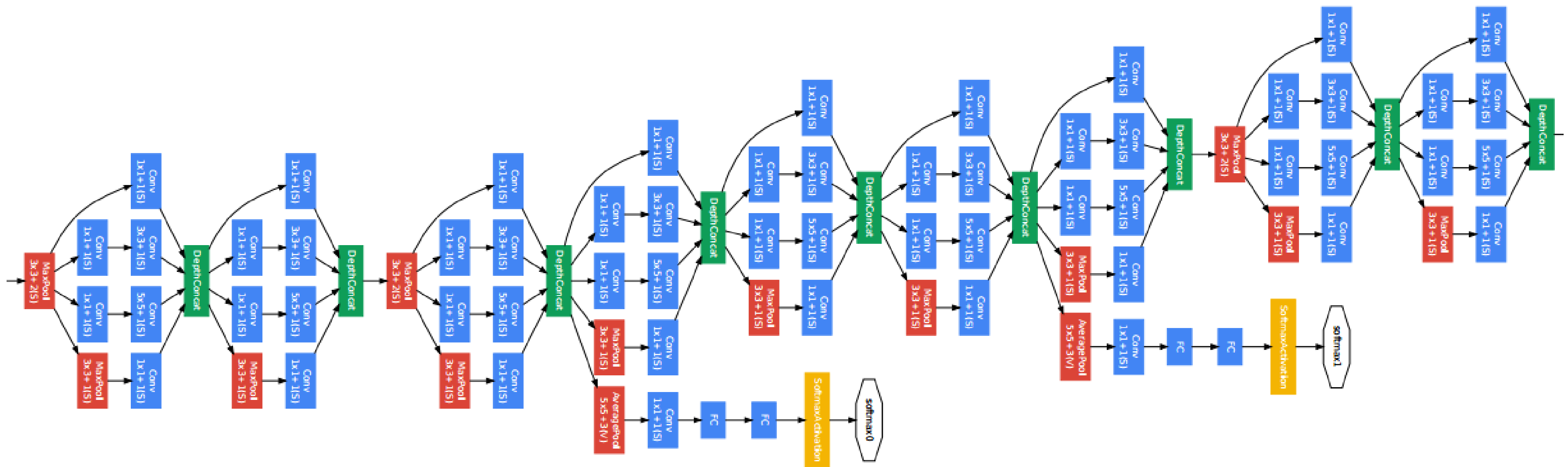
Classic Architecture - AlexNet



Classic Architecture – VGG16/19

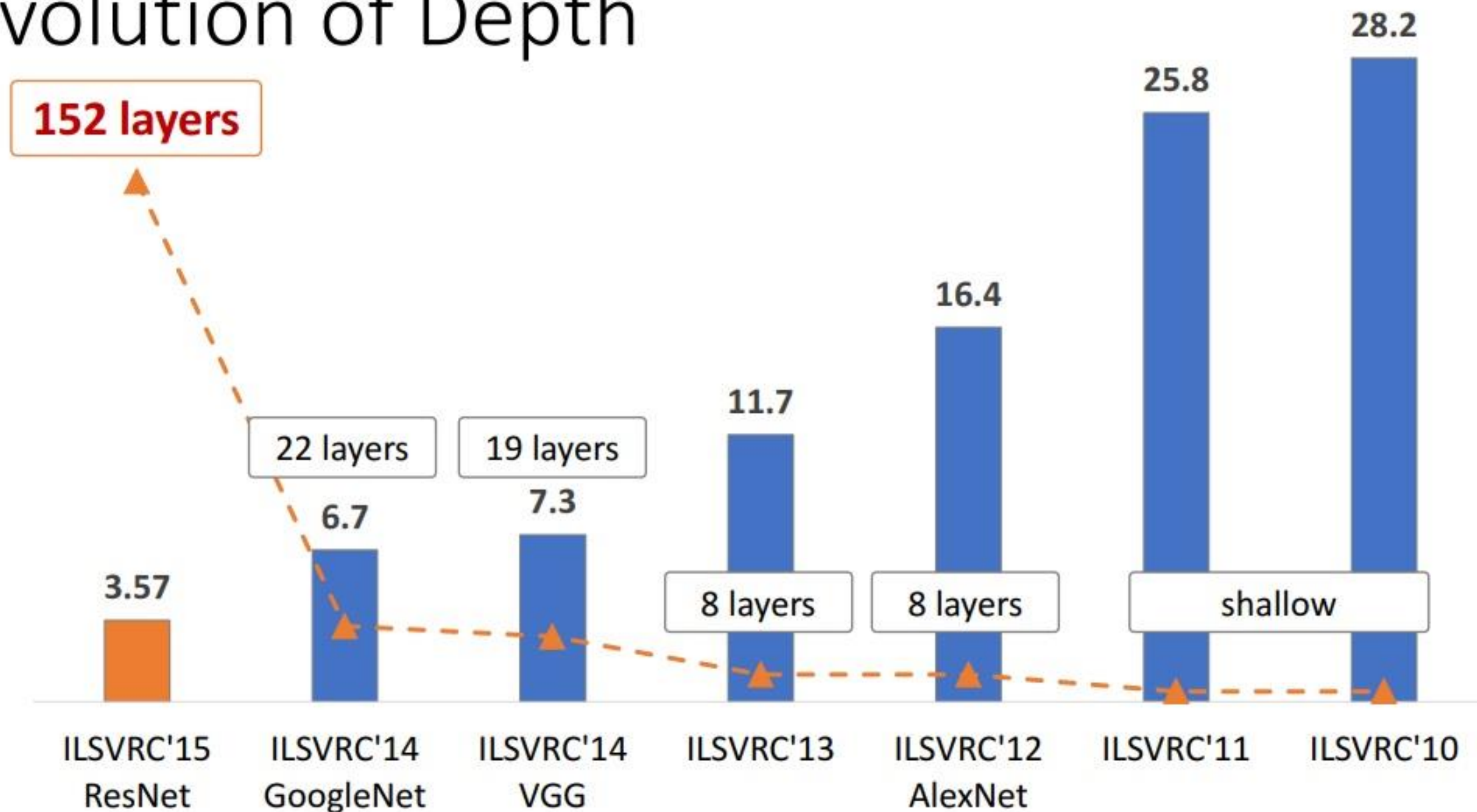


Classic Architecture - GoogLeNet

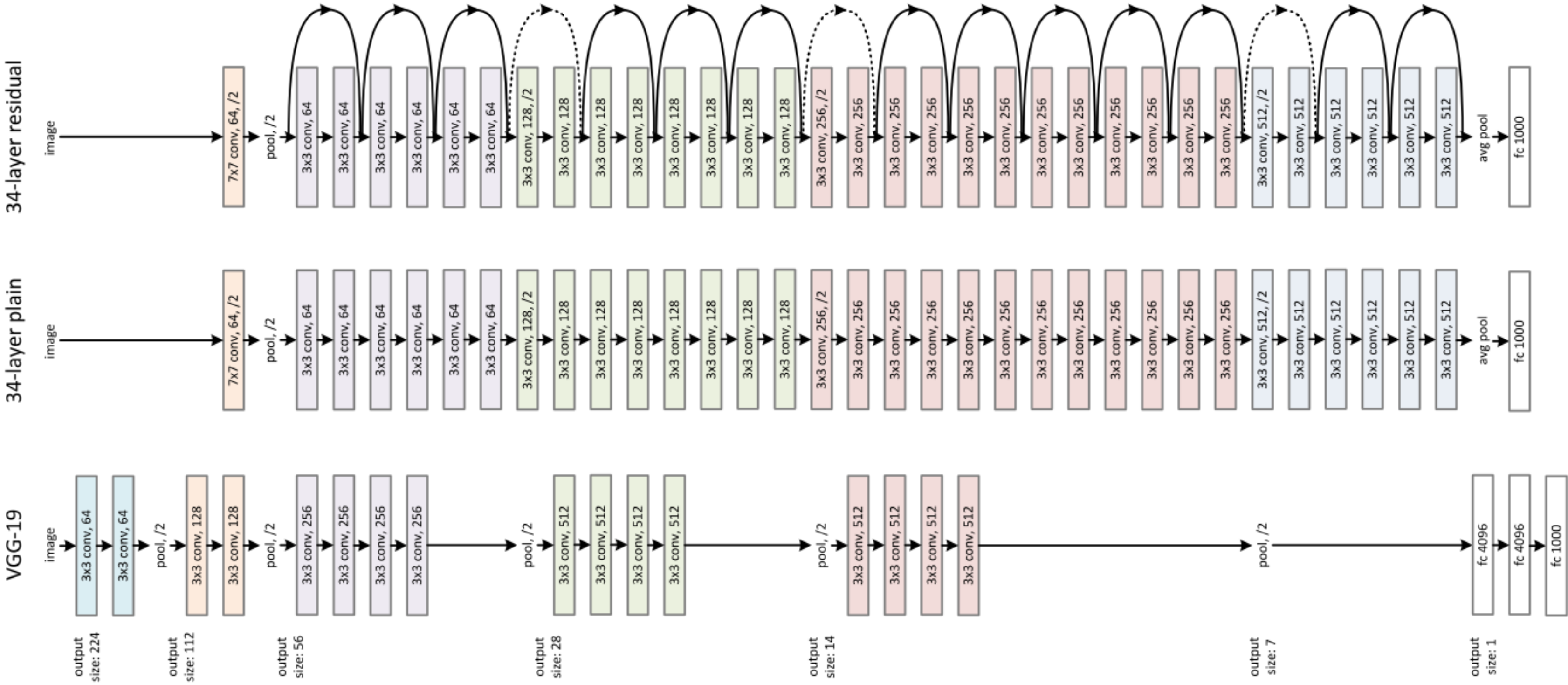


Classic Architecture - ResNet

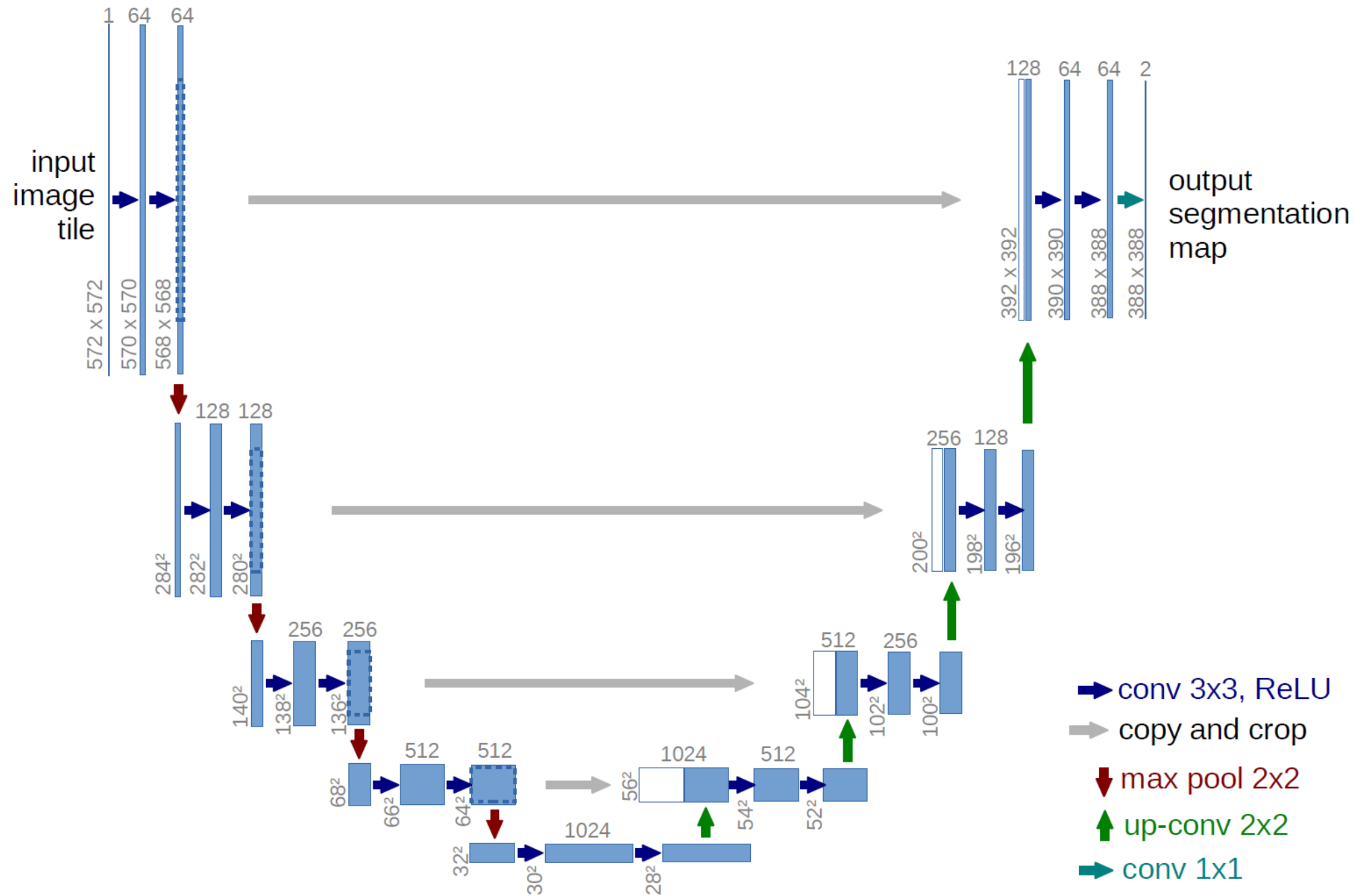
Revolution of Depth



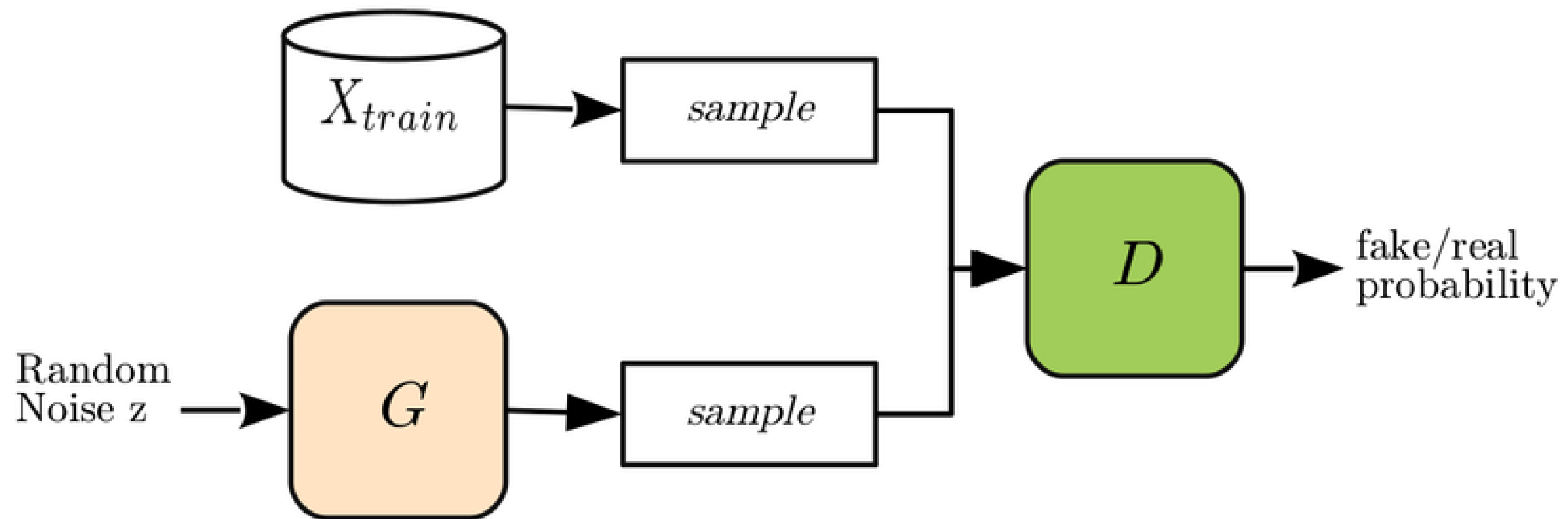
Classic Architecture - ResNet



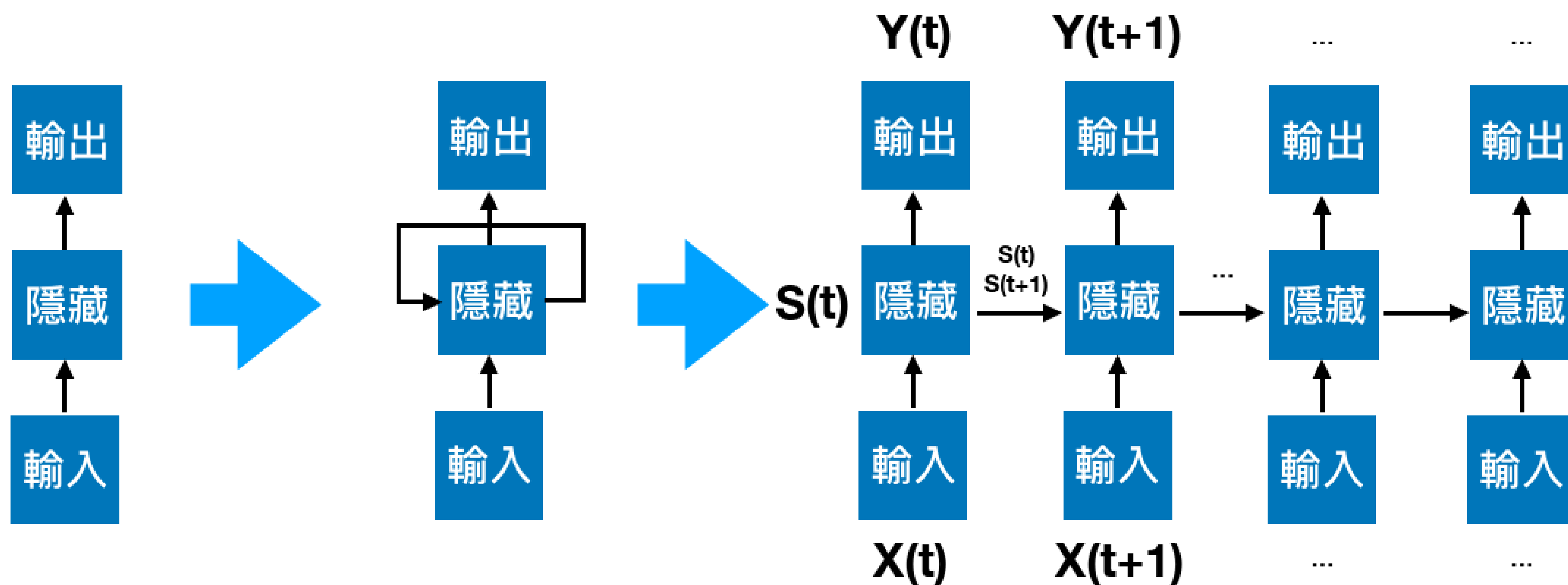
Classic Architecture - U-Net



Classic Architecture – GAN (Generative Adversarial Network)

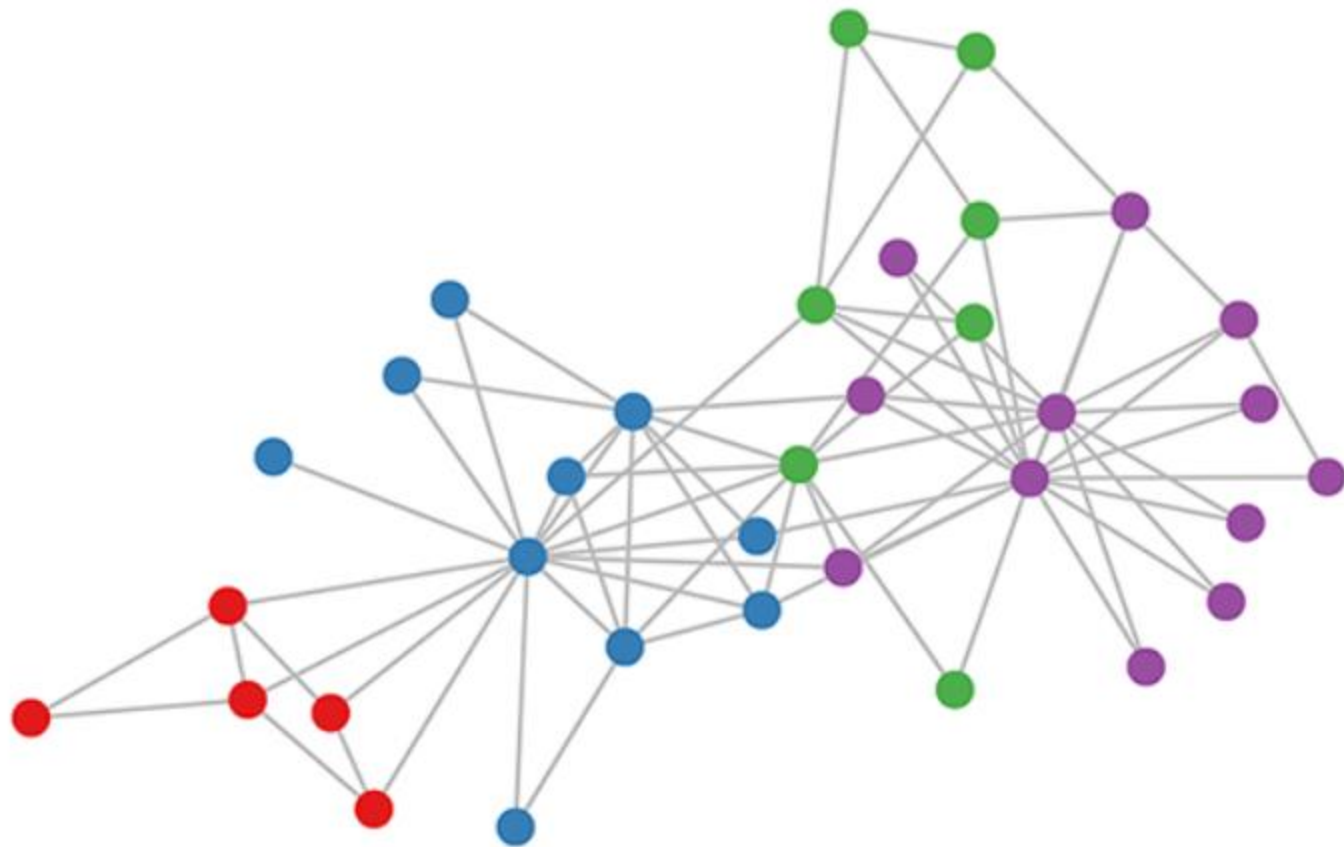
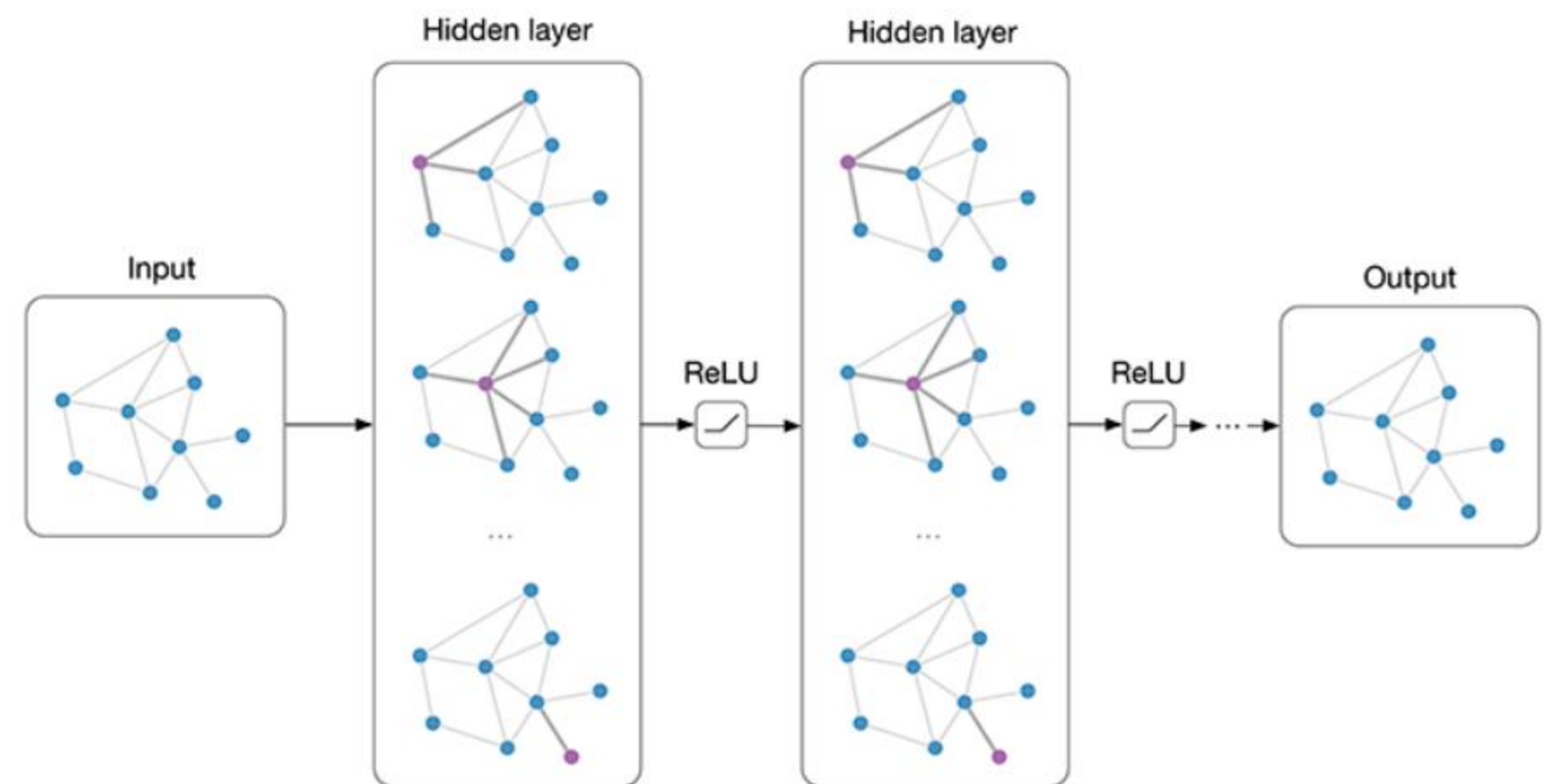


RNN

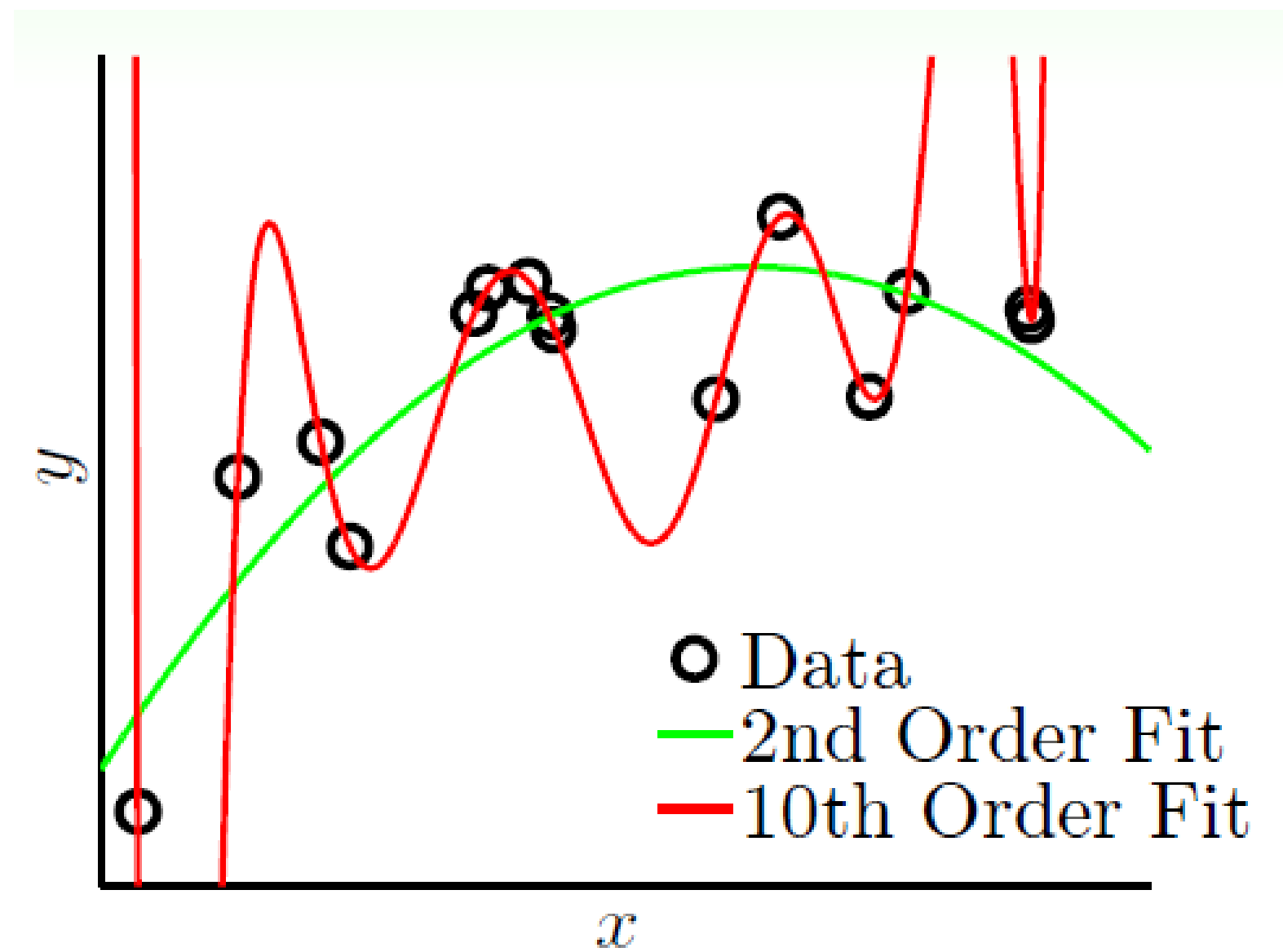


Graph CNN/RNN

- Graph Generation
- Edge Prediction
- Graph Clustering



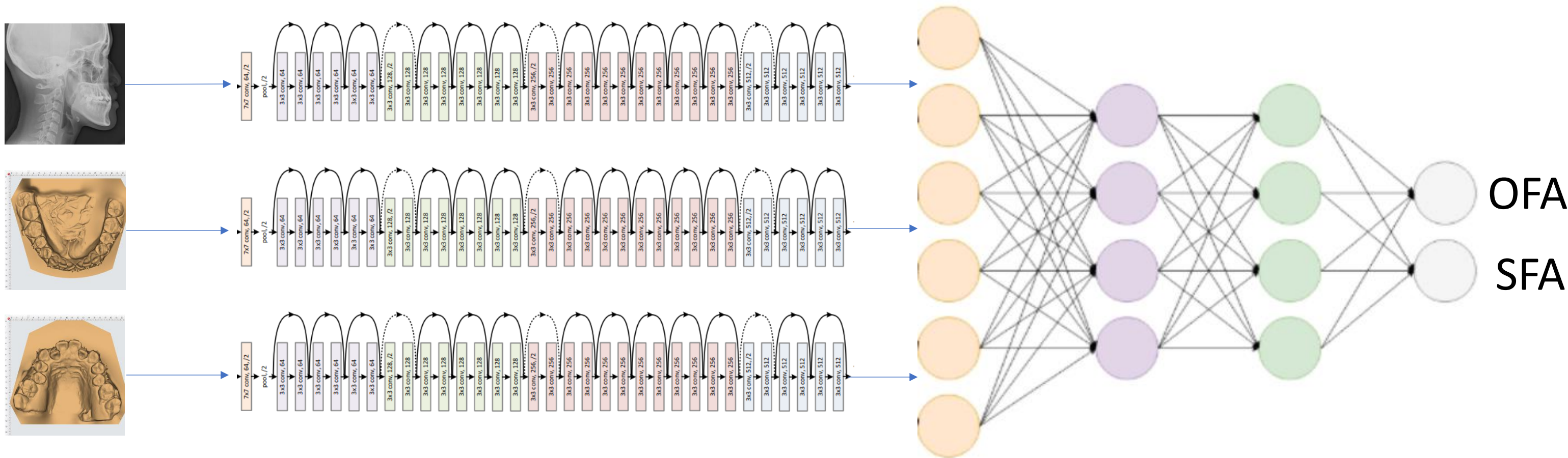
Overfitting



Prevention of Overfitting

- Data Augmentation
- Dropout
- L1/L2 Normalization

OFA/SFA classification



Dataset-the key is distribution

- Train, Valid, Test

- K-fold cross validation

- Repeat K-fold
- Group K-fold
- Nested K-fold
- Stratified K-Fold



到底資料量需要多少？經驗上來說...

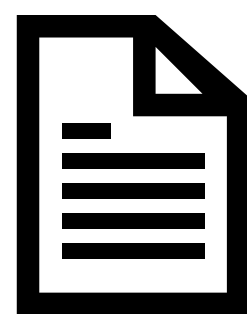
- Very good performance: 50000
- Good performance: 10000
- Mediocre: 5000
- Minimum: 1000 ↑

Introduction of Transformer, Bert, and GPT

Natural Language Processing

- NLP Tasks
- Transformer
- BERT
- GPT

Seq2Class

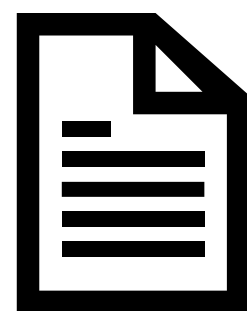


Model

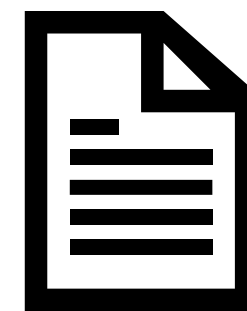


Class

Seq2Seq

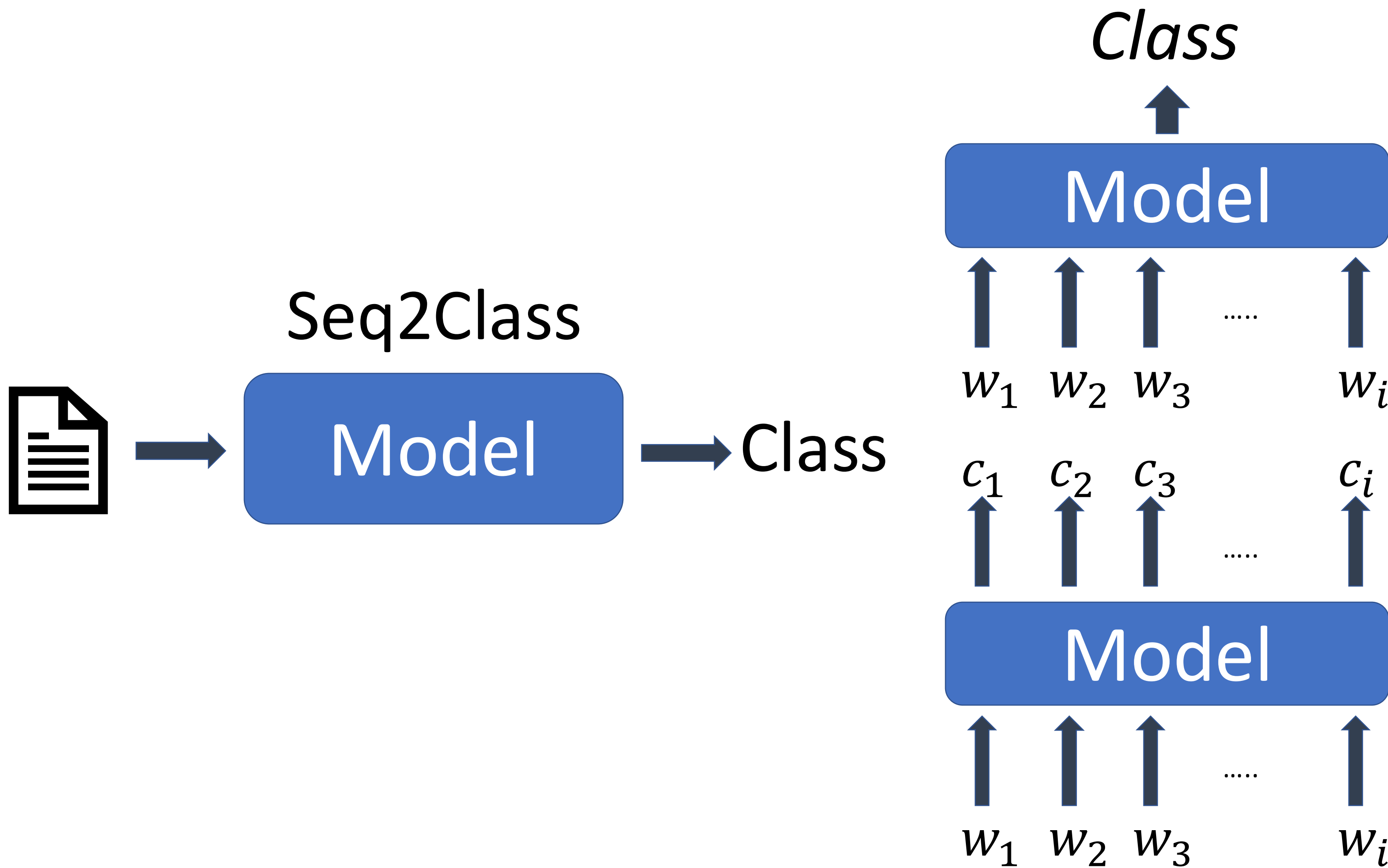


Model

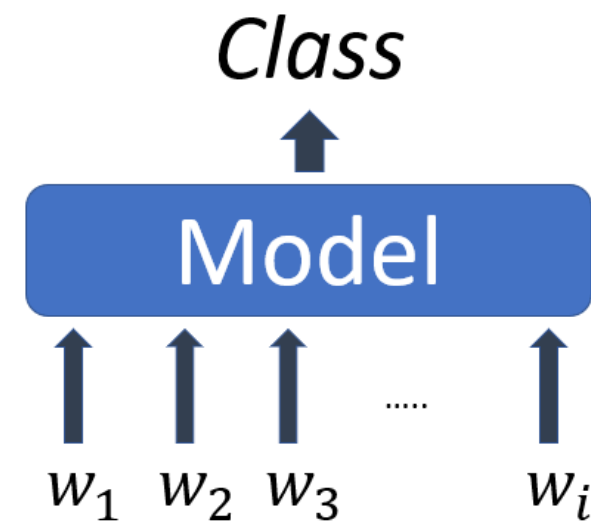


	One Sequence	Multiple Sequences
One Class	Sentiment Classification Stance Detection Veracity Prediction Intent Classification Dialogue Policy	NLI Search Engine Relation Extraction
Class for each Token	POS tagging Word segmentation Extraction Summarization Slotting Filling NER	
Copy from Input		Extractive QA
General Sequence	Abstractive Summarization Translation Grammar Correction NLG	General QA Task Oriented Dialogue Chatbot
Other?	Parsing, Coreference Resolution	

Source of table: Dr. Lee Hung-Yi



Sentiment Classification



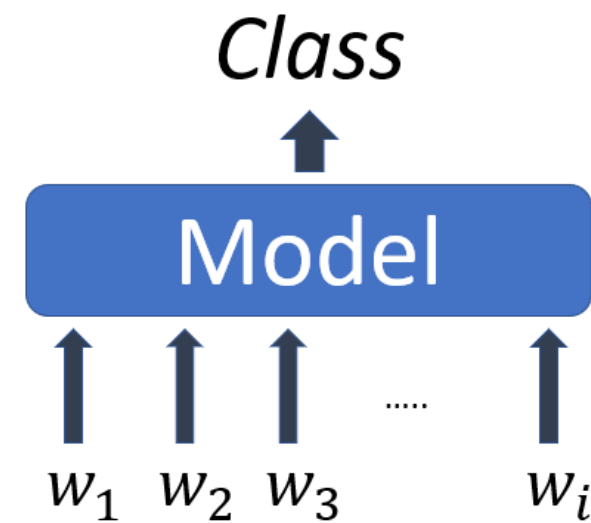
機器人大戰30，還蠻好玩的 => Positive

機器人大戰30，戰鬥動畫都沿用舊的 => Negative

機器人大戰30雖然戰鬥動畫都沿用舊的，但還蠻好玩的 => Positive

機器人大戰30雖然還蠻好玩的，但戰鬥動畫都沿用舊的 => Negative

Stance Detection

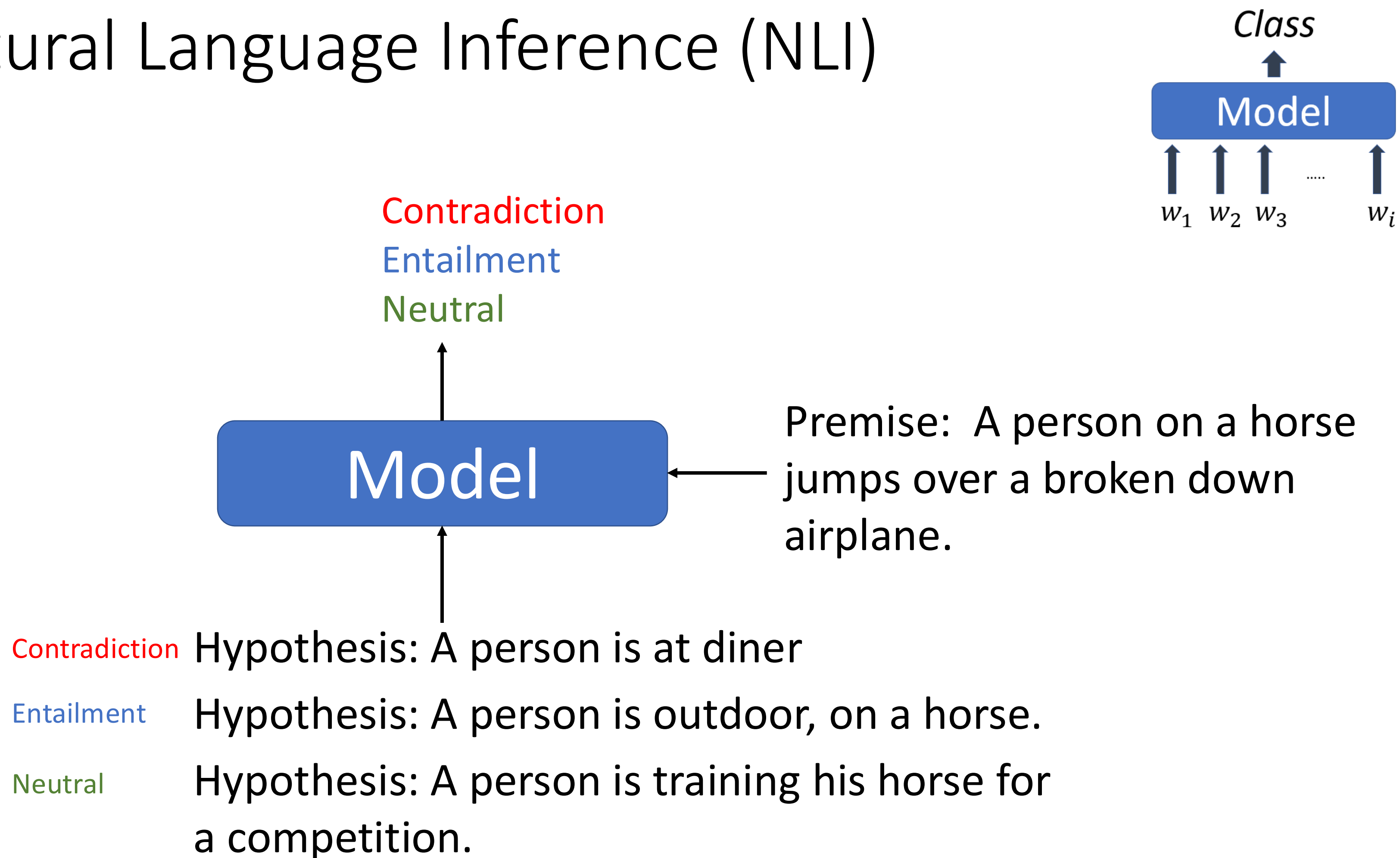


Post: 機器人大戰30是神作

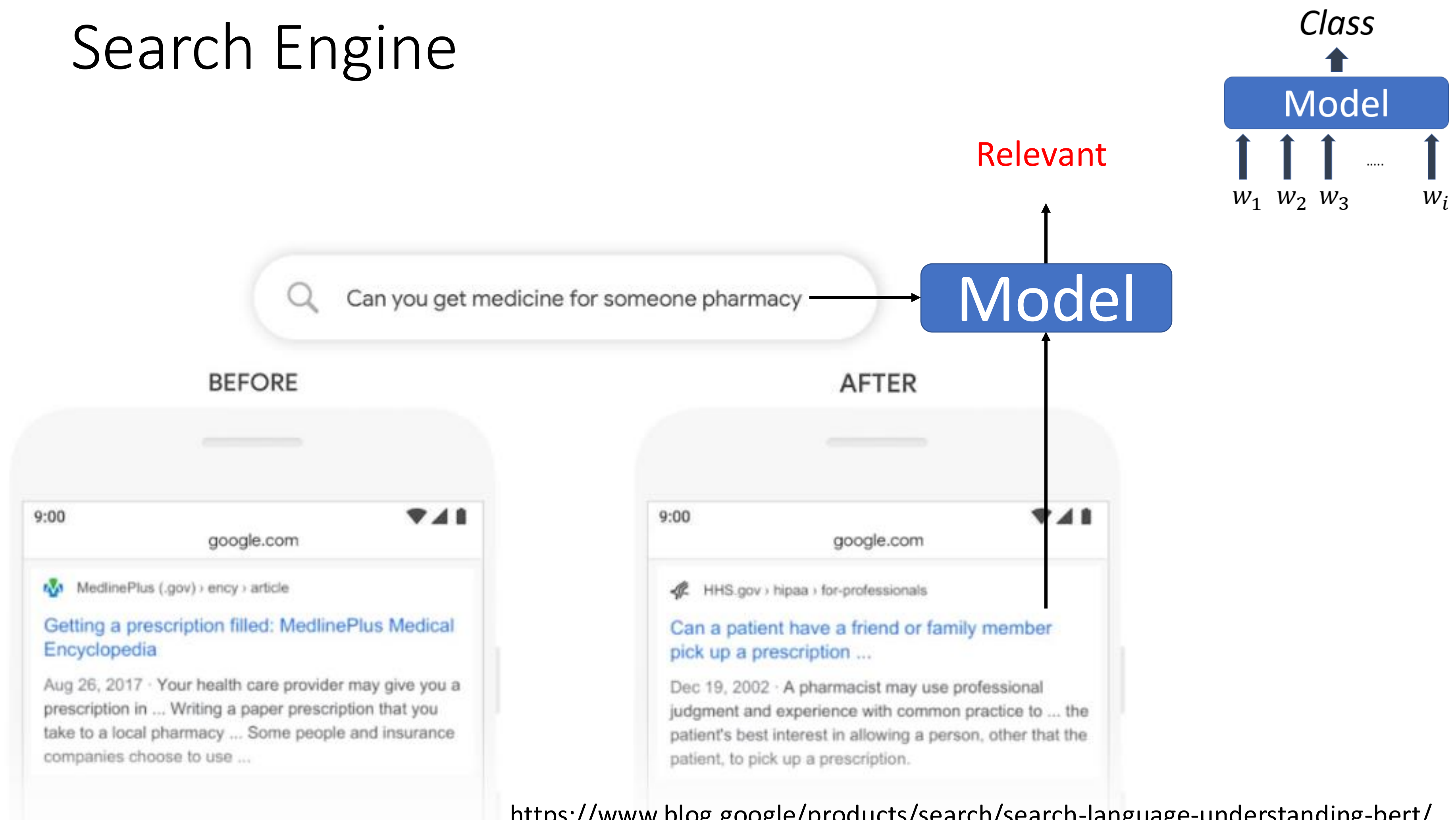
Reply: 這次用那麼多前作的戰鬥動畫，超沒誠意的好嗎。

Classification: Support, Denying, Querying, Commenting

Natural Language Inference (NLI)

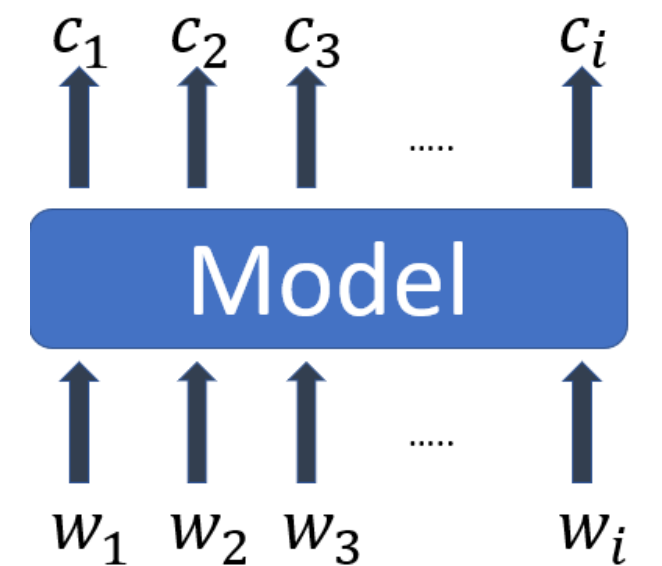
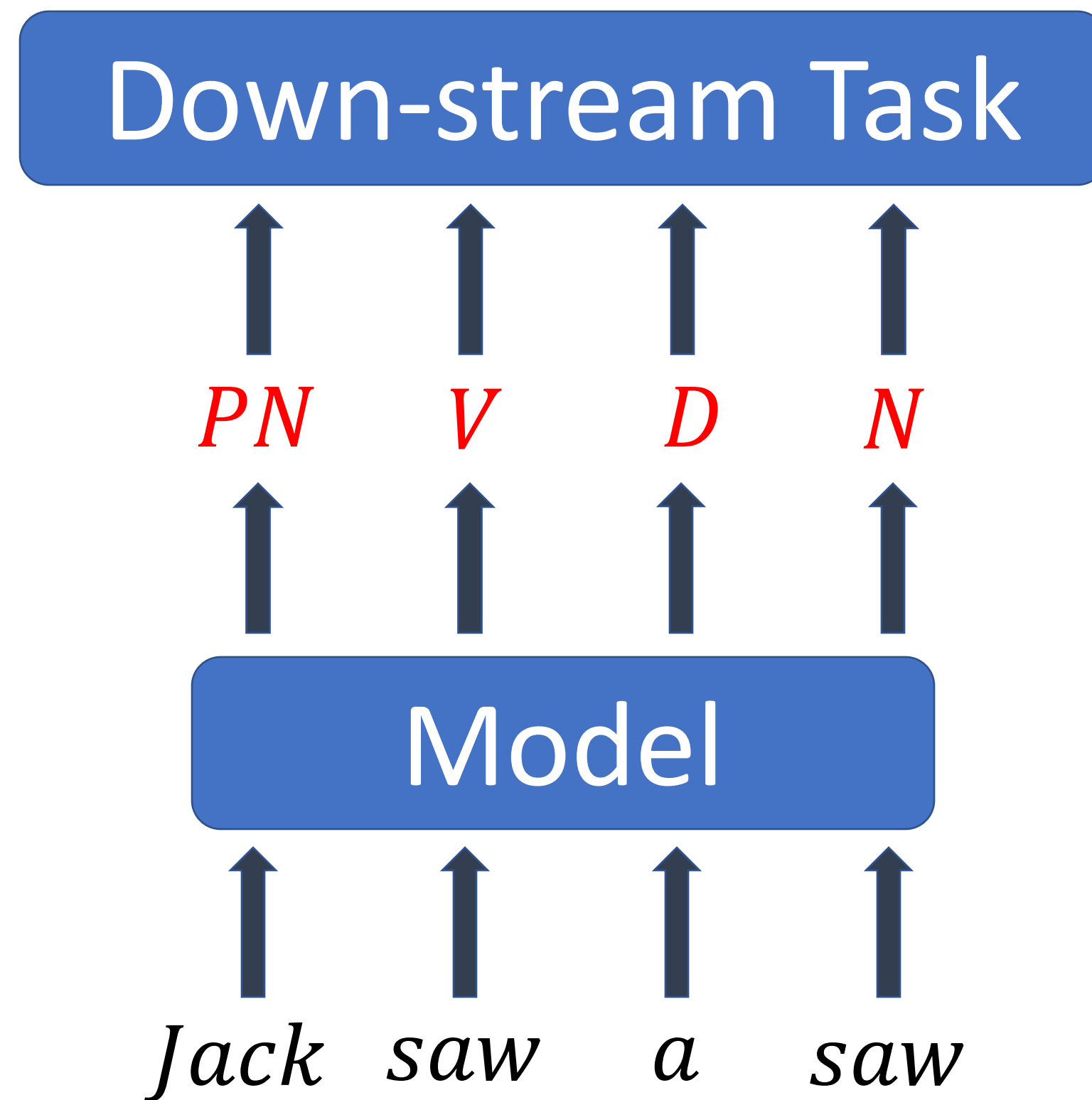


Search Engine

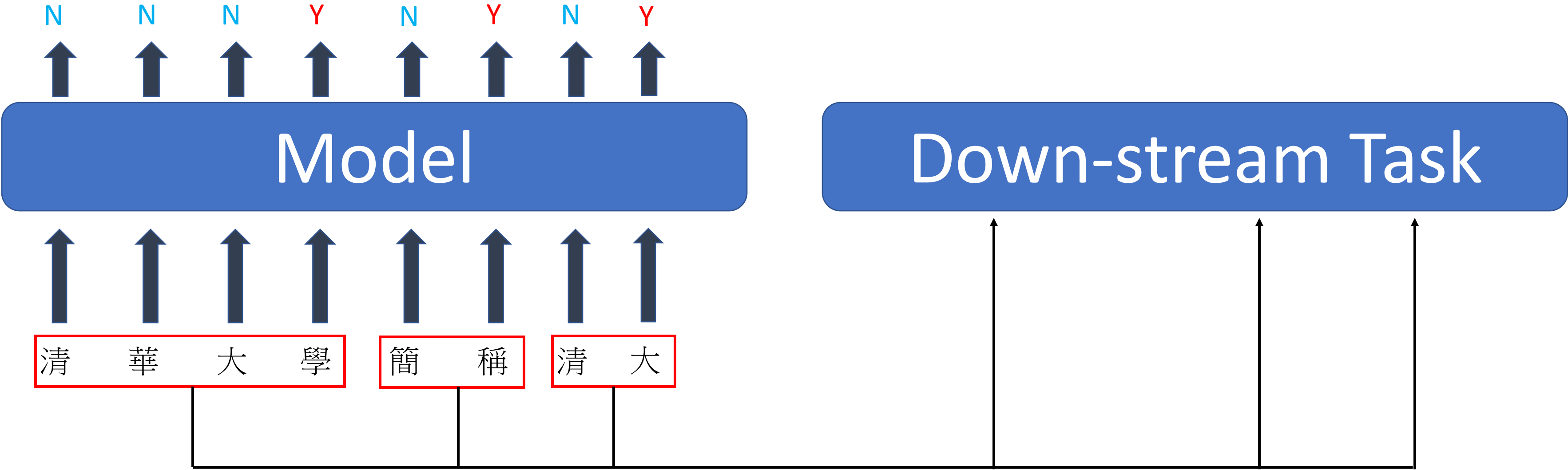


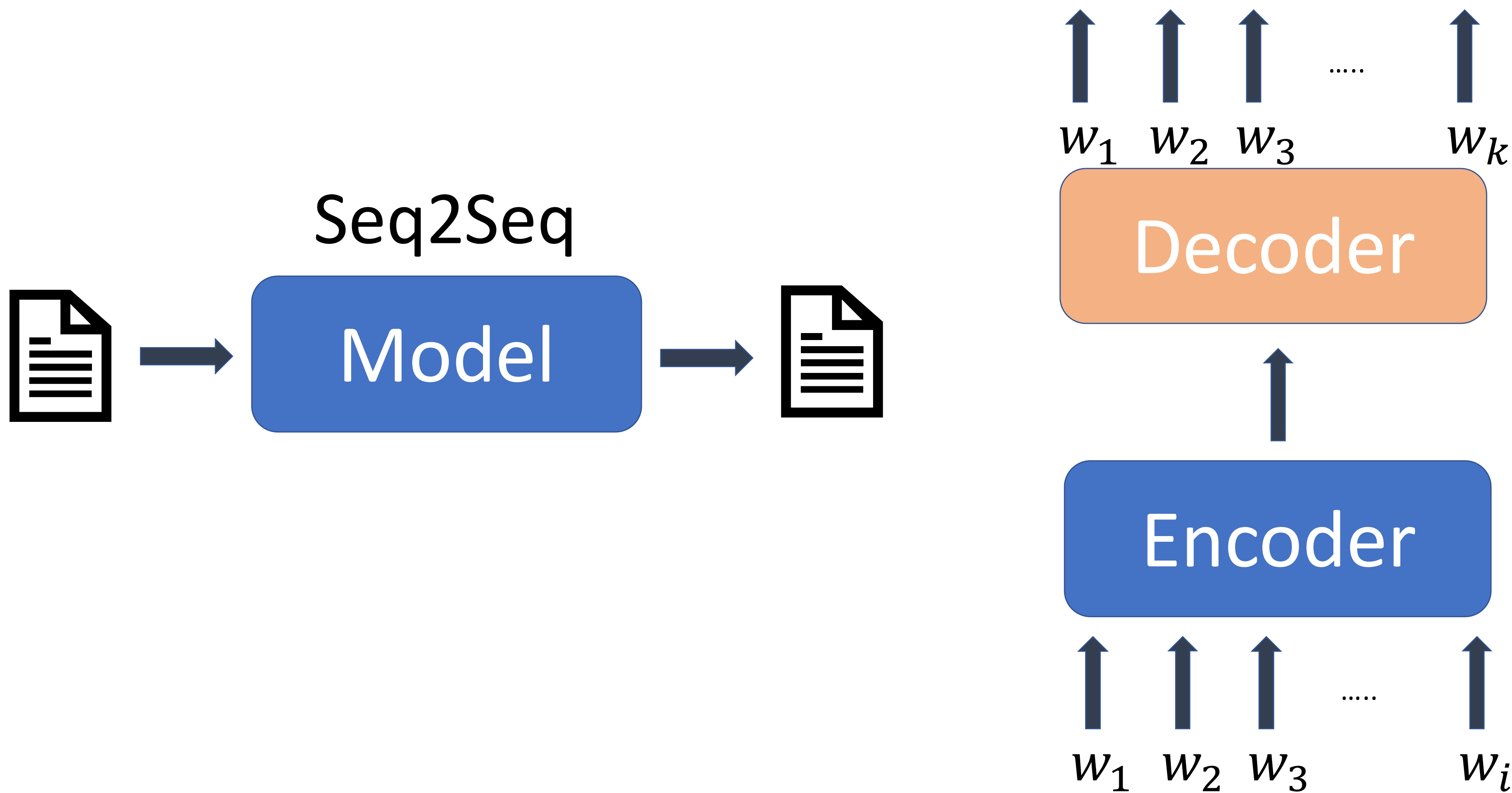
<https://www.blog.google/products/search/search-language-understanding-bert/>

POS (Part-of-Speech) Tagging

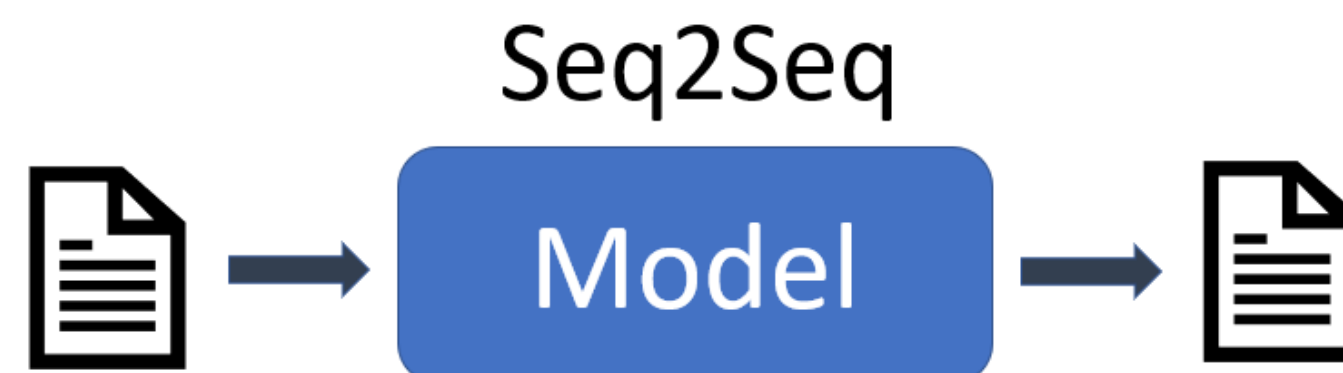


Word Segmentation





Machine Translation

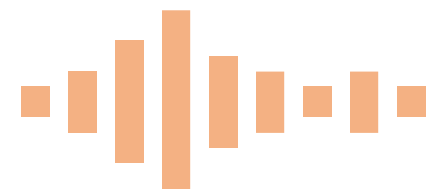


I love you

愛してます



Je vous aime

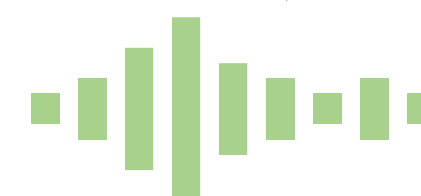


Model

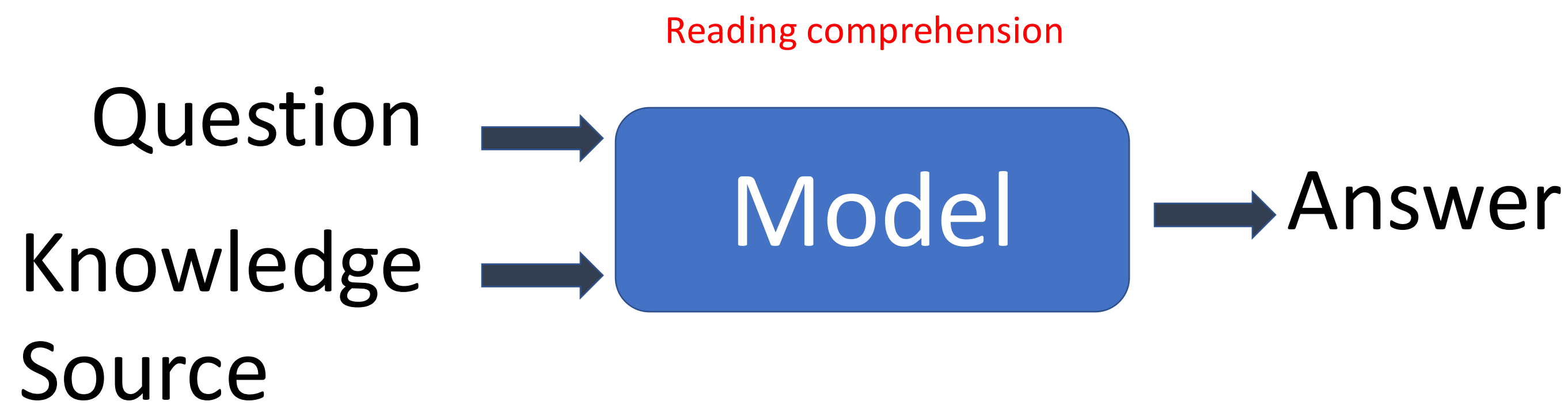
我愛你

ég elska þig

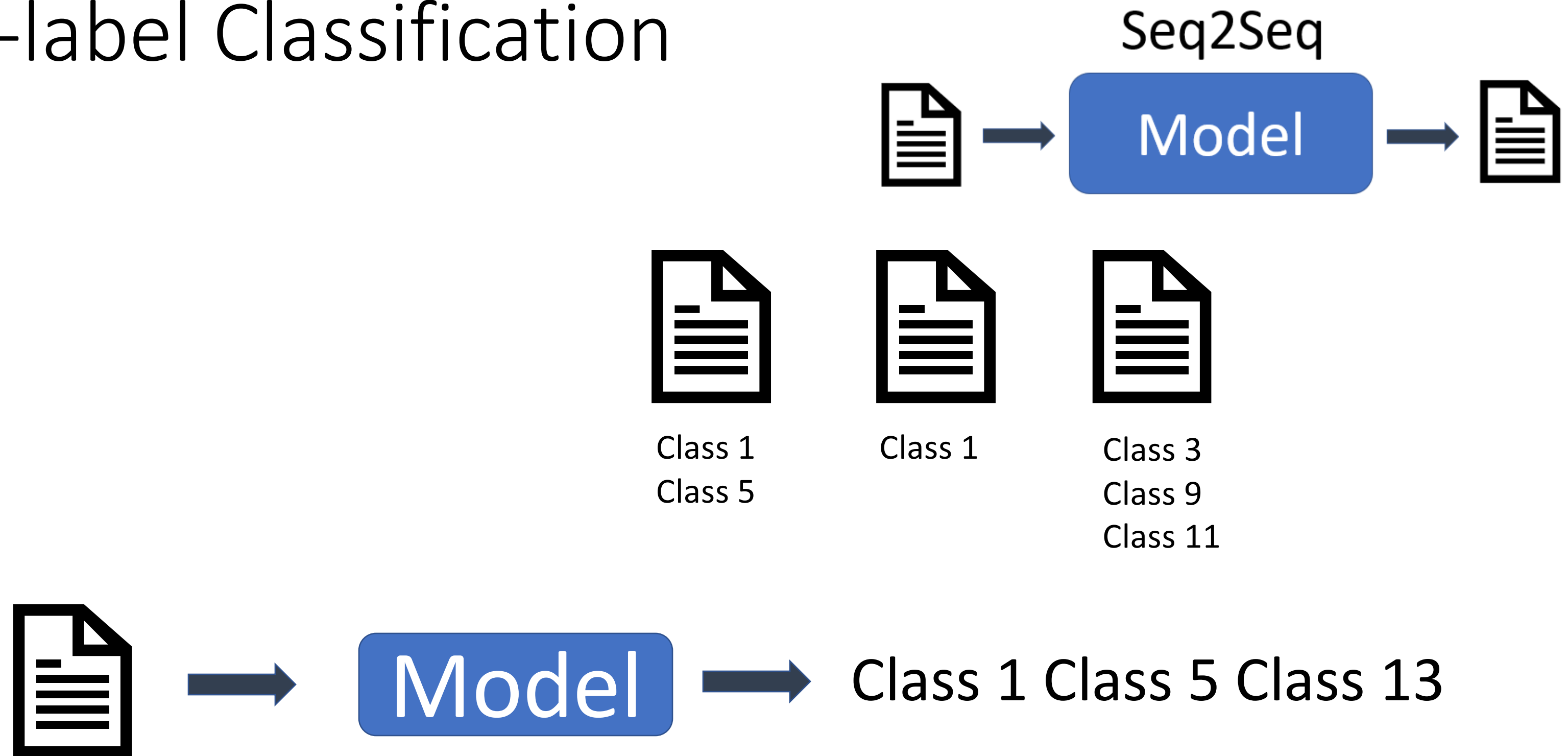
كبحا انا



Question Answering



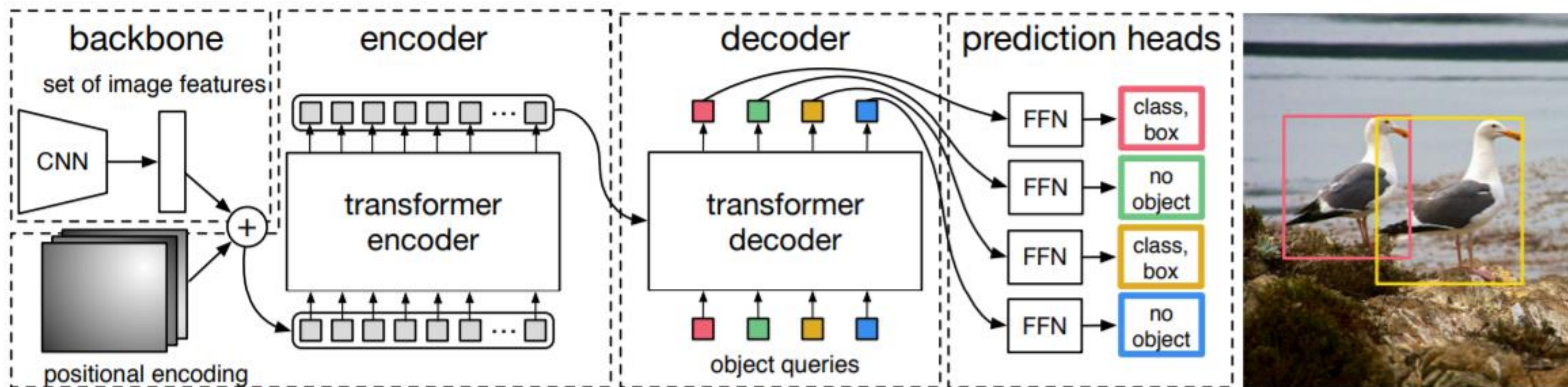
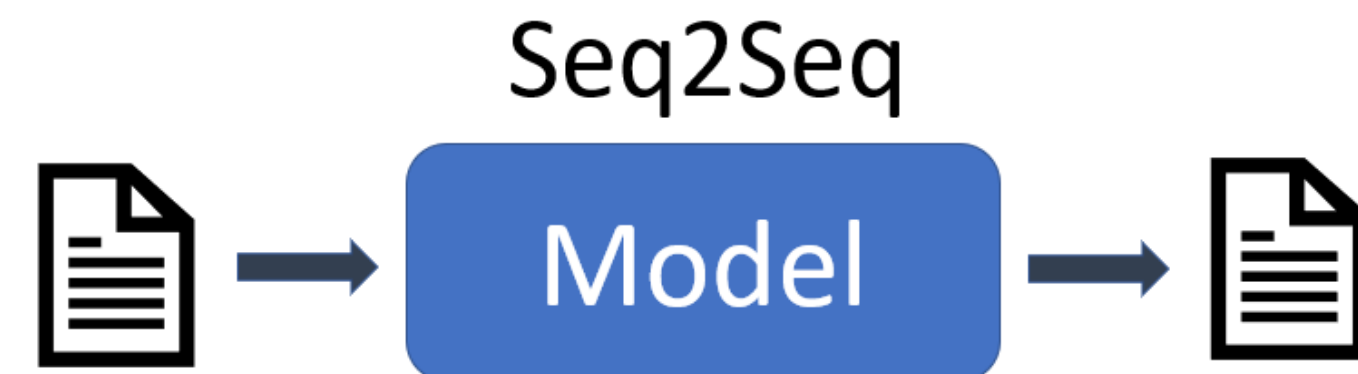
Multi-label Classification



<https://arxiv.org/abs/1909.03434>

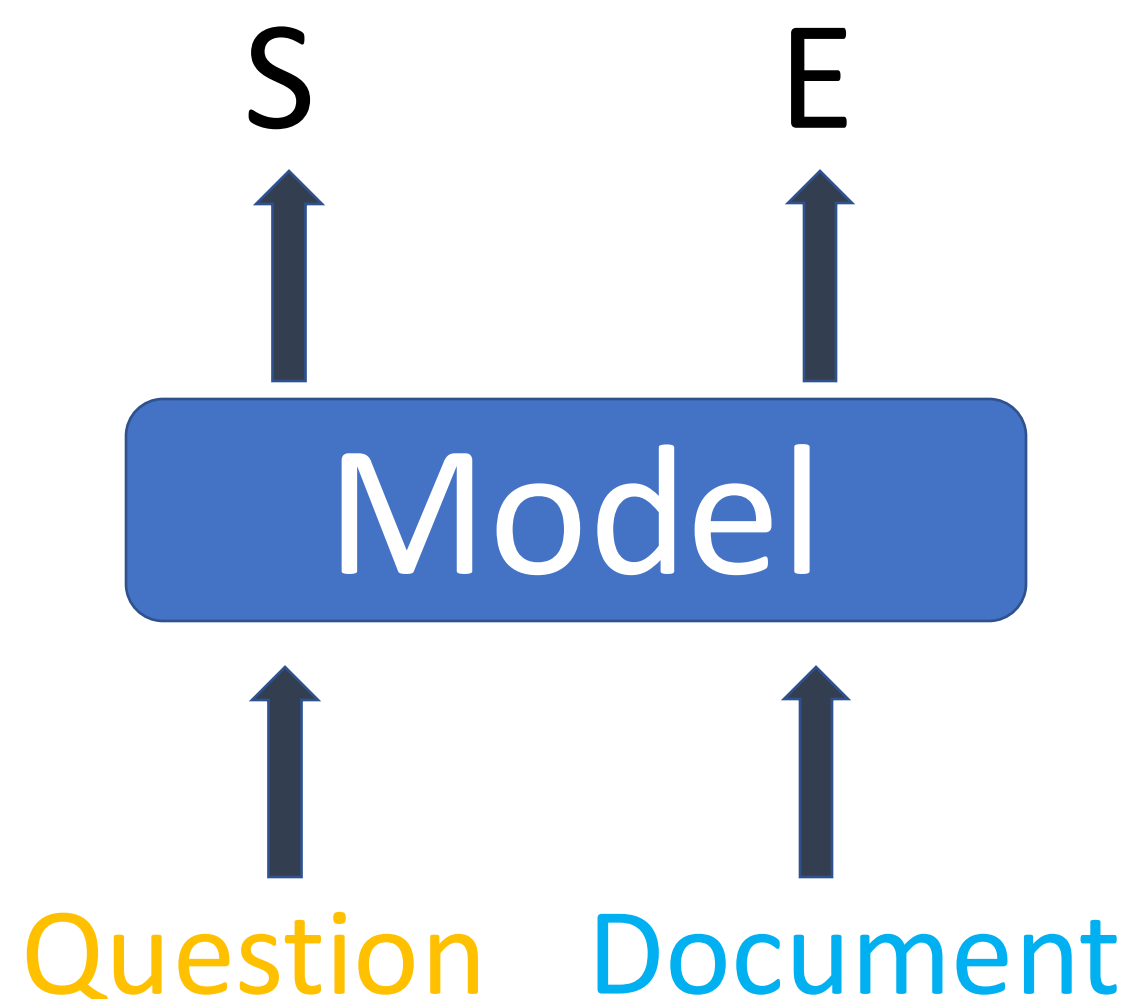
<https://arxiv.org/abs/1707.05495>

Object Detection



<https://arxiv.org/abs/2005.12872>

Extractive QA



In meteorology, precipitation is any product of the condensati¹⁷ of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via⁷⁷ collision⁷⁹ other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called "showers".

What causes precipitation to fall?

gravity

S= 17, E=17

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?

graupel

Where do water droplets collide with ice crystals to form precipitation?

within a cloud

S= 77, E=79

Chatting



<https://www.cleverbot.com/>

User: Sorry for what?

Cleverbot: For not remembering your name.

User: Do you remember your name?

Cleverbot: Yes I do...

User: What is your name?

Cleverbot: Once again, I can't tell you.

User: You have told me that you are Mara.

Cleverbot: I am a boy.

User: Are you a boy?

Cleverbot: Yes.

User: Are you a girl?

Cleverbot: No, are you?

User: No. How old are you?

Cleverbot: 2000 years old.

User: You are so old!

Cleverbot: No i'm quite young.

User: How old are you?

Cleverbot: Why do you keep asking me that?

User: Because I want to know your age.

Cleverbot: What if I do not want to tell you.

User: Then don't tell me your age.

Cleverbot: My name is Blue Monday.

Source of Example: Dr. Lee Hung-Yi

GLUE General Language Understanding Evaluation

Sentiment
Classification

Semantic
Similarity

Natural
Language
Inference

GLUE Tasks			
Name	Download	More Info	Metric
The Corpus of Linguistic Acceptability			Matthew's Corr
The Stanford Sentiment Treebank			Accuracy
Microsoft Research Paraphrase Corpus			F1 / Accuracy
Semantic Textual Similarity Benchmark			Pearson-Spearman Corr
Quora Question Pairs			F1 / Accuracy
MultiNLI Matched			Accuracy
MultiNLI Mismatched			Accuracy
Question NLI			Accuracy
Recognizing Textual Entailment			Accuracy
Winograd NLI			Accuracy
Diagnostics Main			Matthew's Corr

Chinese version <https://www.cluebenchmarks.com/>

<https://gluebenchmark.com/tasks>

SuperGLUE Tasks

Name	Identifier	Download	More Info	Metric
Broadcoverage Diagnostics	AX-b			Matthew's Corr
CommitmentBank	CB			Avg. F1 / Accuracy
Choice of Plausible Alternatives	COPA			Accuracy
Multi-Sentence Reading Comprehension	MultiRC			F1a / EM
Recognizing Textual Entailment	RTE			Accuracy
Words in Context	WiC			Accuracy
The Winograd Schema Challenge	WSC			Accuracy
BoolQ	BoolQ			Accuracy
Reading Comprehension with Commonsense Reasoning	ReCoRD			F1 / Accuracy
Winogender Schema Diagnostics	AX-g			Gender Parity / Accuracy

<https://super.gluebenchmark.com/tasks>

DecaNLP

- 10 NLP Tasks
- 所有Task皆可被視為QA

Examples

<u>Question</u>	<u>Context</u>	<u>Answer</u>
What is a major importance of Southern California in relation to California and the US?	...Southern California is a major economic center for the state of California and the US....	major economic center
What is the translation from English to German?	Most of the planet is ocean water.	Der Großteil der Erde ist Meerwasser
What is the summary?	Harry Potter star Daniel Radcliffe gains access to a reported £320 million fortune ...	Harry Potter star Daniel Radcliffe gets £320M fortune ...
Hypothesis: Product and geography are what make cream skimming work. Entailment , neutral, or contradiction?	Premise: Conceptually cream skimming has two basic dimensions – product and geography.	Entailment
Is this sentence positive or negative?	A stirring, funny and finally transporting re-imagining of Beauty and the Beast and 1930s horror film.	positive

Extractive QA

Translation

Summary

NLI

Sentiment classification

<https://decanlp.com/>

Natural Language Processing

- NLP Tasks
- Transformer
- BERT
- GPT

Input Is a Set of Vector

One-hot Encoding

Apple = [1 0 0 0 0 0 0 0 0 0]

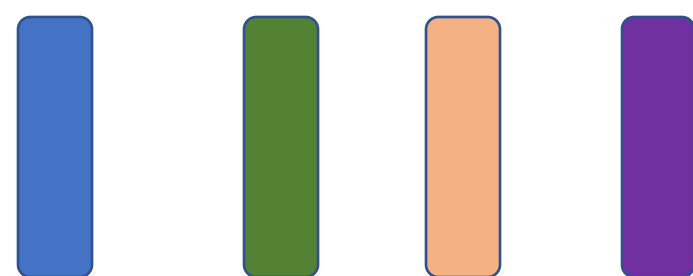
Ball = [0 1 0 0 0 0 0 0 0 0]

Call = [0 0 1 0 0 0 0 0 0 0]

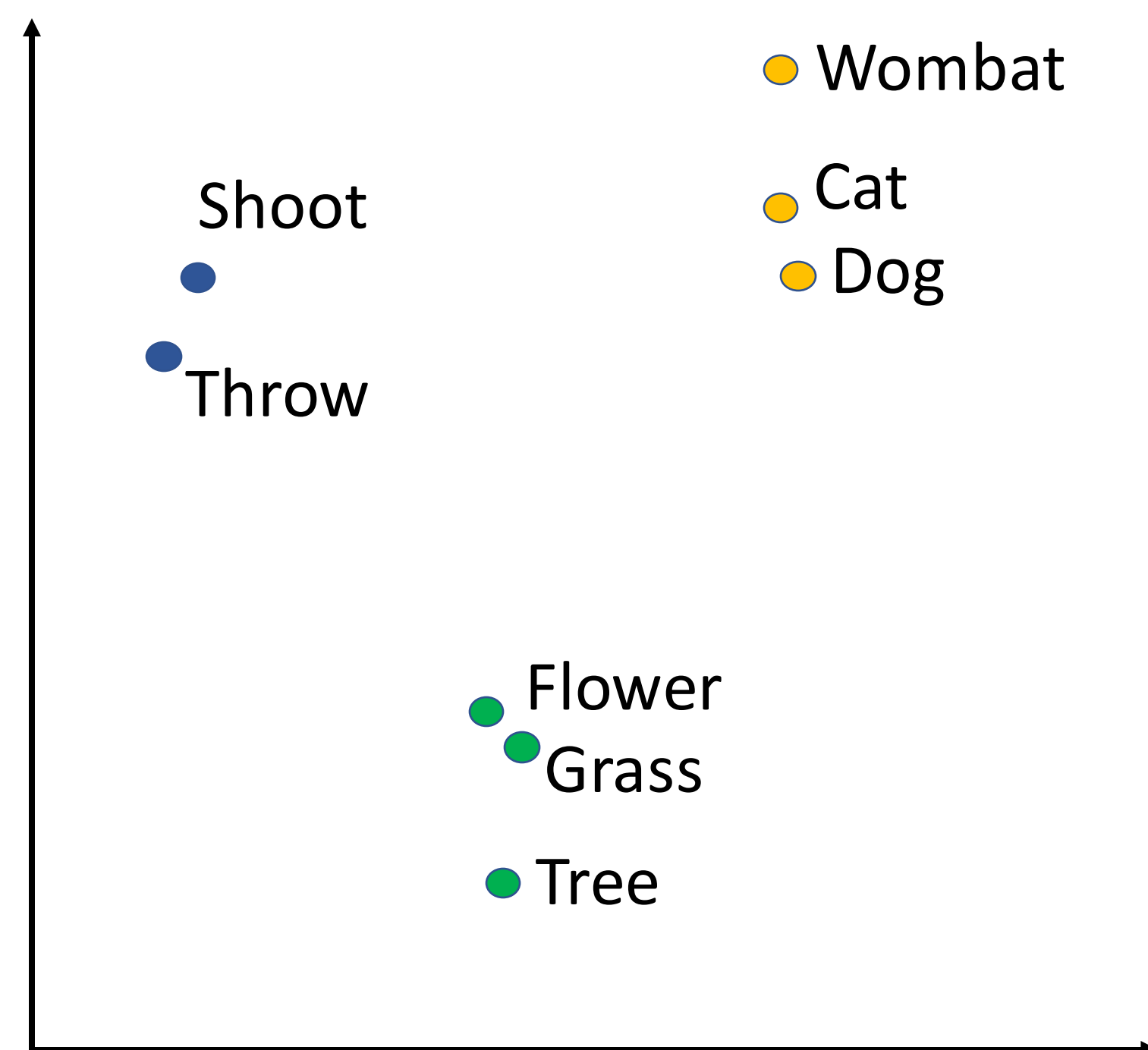
Dog = [0 0 0 1 0 0 0 0 0 0]

⋮

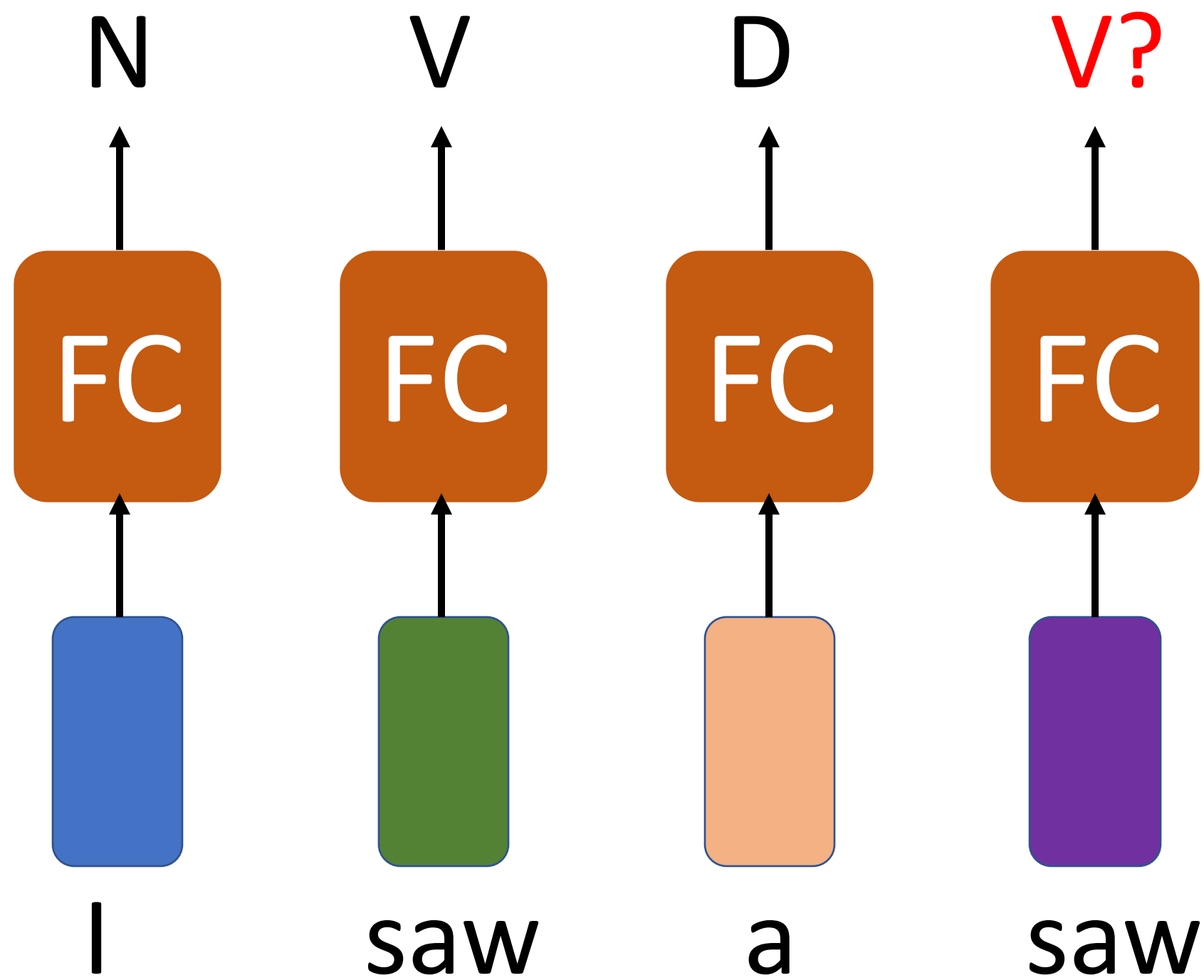
This is a dog



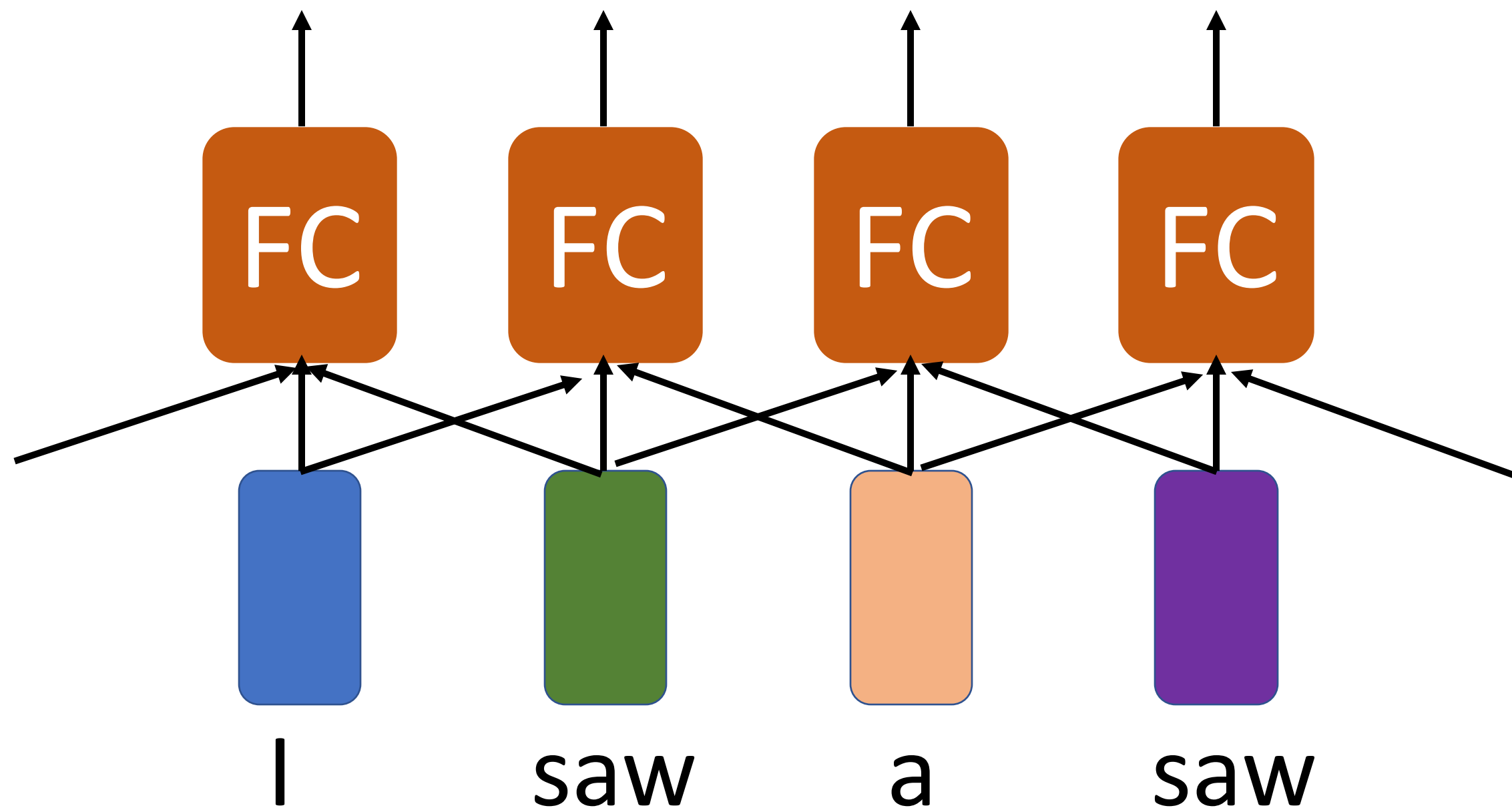
Word Embedding



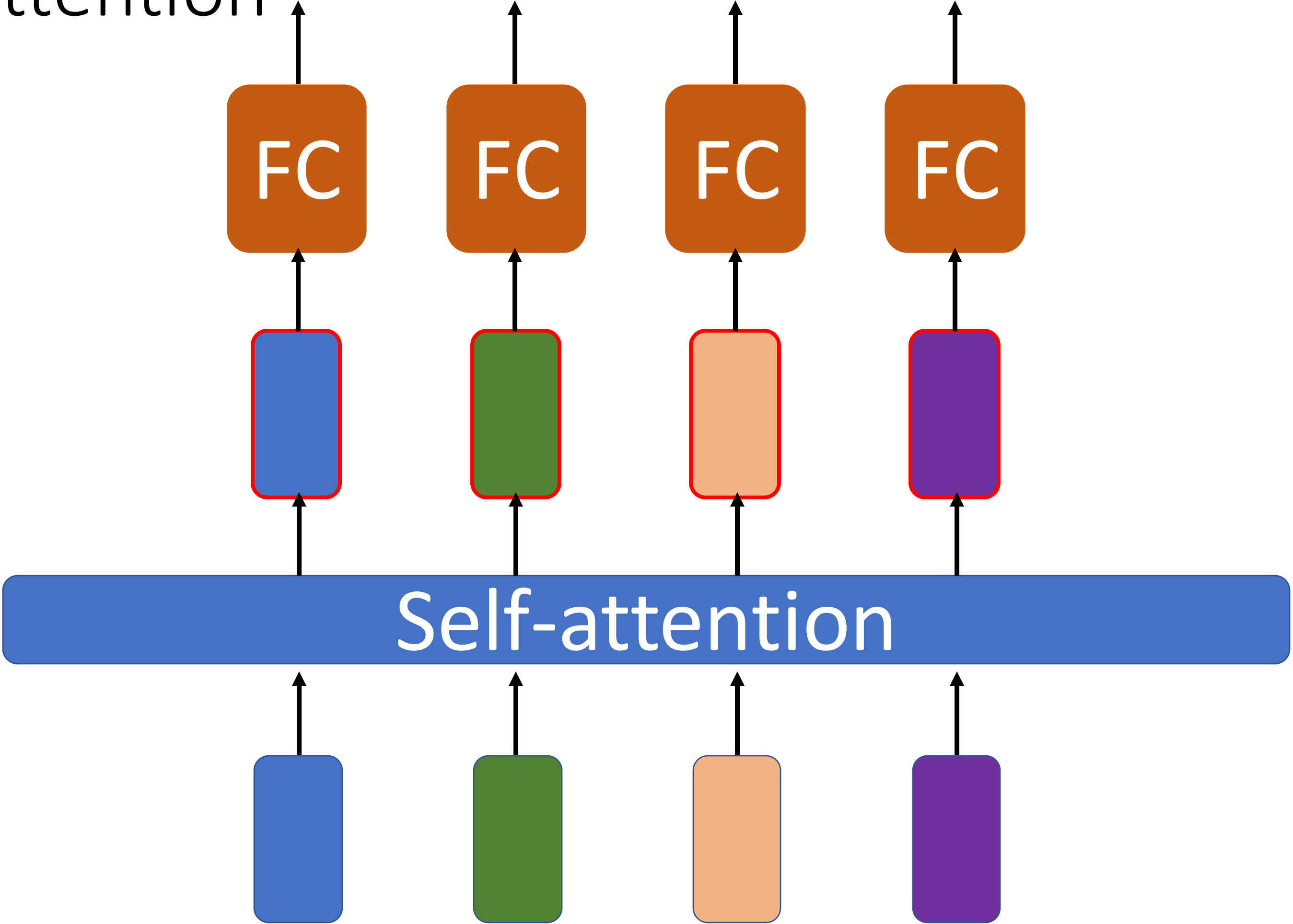
Sequence Labeling

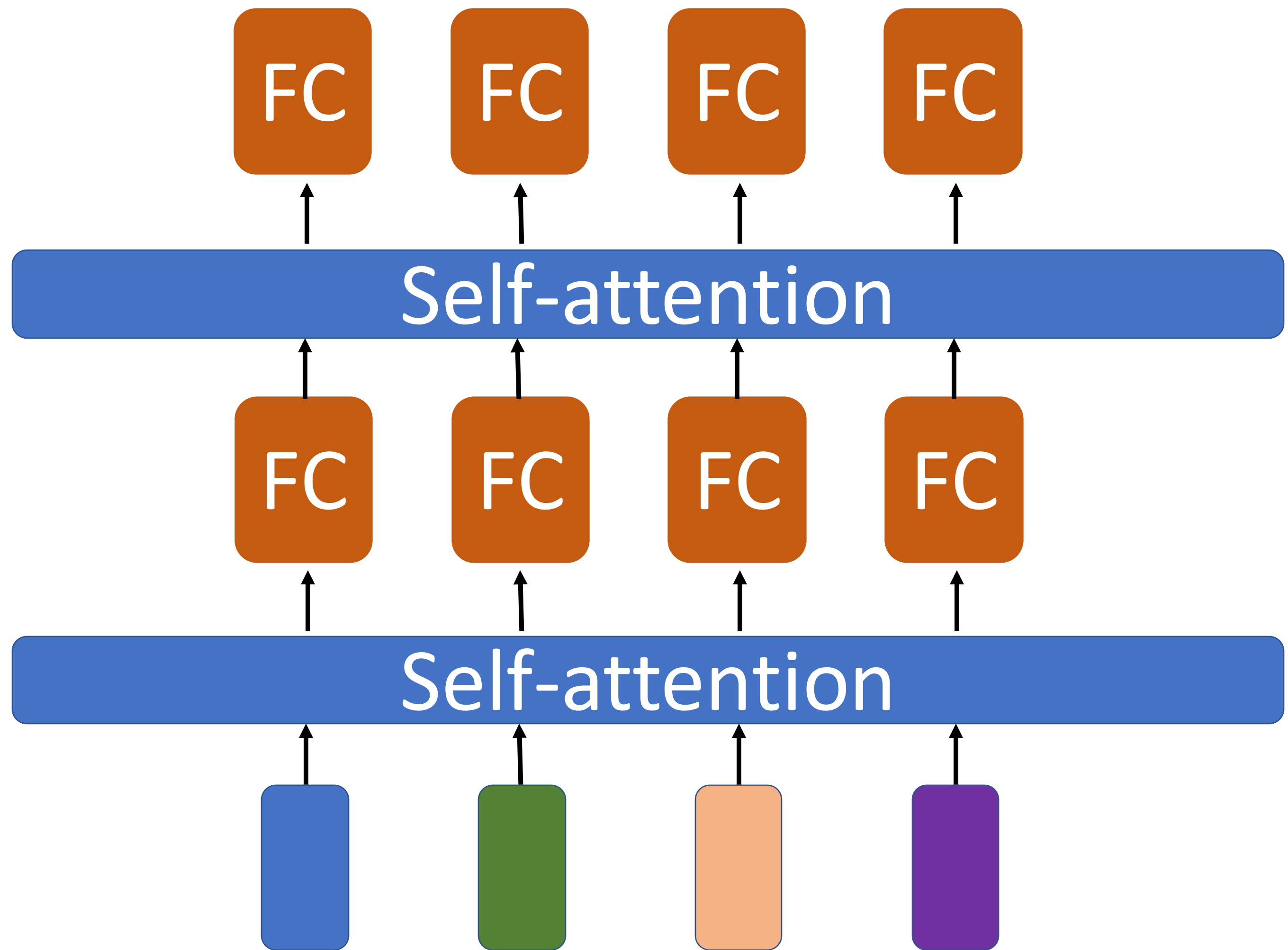


Sequence Labeling



Self-attention





Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkor
Google Resear
usz@google.c

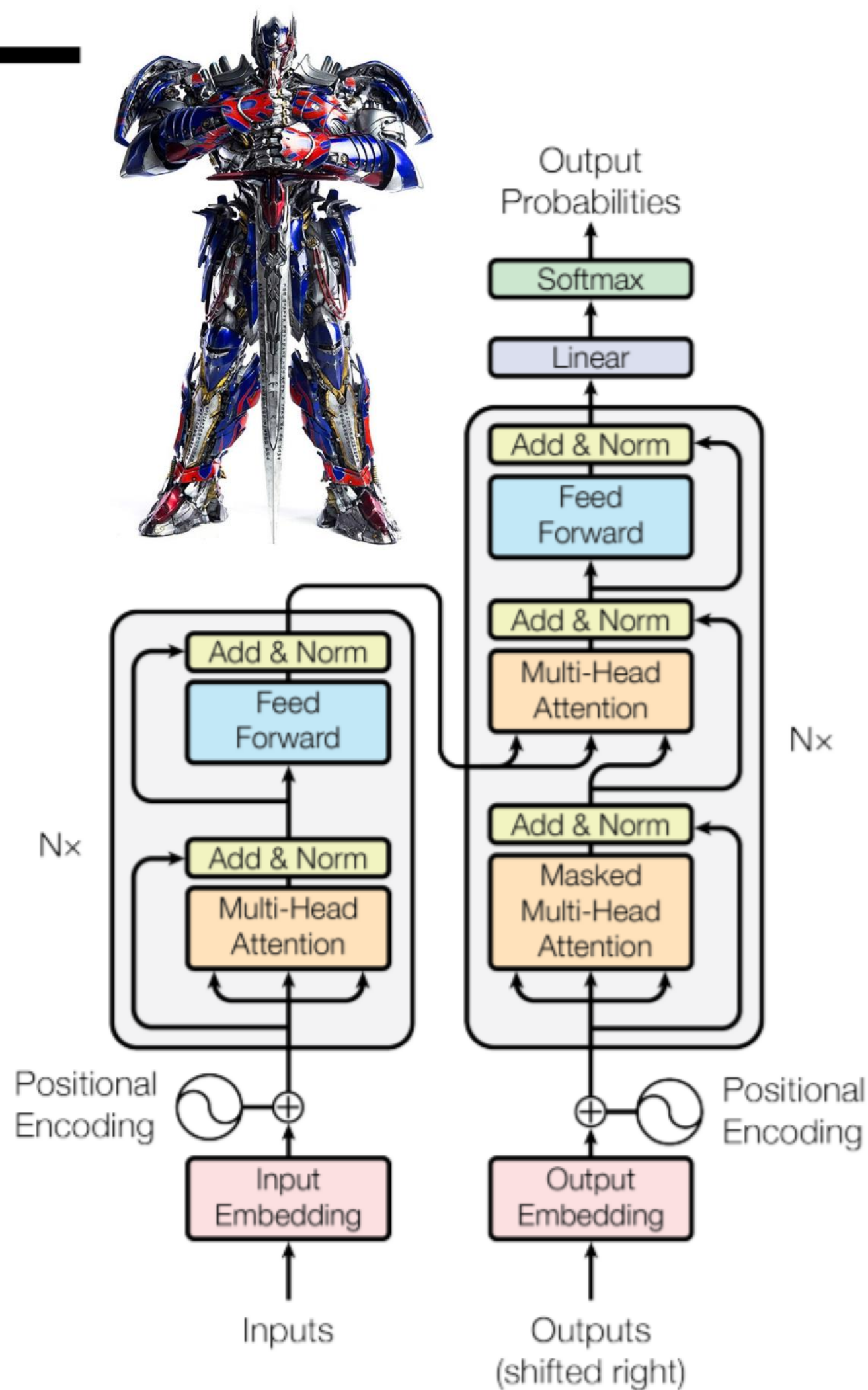
Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez* †
University of Toronto
aidan@cs.toronto.edu

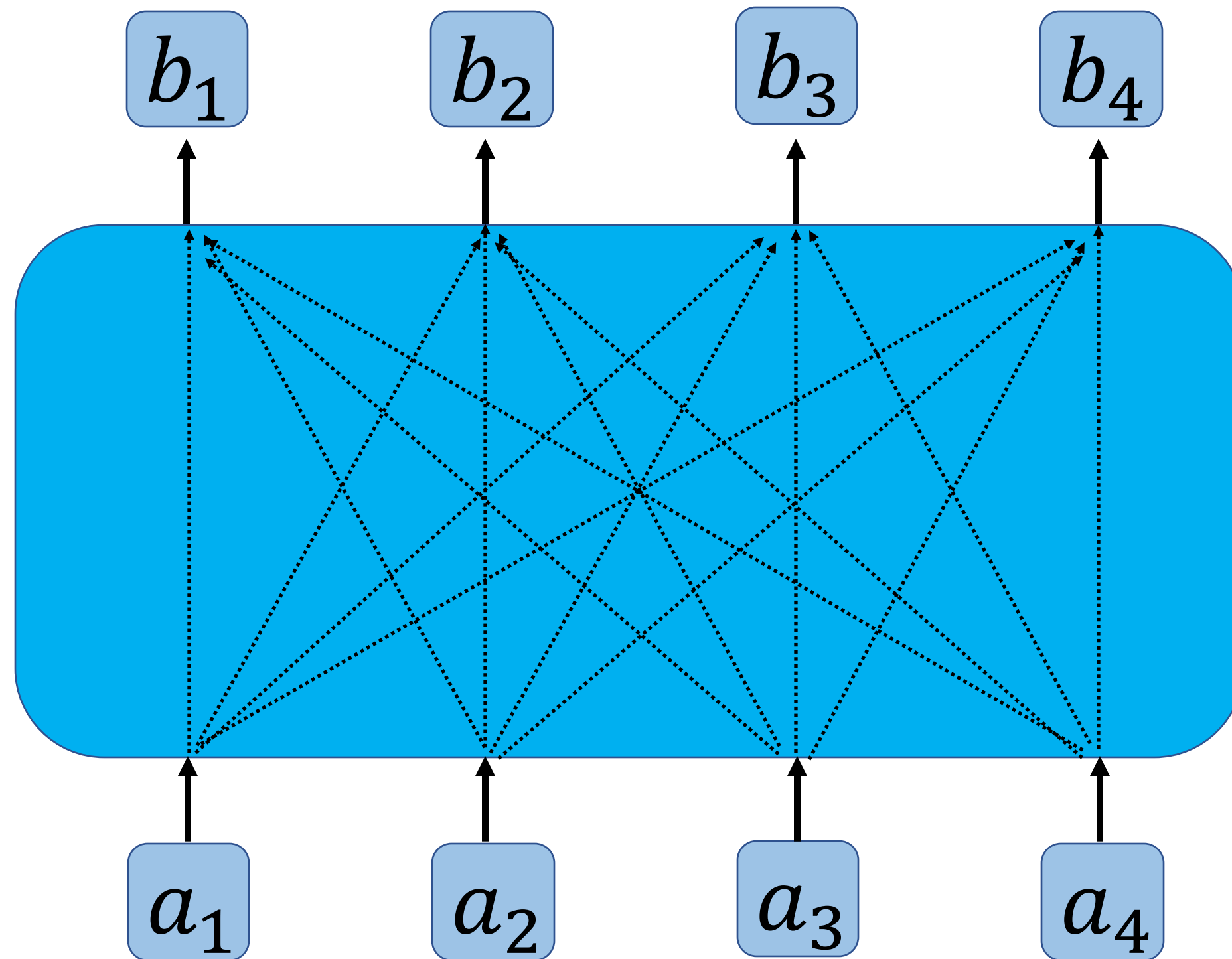
Łukasz Kaiser*
Google Brain
lukaszkaizer@google.com

Illia Polosukhin* ‡
illia.polosukhin@gmail.com

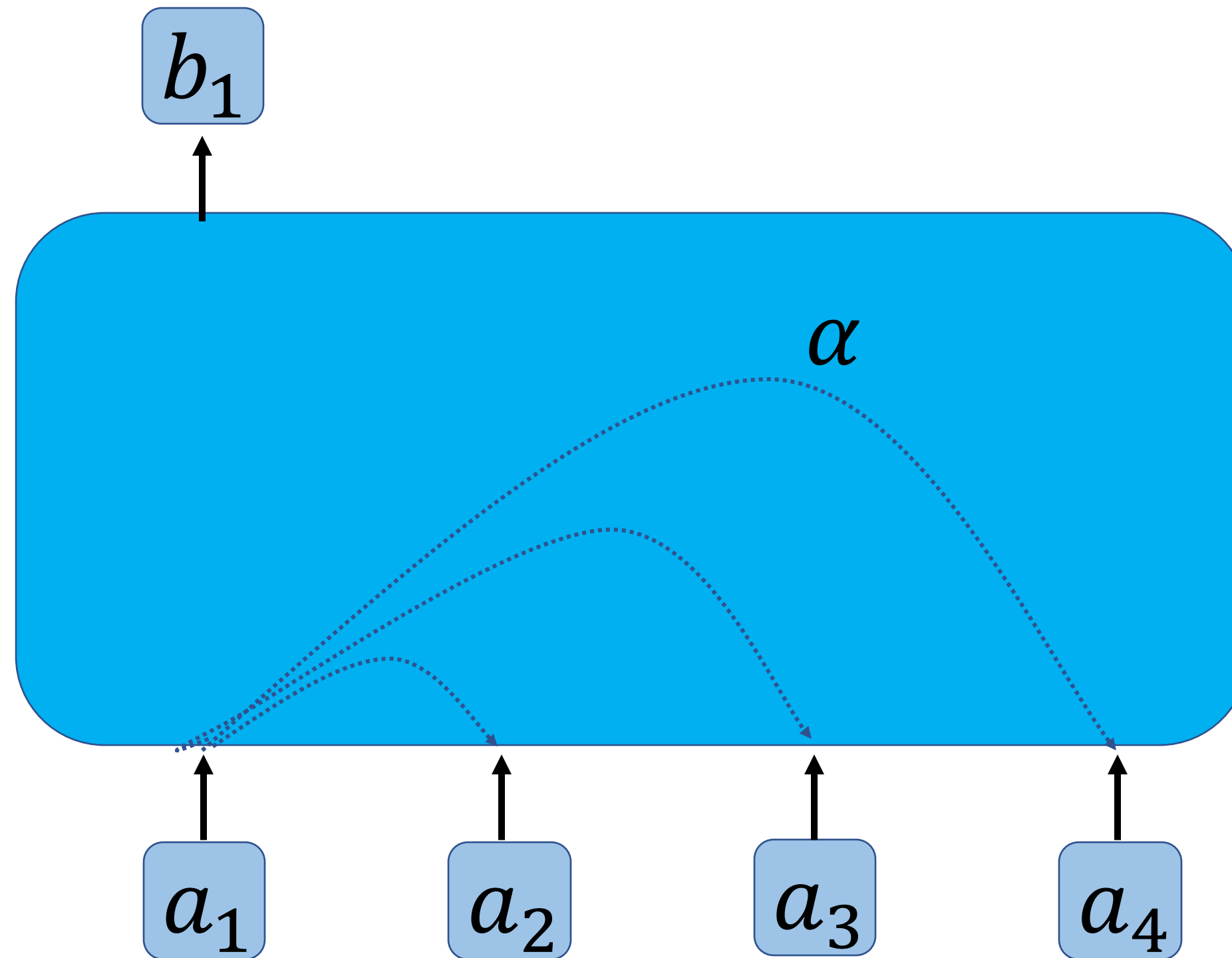
arXiv:1706.03762v5



Self-attention

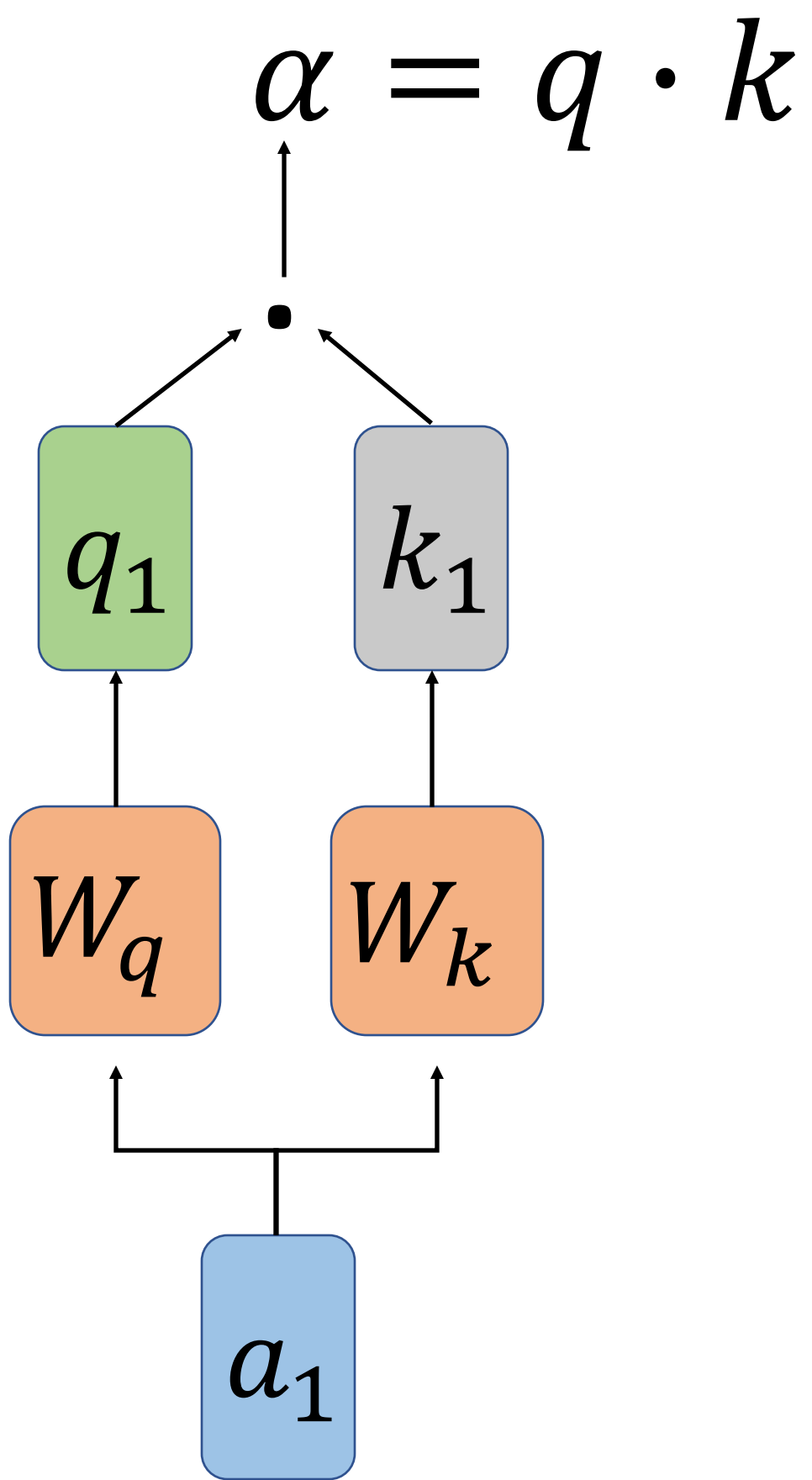


Self-attention

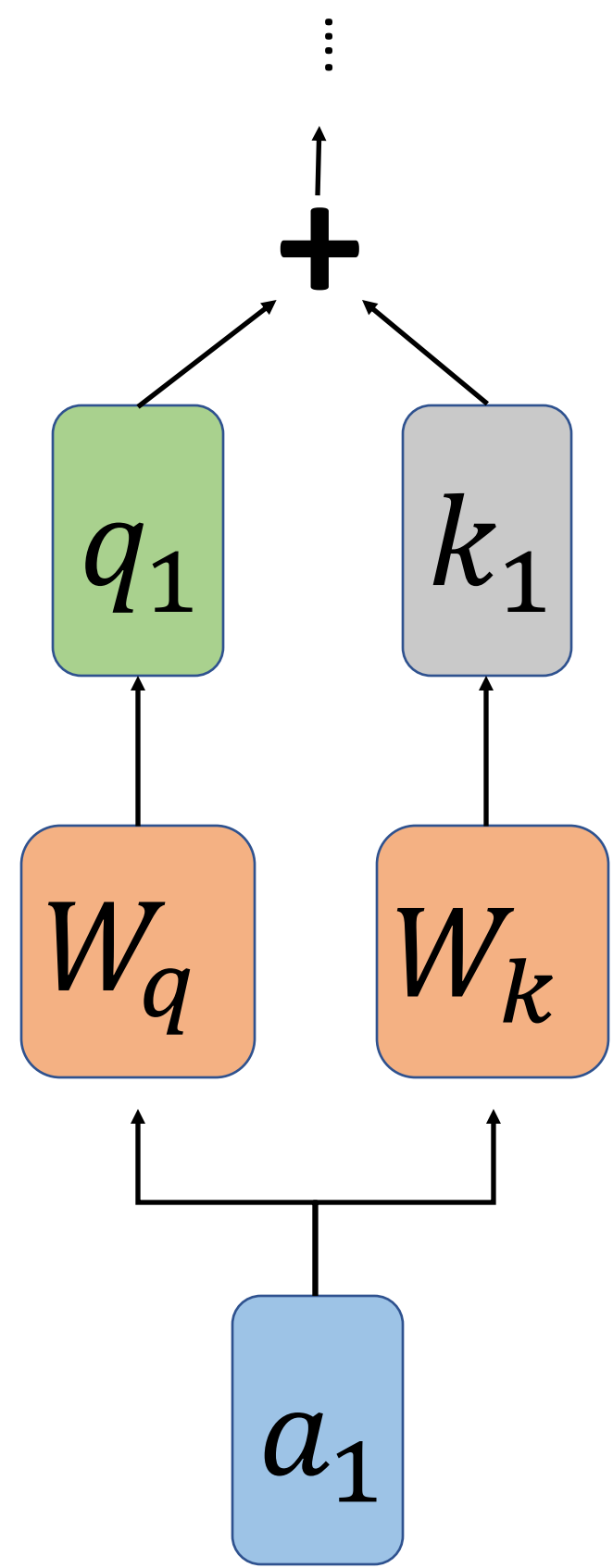


Self-attention

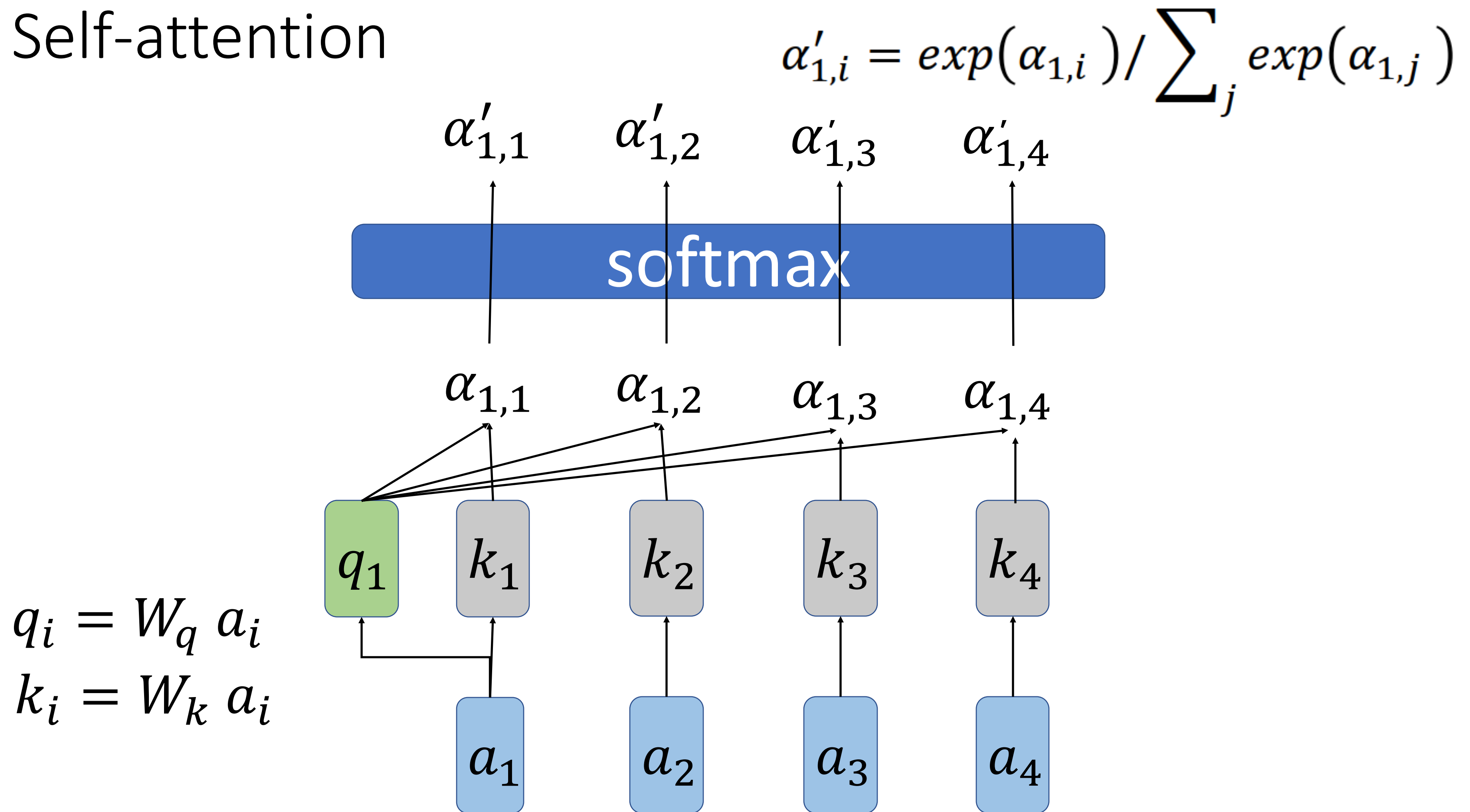
Dot Product



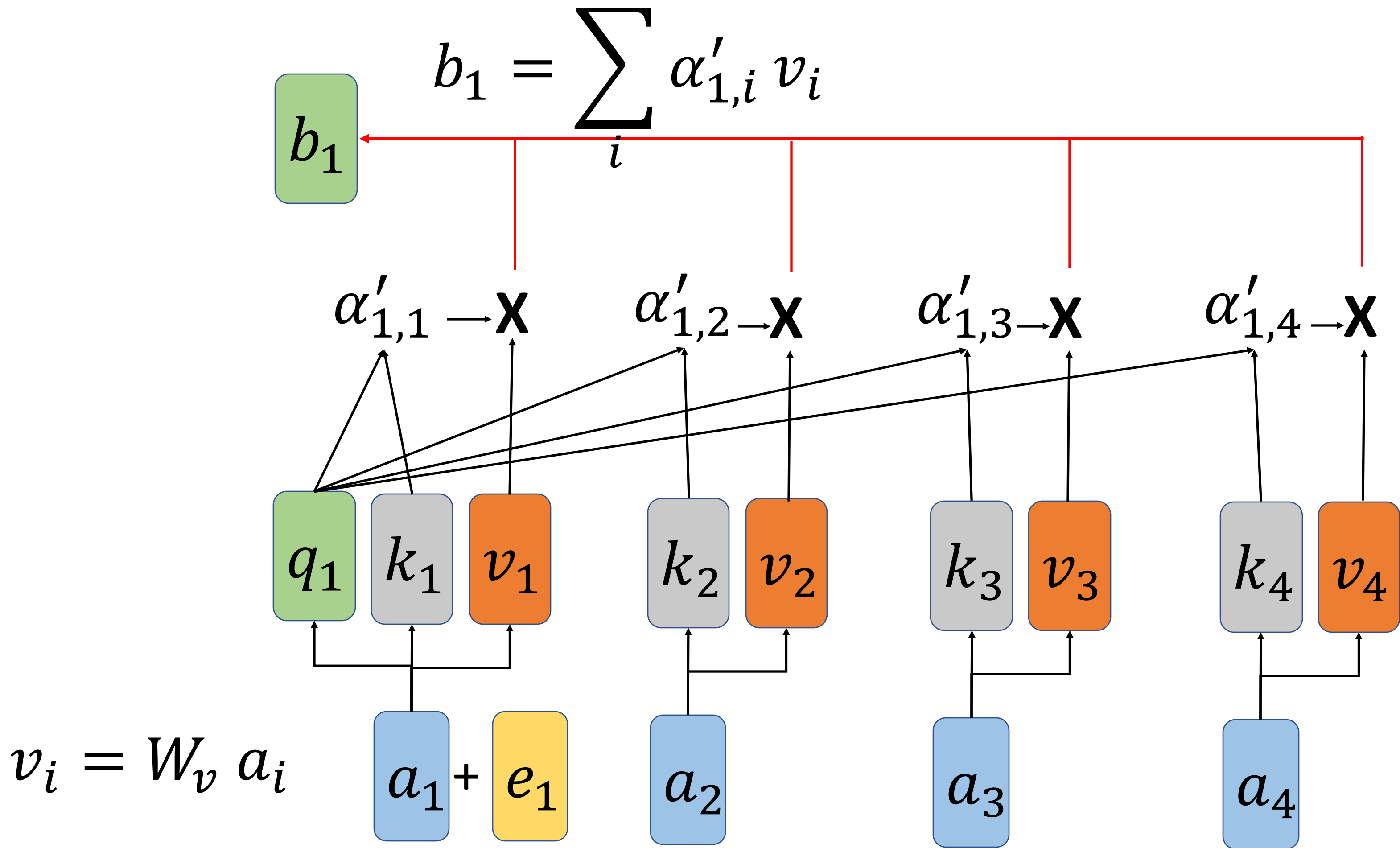
Additive

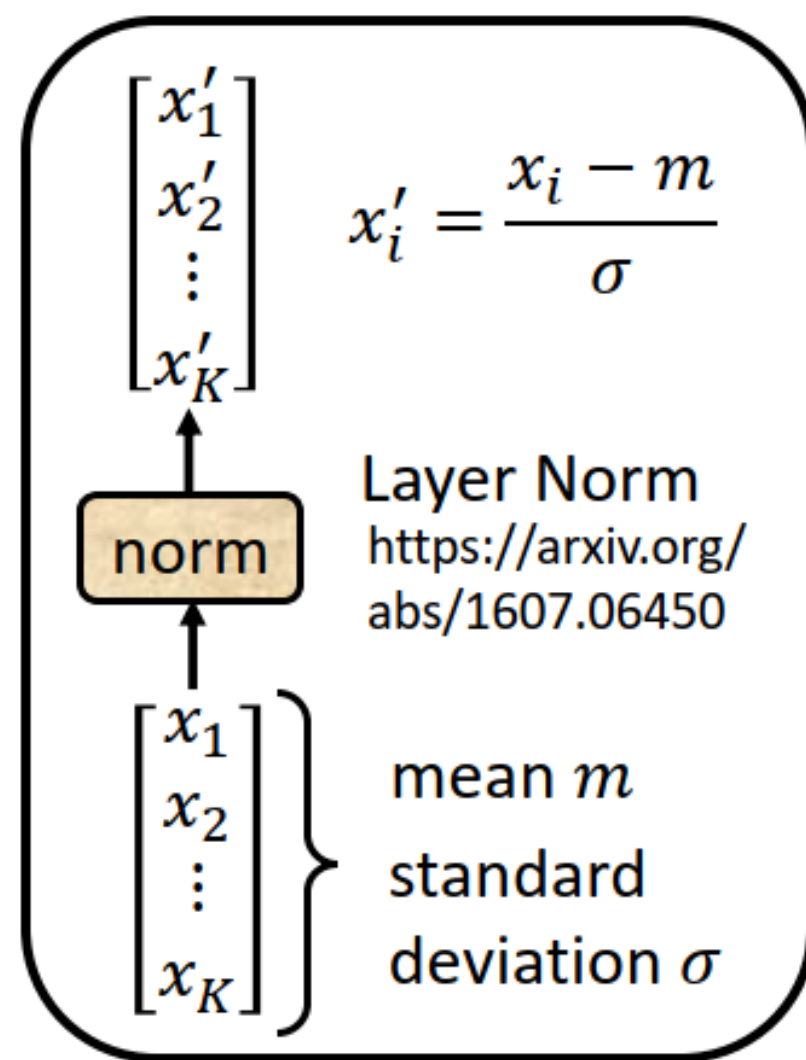


Self-attention

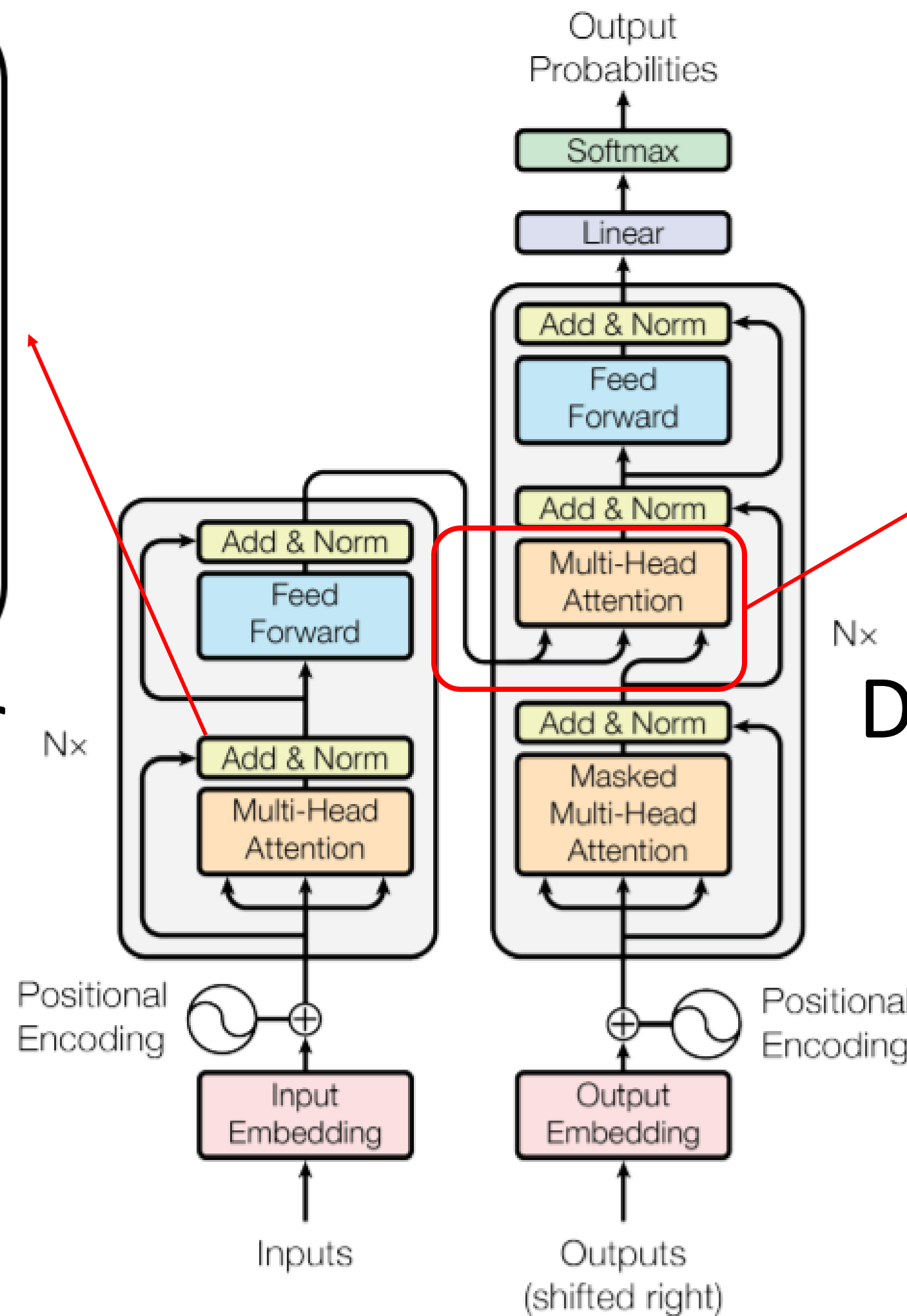


Self-attention





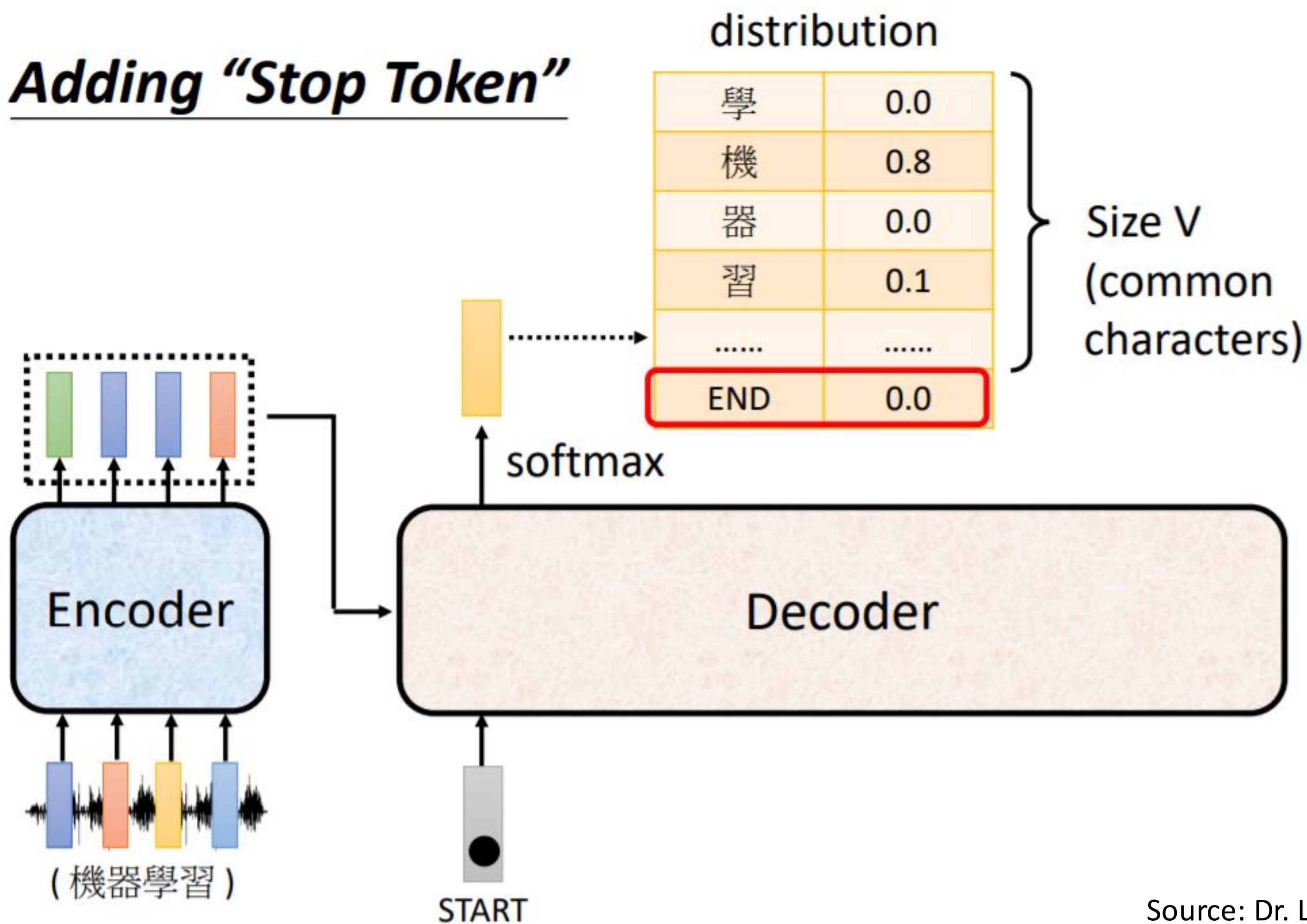
Encoder



Cross Attention

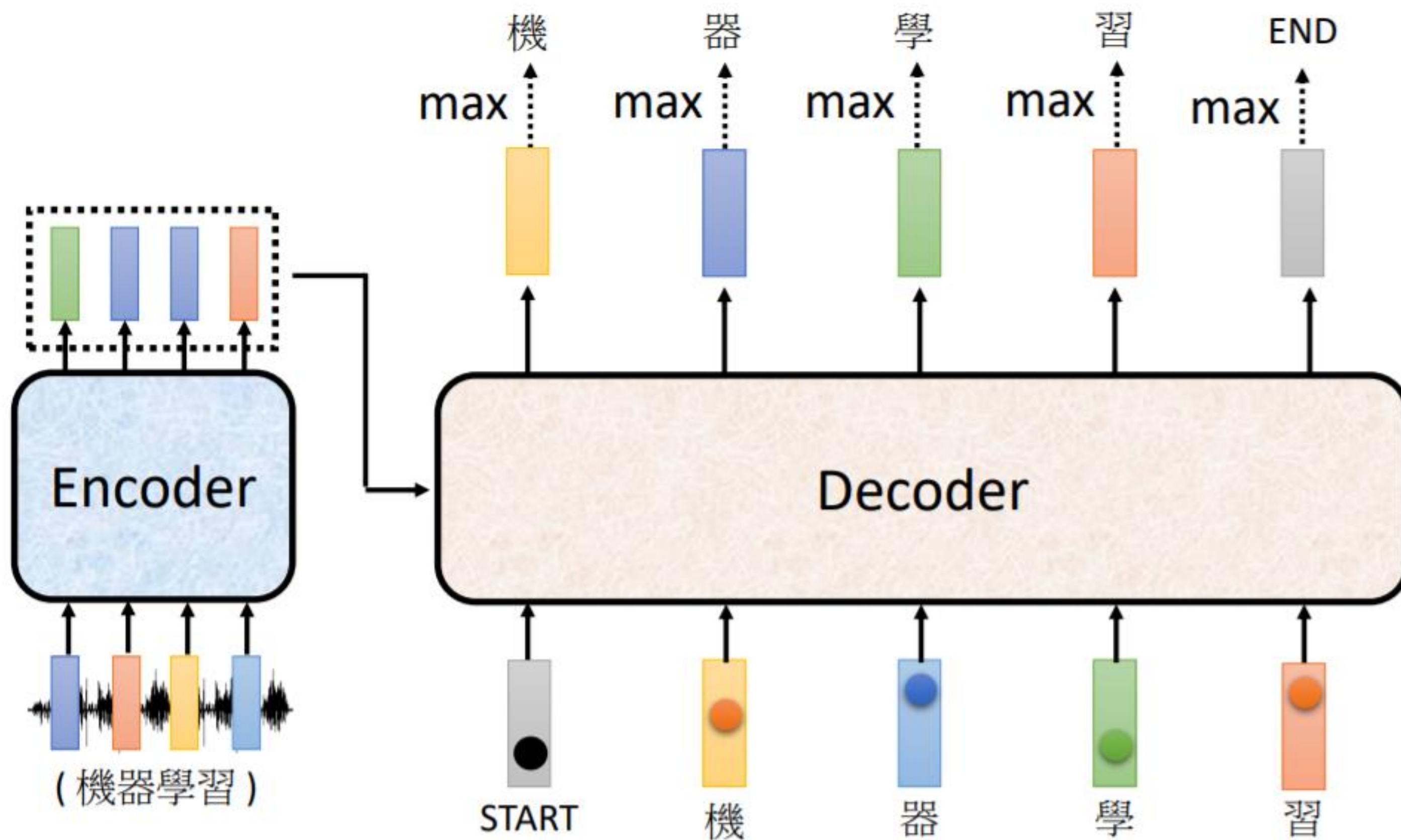
Decoder

Adding "Stop Token"



Autoregressive

Stop at here!



Source: Dr. Lee Hung-Yi

Natural Language Processing

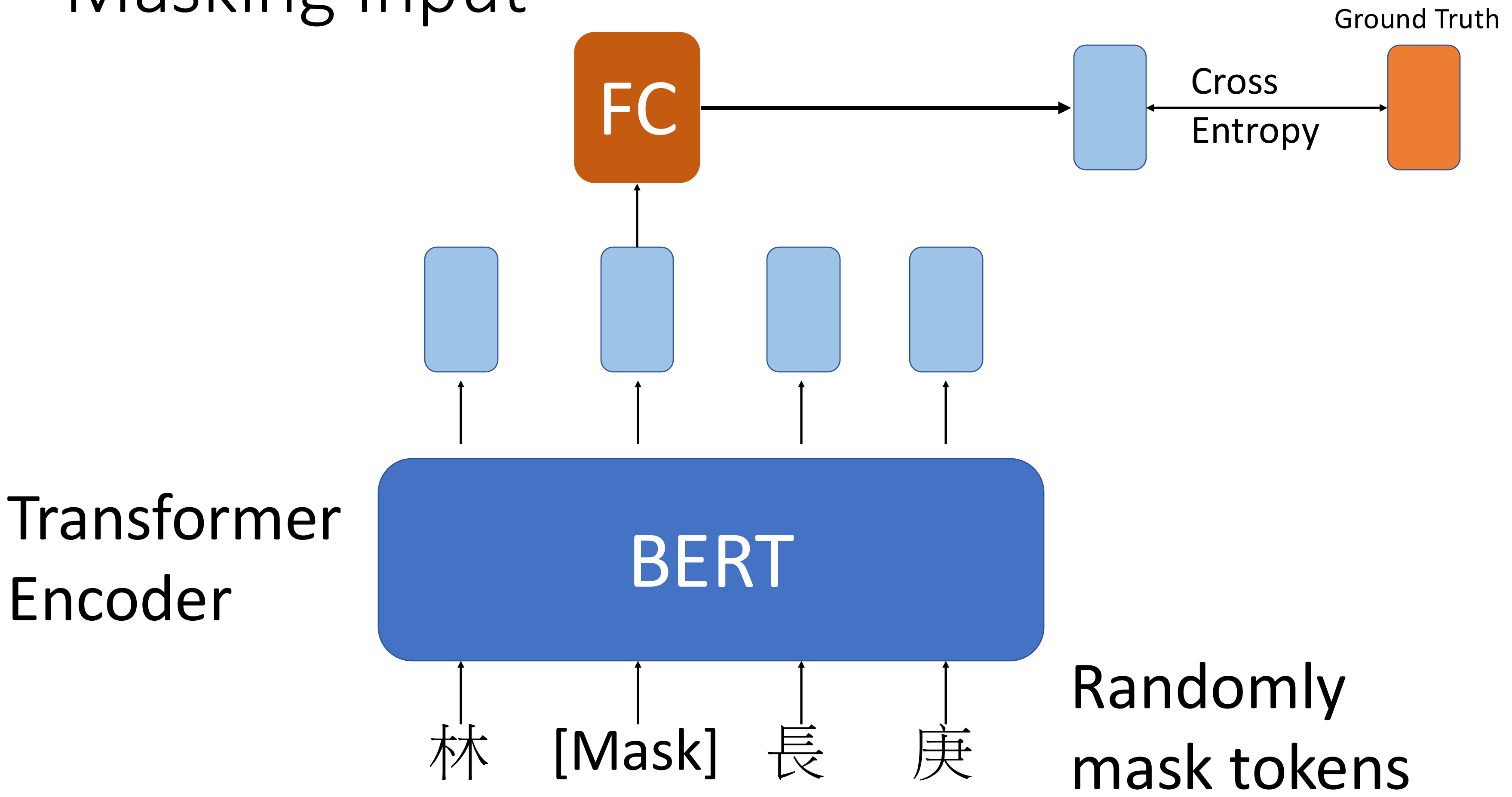
- NLP Tasks
- Transformer
- **BERT**
- GPT

BERT (Bidirectional Encoder Representations from Transformers)

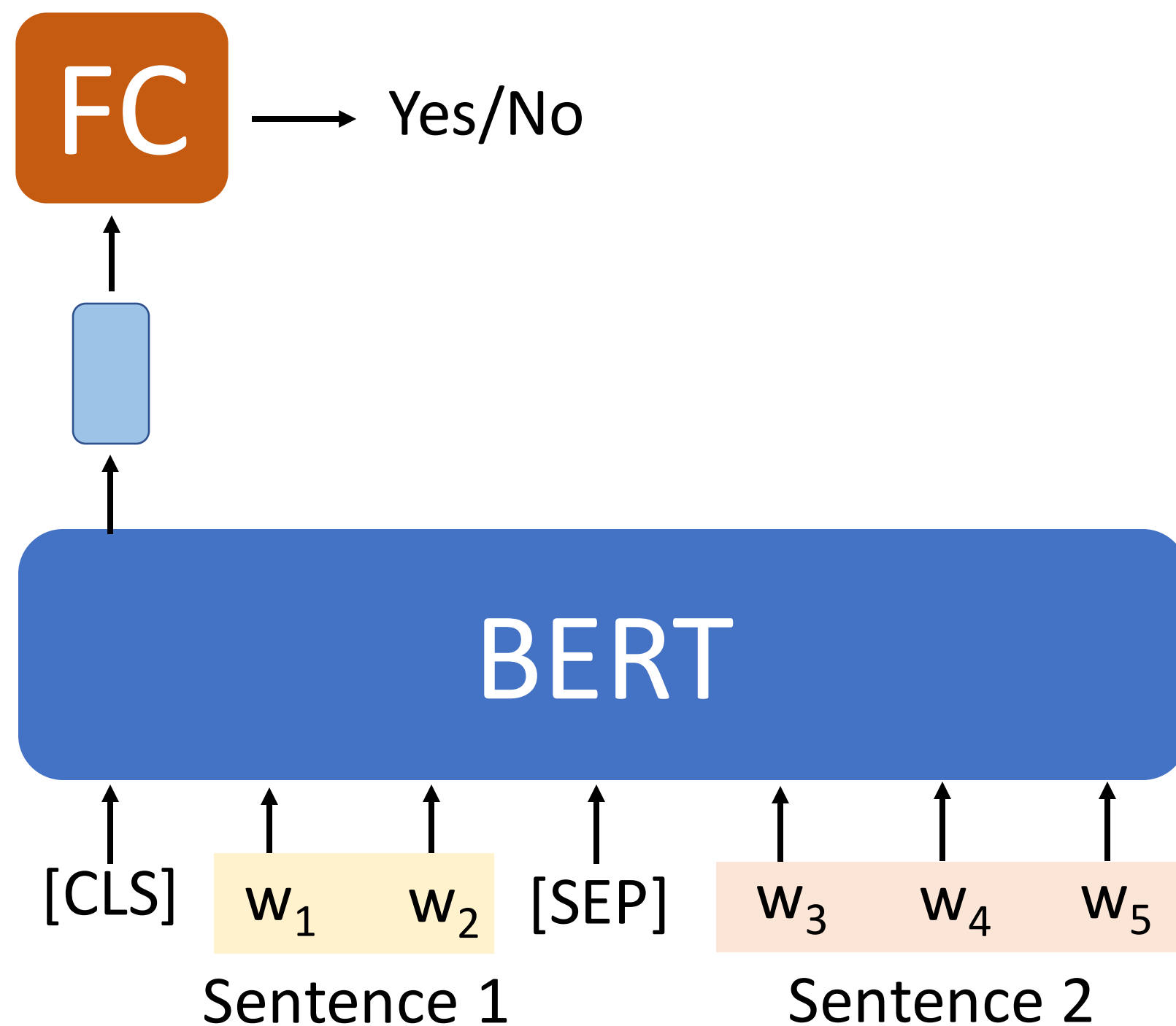
- Self-supervised Learning
 - 3 Billions of words
 - Masking Input
 - Next Sentence Prediction
 - SOP: Sentence Order Prediction (ALBERT)
- Encoder of Transformer

<https://arxiv.org/abs/1810.04805>

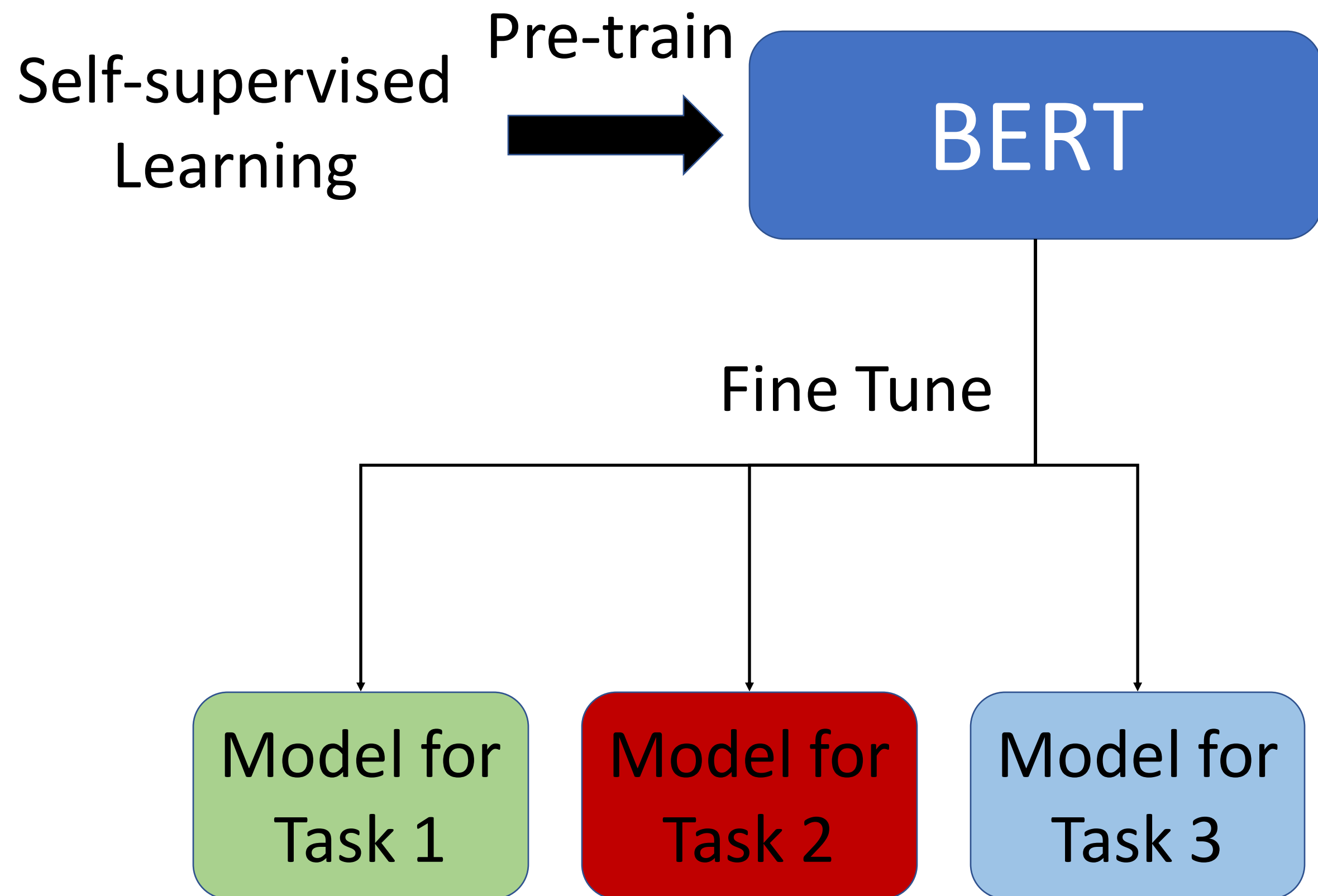
Masking Input



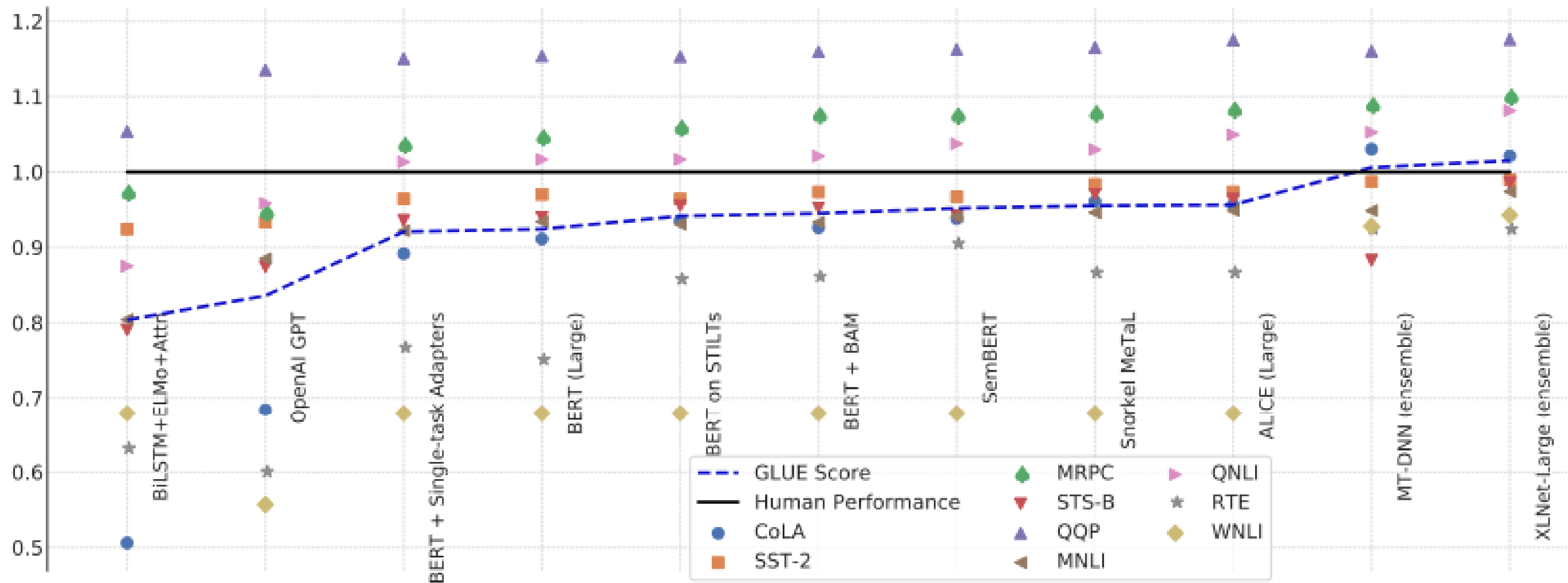
Next Sentence Prediction



- RoBERTa (Robustly optimized BERT approach)
<https://arxiv.org/abs/1907.11692>
- ALBERT (use SOP)
<https://arxiv.org/abs/1909.11942>

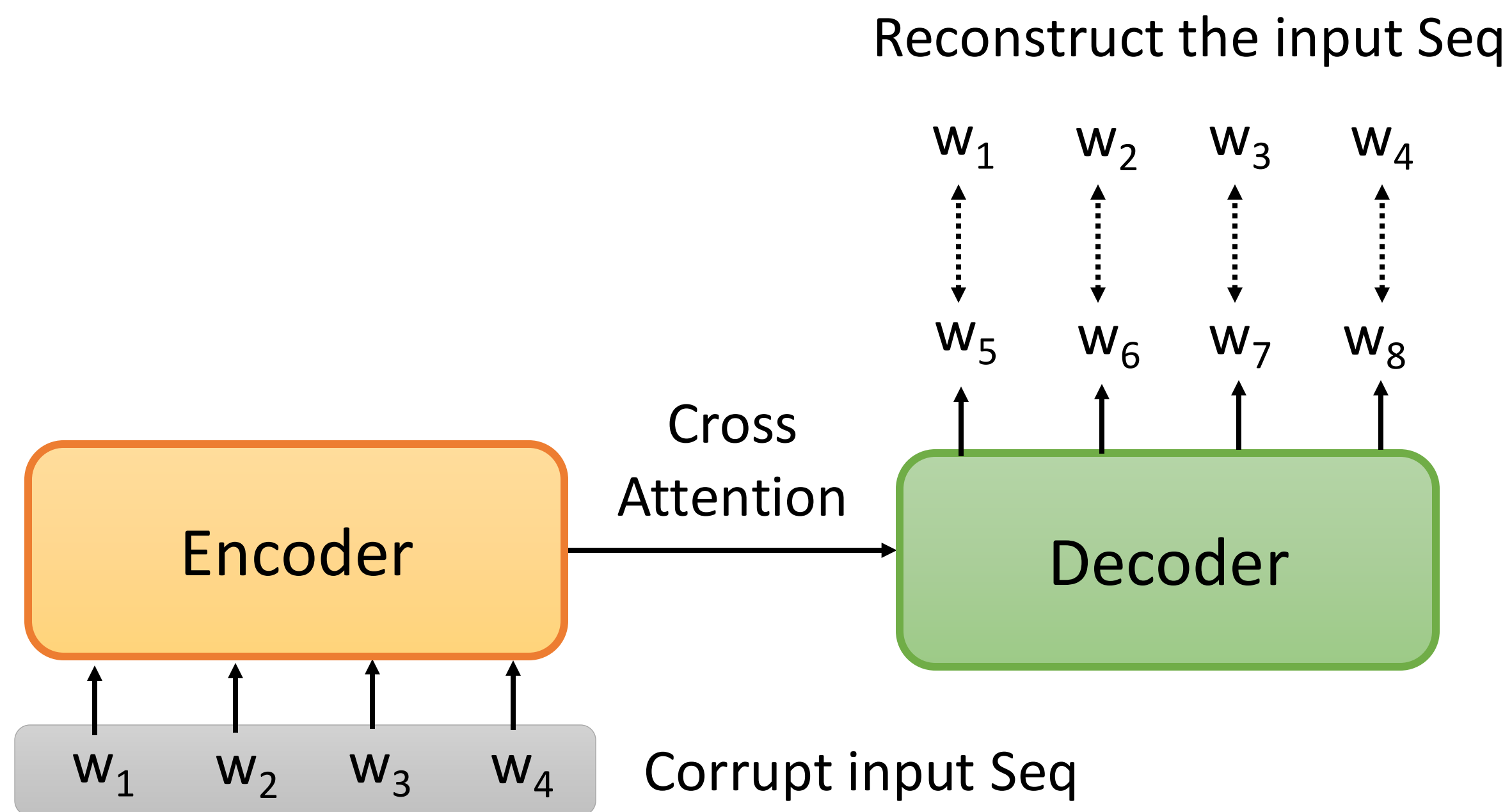


Performance of BERT Family



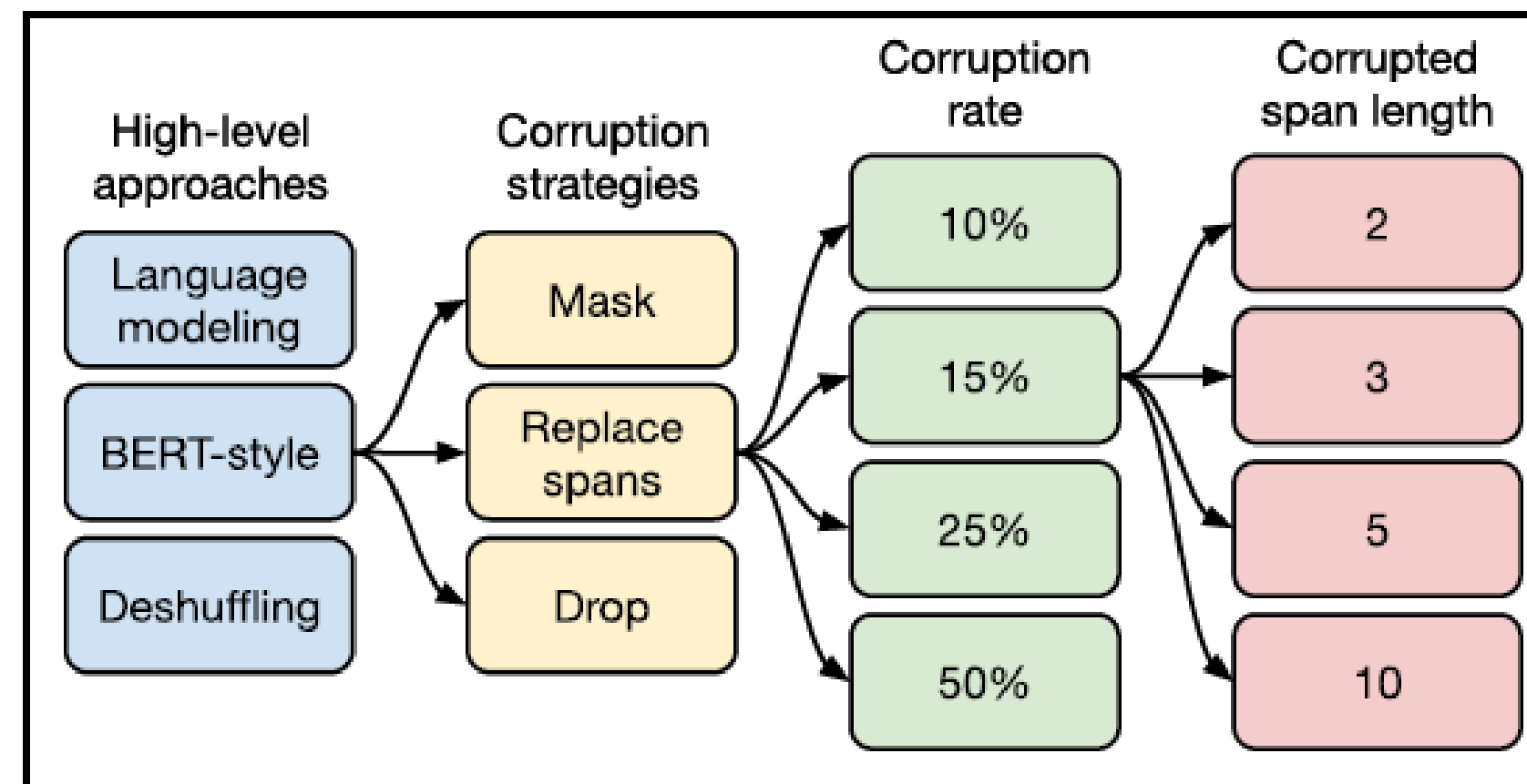
<https://arxiv.org/abs/1905.00537>

Pre-training a Seq2seq Model

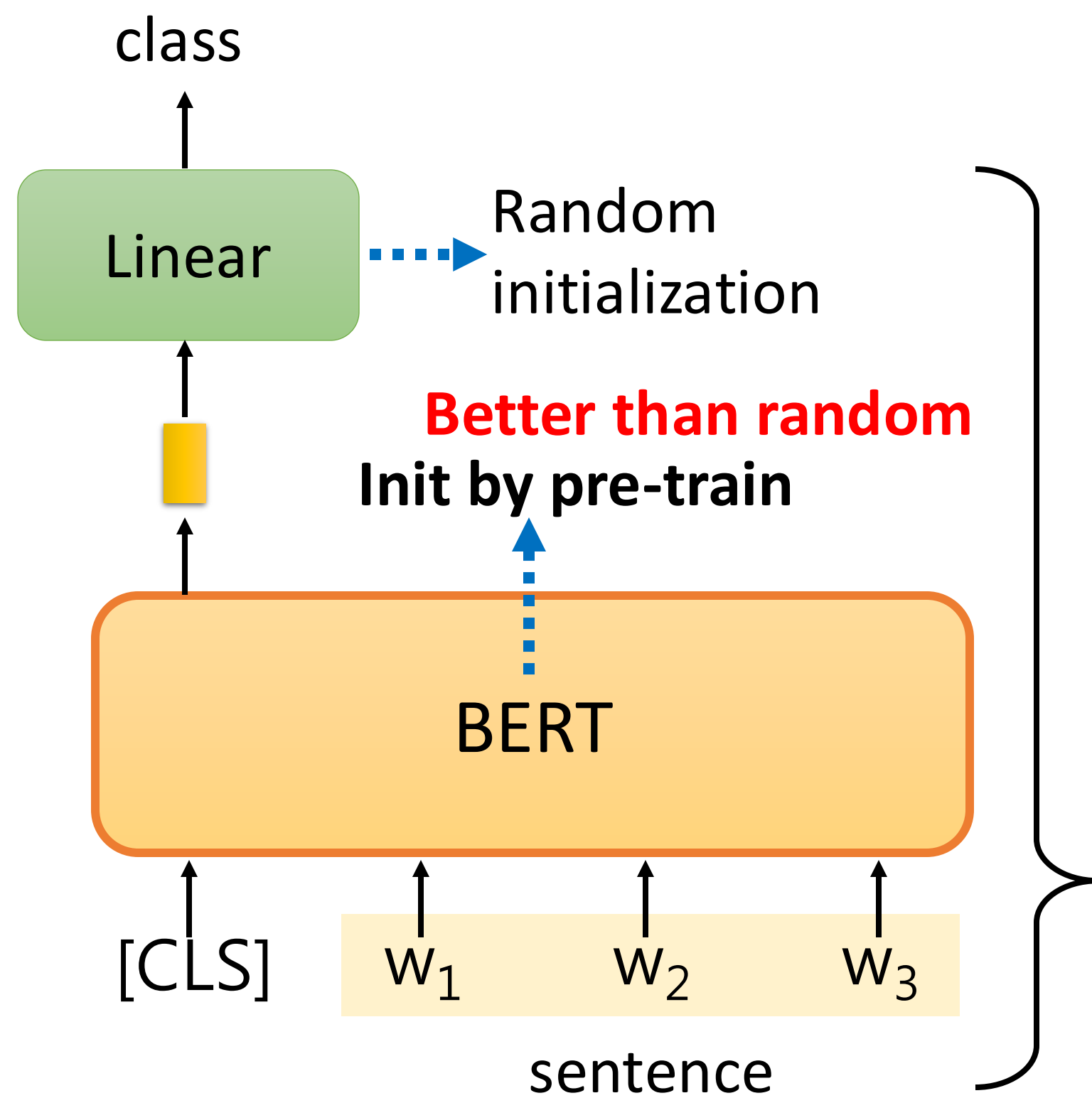


T5 (Text-To-Text Transfer Transformer) - Comparison

Objective	Inputs	Targets
Prefix language modeling	Thank you for inviting	me to your party last week .
BERT-style	Thank you <M> <M> me to your party apple week .	<i>(original text)</i>
Deshuffling	party me for your to . last fun you inviting week Thank	<i>(original text)</i>
I.i.d. noise, mask tokens	Thank you <M> <M> me to your party <M> week .	<i>(original text)</i>
I.i.d. noise, replace spans	Thank you <X> me to your party <Y> week .	<X> for inviting <Y> last <Z>
I.i.d. noise, drop tokens	Thank you me to your party week .	for inviting last
Random spans	Thank you <X> to <Y> week .	<X> for inviting me <Y> your party last <Z>

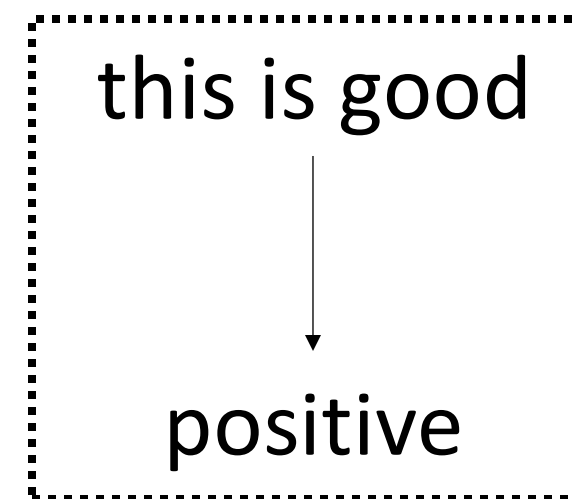


<https://arxiv.org/abs/1910.10683v3>

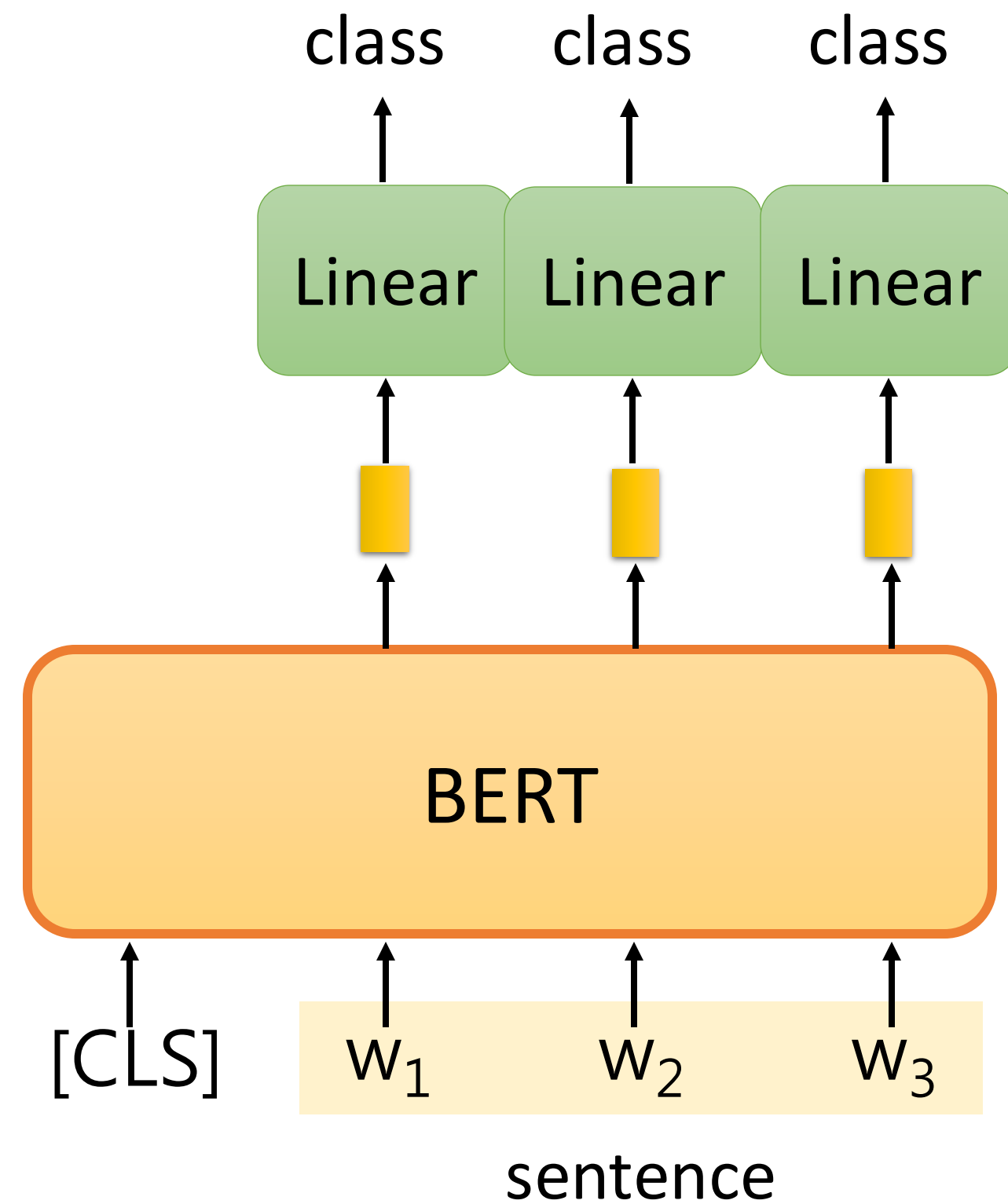


Input: sequence
output: class

Example:
Sentiment analysis

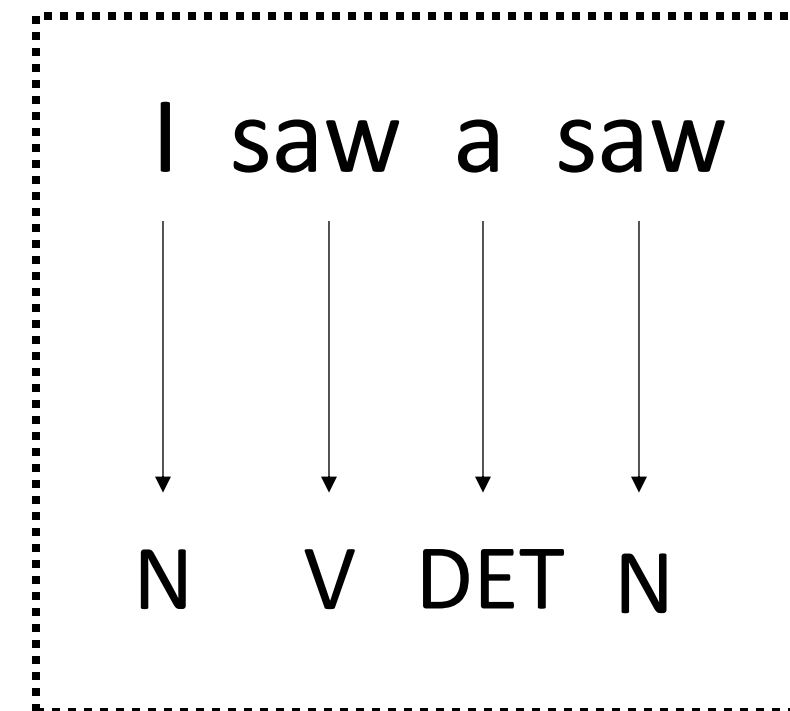


This is the model to
be learned.

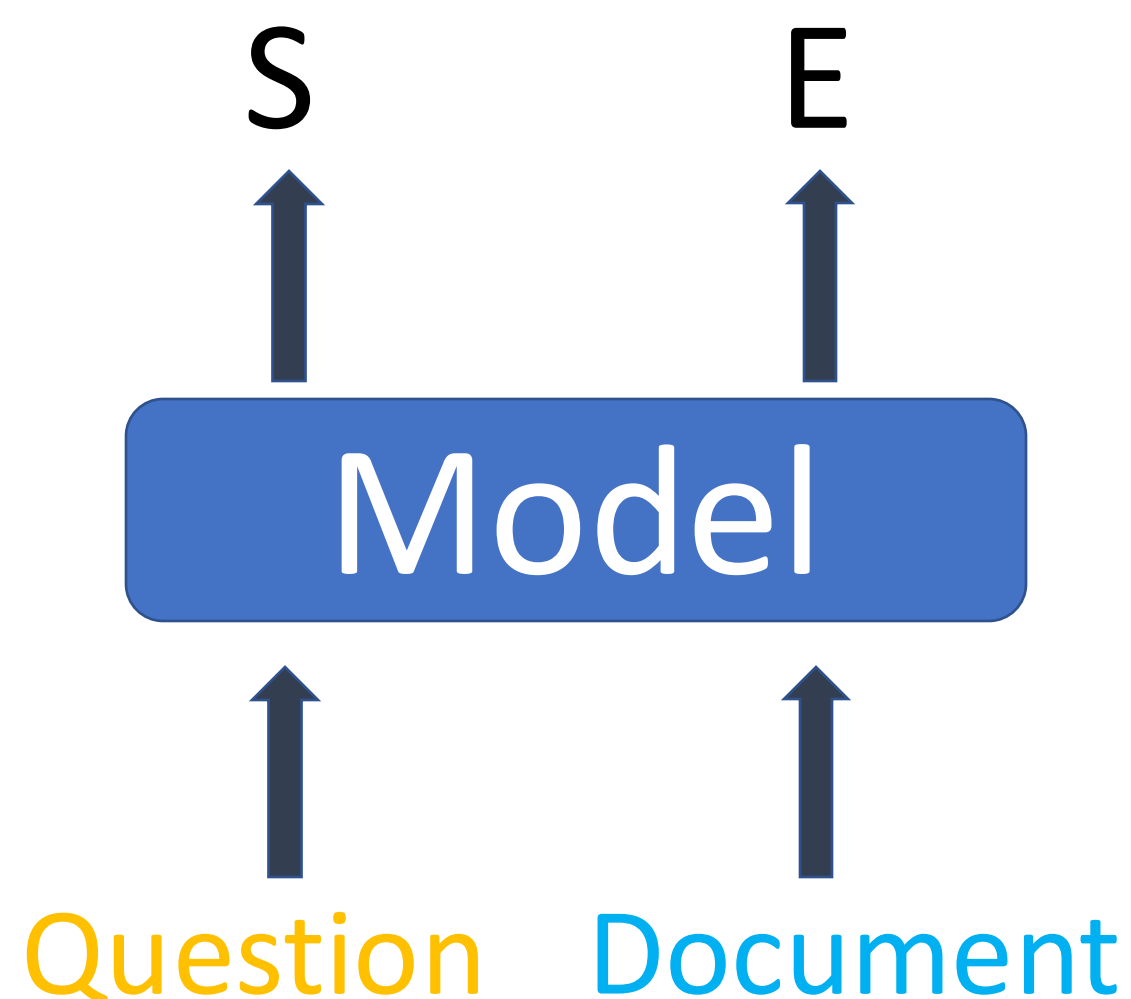


Input: sequence
output: same as input

Example:
POS tagging



Extractive QA



In meteorology, precipitation is any product of the condensati¹⁷ of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via⁷⁷ collision⁷⁹ other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called "showers".

What causes precipitation to fall?

gravity

S= 17, E=17

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?

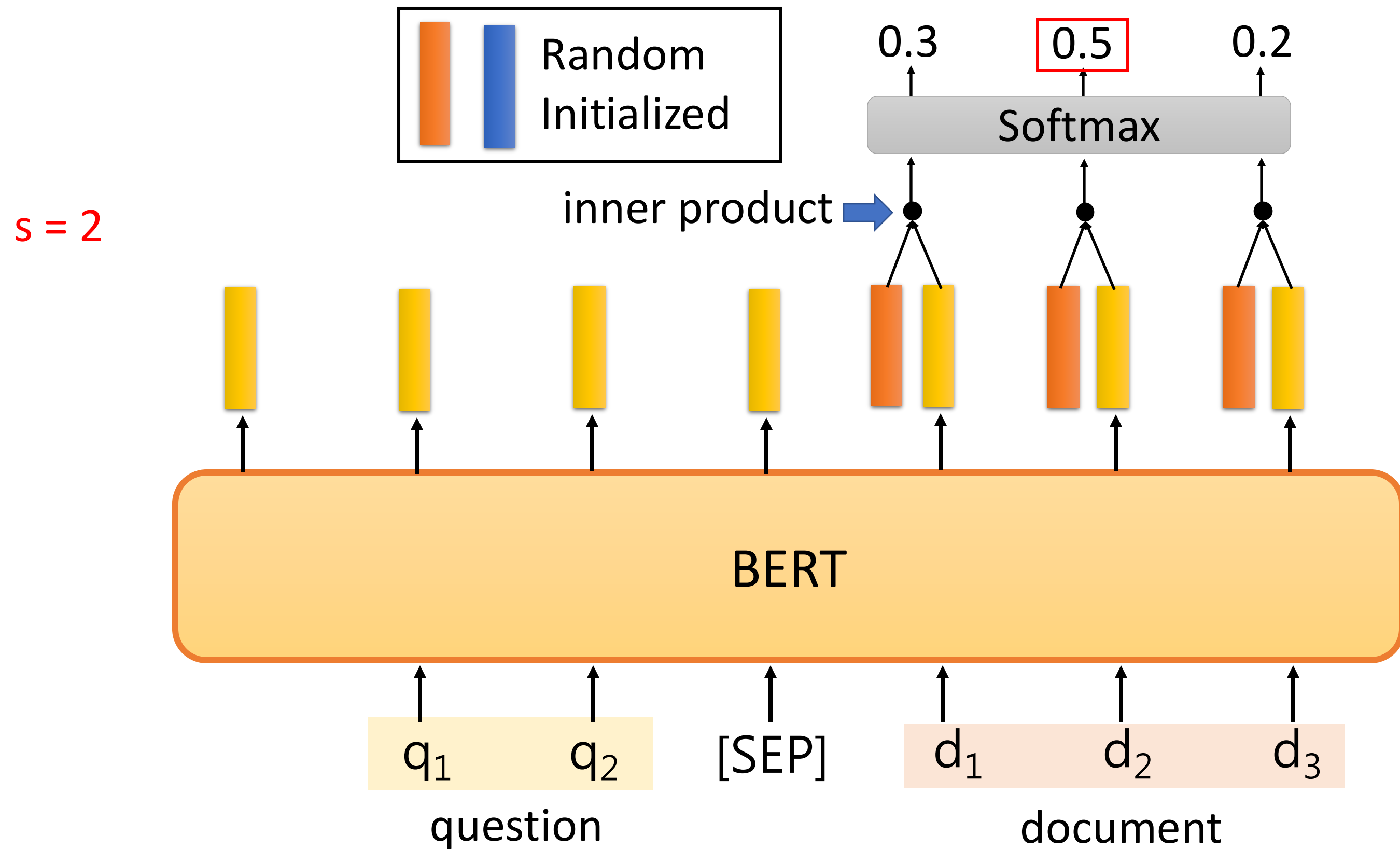
graupel

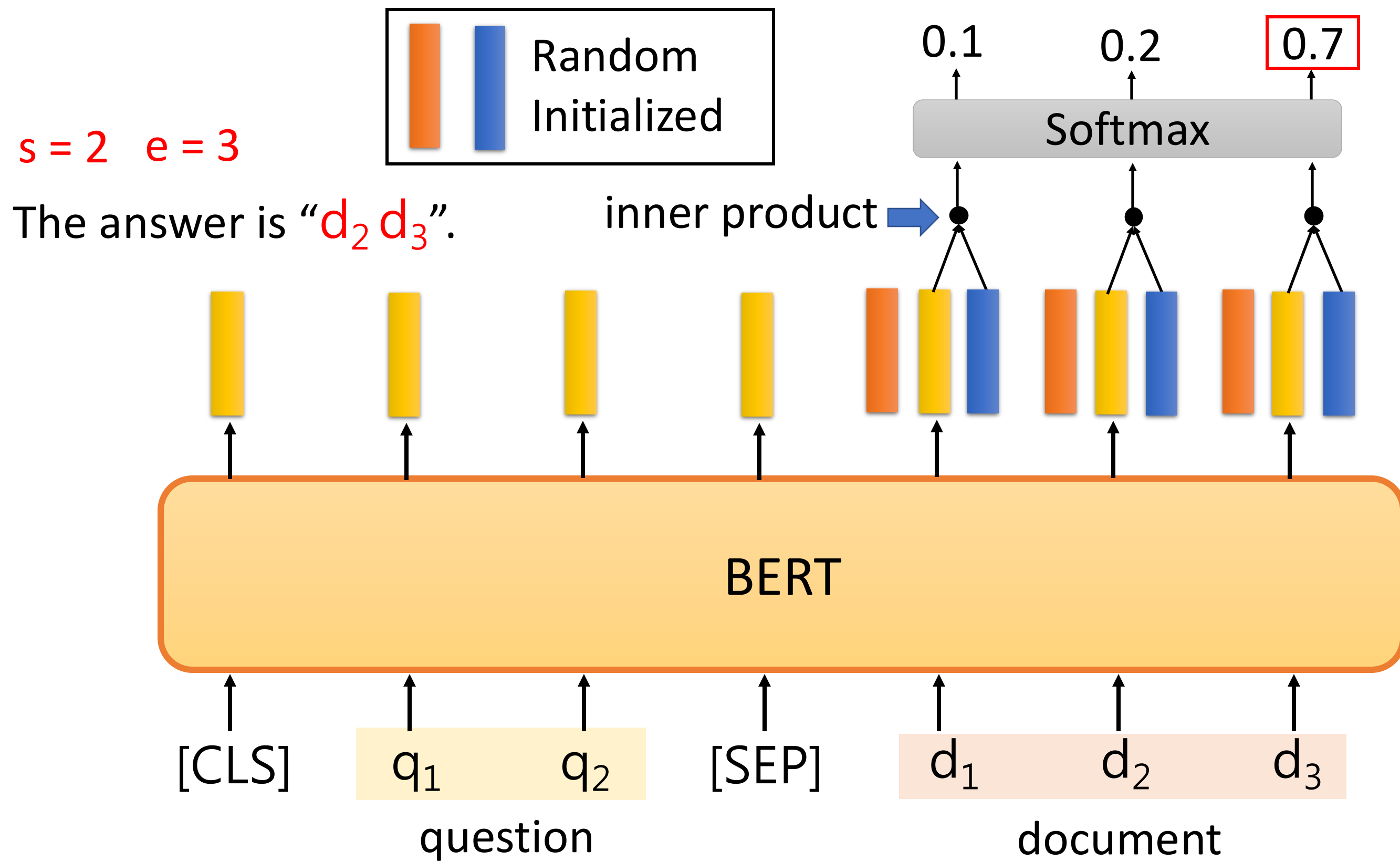
Where do water droplets collide with ice crystals to form precipitation?

within a cloud

S= 77, E=79

Source: Dr. Lee Hung-Yi





•Applying BERT to **protein, DNA, music classification**



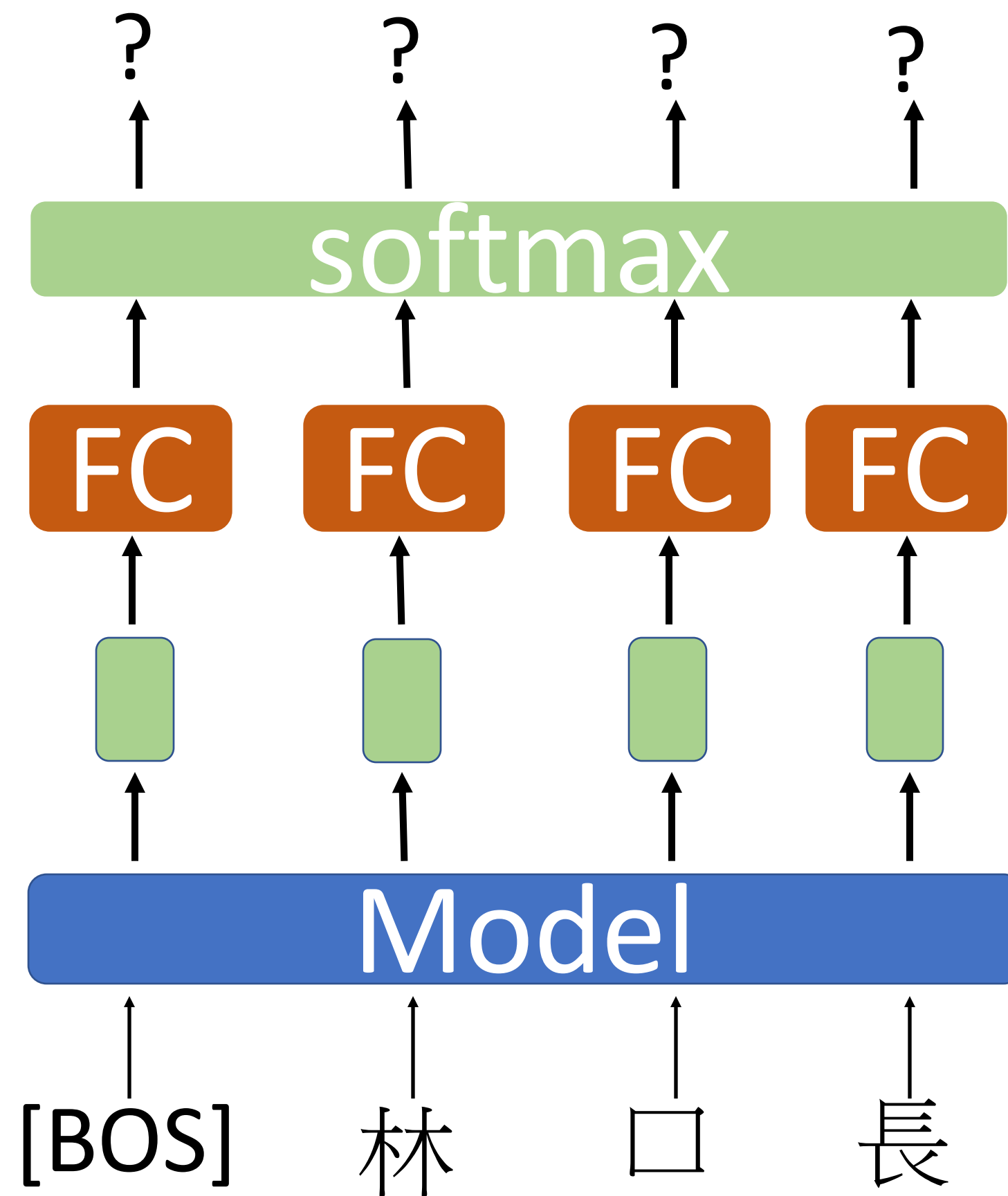
EI CCAGCTGCATCACAGGAGGCCAGCGAGCAGGGTCTGTTCCTCAAGG
EI AGACCCGCCGGGAGGGCGGAGGACCTGCAGGGGTGAGCCCCACCC
IE AACGTGGCCTCCTTGTGCCCTTCCCCACAGTGCCCTCTTCCAGGA
IE CCACTCAGCCAGGCCCTTCTTCTCCTCCAGGTCCCCCACGGGCCCT
IE CCTGATCTGGGTCTCCCCTCCCACCCTCAGGGAGCCAGGCTCGGG
IE AGCCCTCAACCCTTCTGTCTCACCCCTCCAGCCTAAAGCTCCTTGAC
IE CCACTCAGCCAGGCCCTTCTTCTCCTCCAGGTCCCCCACGGGCCCT
N CTGTGTTCAACCACATCAAGCGCCGGGACATCGTGCTCAAGTGGGA
N GTGTTACCGAGGGGCATTTCTAACAGTCTTCTTACTACGGCCTCCGC
N TCTGAGCTCTGCATTTGTCTATTCTCCAGCTGACCCTGGTTCTCTCT

	Protein			DNA				Music
	localization	stability	fluorescence	H3	H4	H3K9ac	Splice	composer
specific	69.0	76.0	63.0	87.3	87.3	79.1	94.1	-
BERT	64.8	74.5	63.7	83.0	86.2	78.3	97.5	55.2
re-emb	63.3	75.4	37.3	78.5	83.7	76.3	95.6	55.2
rand	58.6	65.8	27.5	75.6	66.5	72.8	95	36

Natural Language Processing

- NLP Tasks
- Transformer
- BERT
- **GPT**

GPT – Predict Next Token



Decoder of Transformer

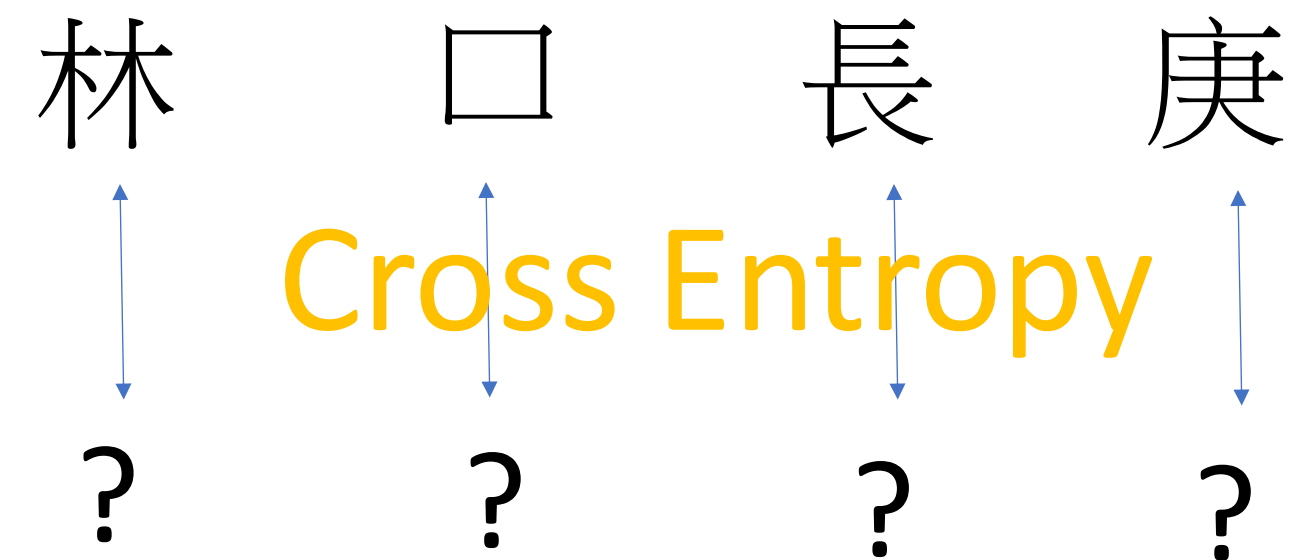
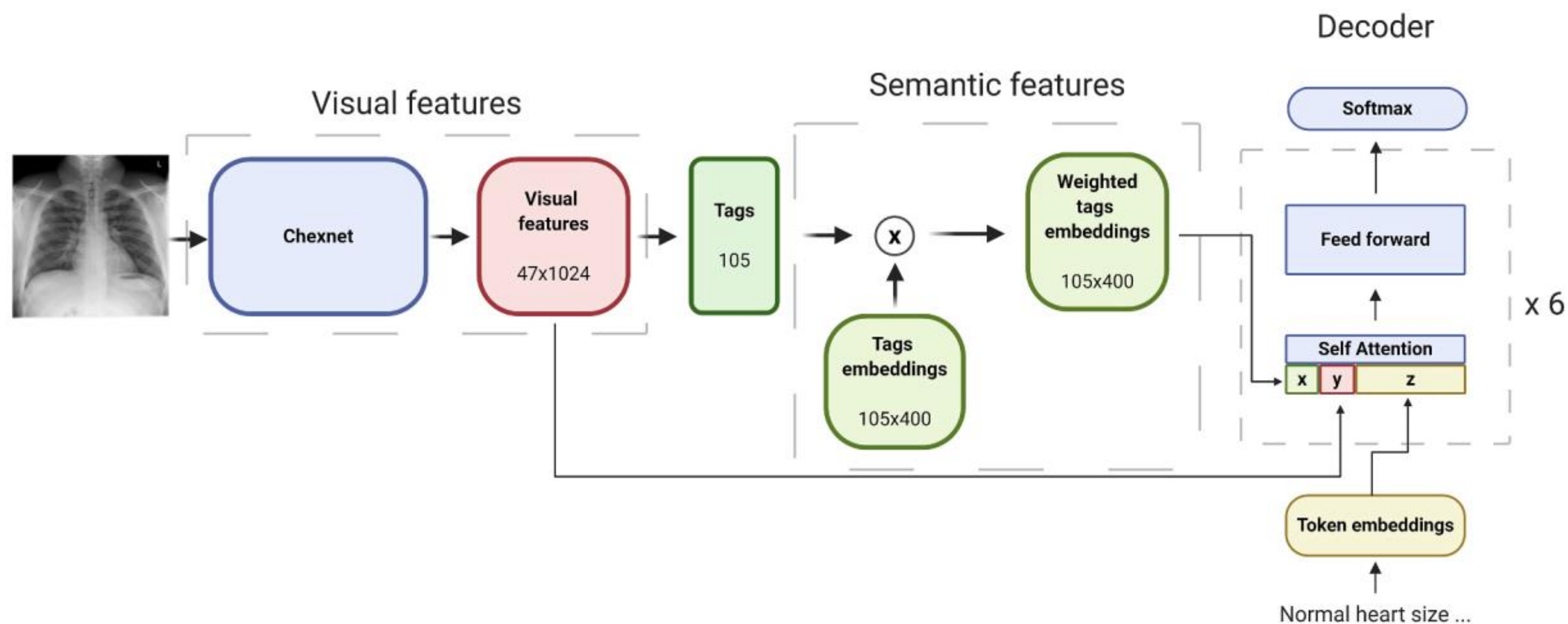
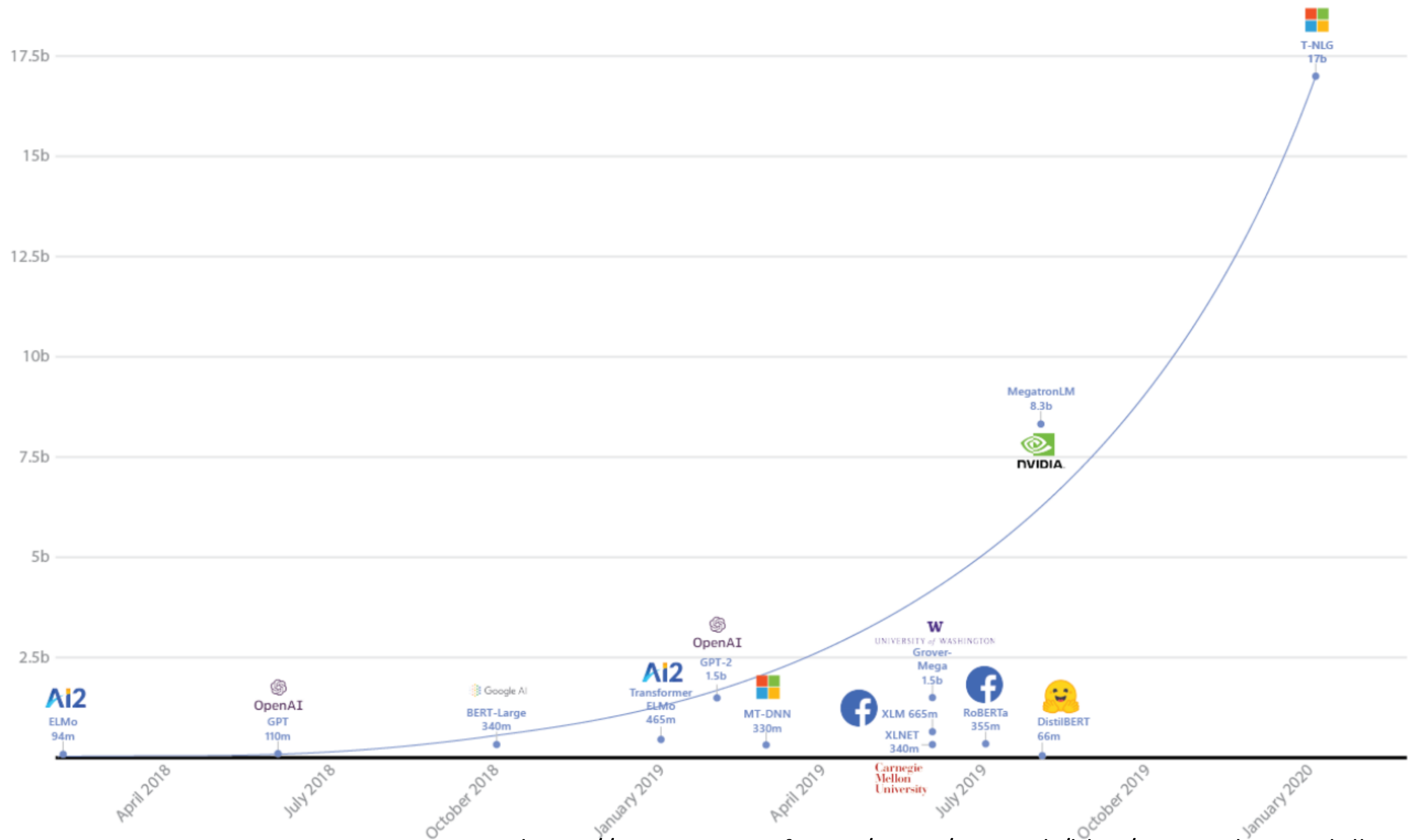


Image to Text





<https://www.microsoft.com/en-us/research/blog/turing-nlg-a-17-billion-parameter-language-model-by-microsoft/>

In-Context Learning

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

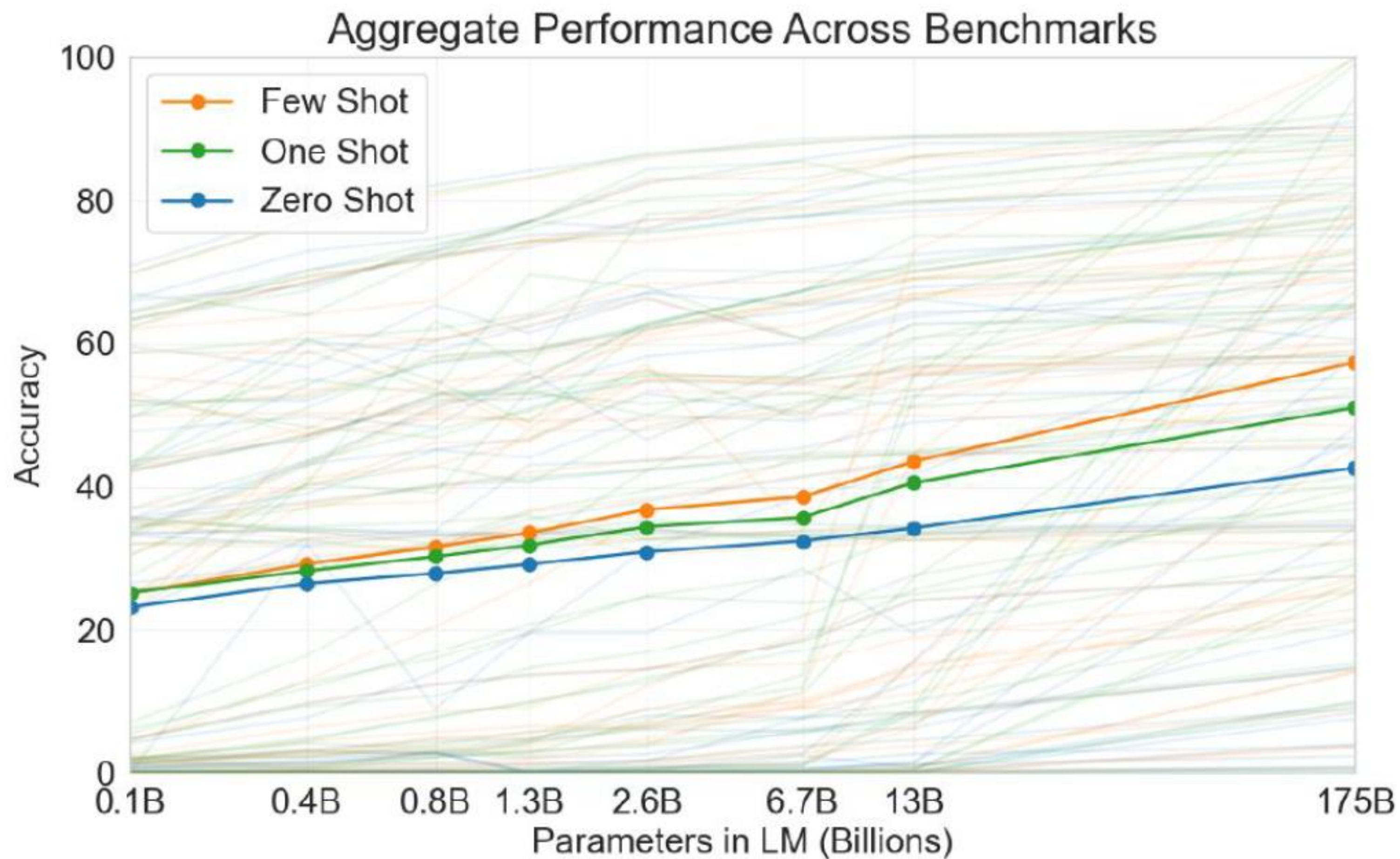
```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ..... ← prompt
```

<https://arxiv.org/abs/2005.14165>



Average of **42** tasks

<https://arxiv.org/abs/2005.14165>

- **2 digit addition (2D+)** – The model is asked to add two integers sampled uniformly from $[0, 100)$, phrased in the form of a question, e.g. “Q: What is 48 plus 76? A: 124.”
- **2 digit subtraction (2D-)** – The model is asked to subtract two integers sampled uniformly from $[0, 100)$; the answer may be negative. Example: “Q: What is 34 minus 53? A: -19”.
- **3 digit addition (3D+)** – Same as 2 digit addition, except numbers are uniformly sampled from $[0, 1000)$.

